
AE4135 – ROTOR/WAKE AERODYNAMICS

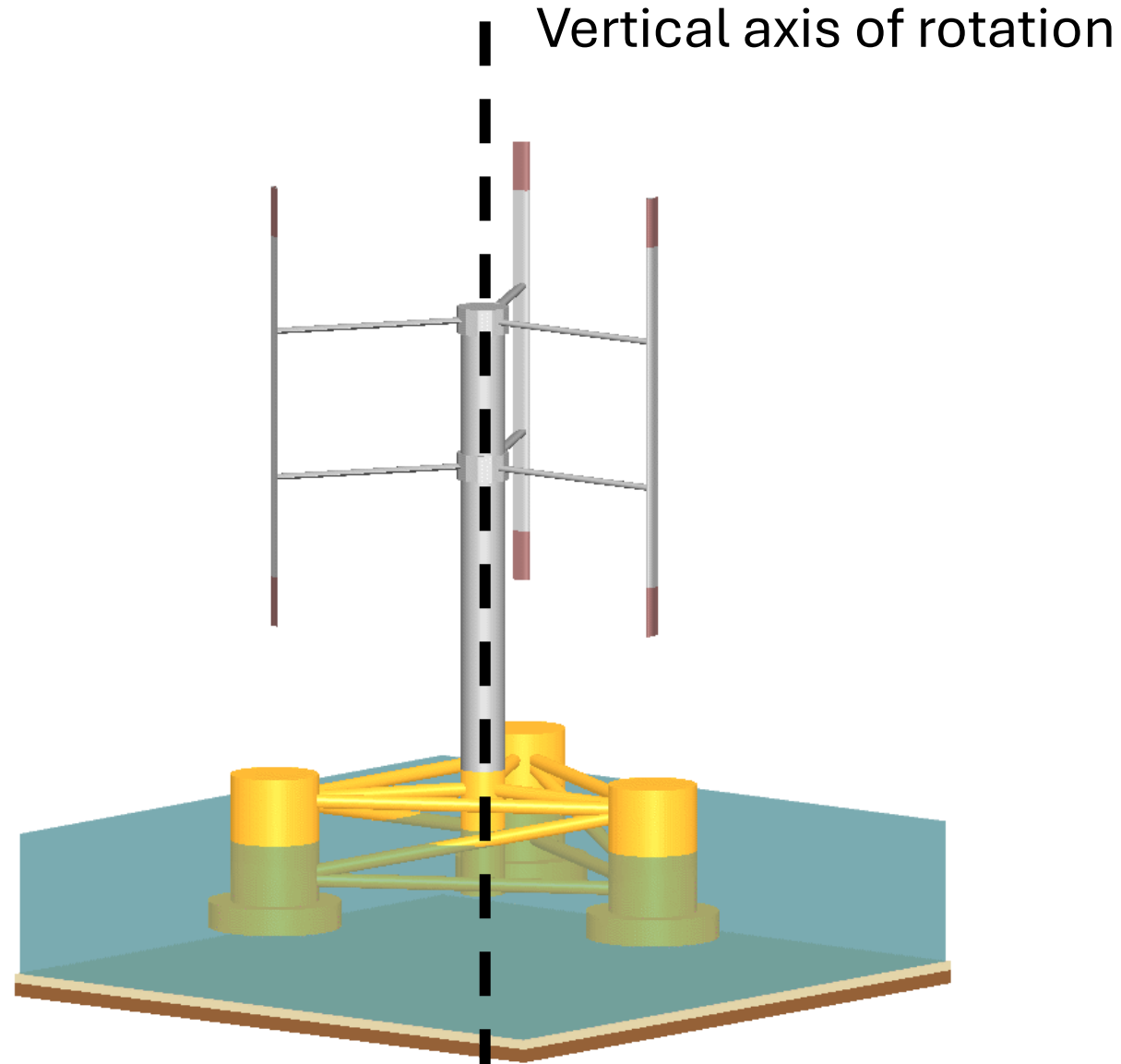
VAWT aerodynamics

LECTURER



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Assistant professor
Large rotor flow physics

VERTICAL-AXIS WIND TURBINES



LEARNING OBJECTIVES

At the end of this lecture, you should be able to:

- Describe the history and future of VAWTs
- Define the basics of VAWT aerodynamics, incl. 2D velocity and force triangle
- Distinguish different modelling techniques and identify the difference between HAWT and VAWT wake dynamics
- Identify the role of unsteady aerodynamics in VAWT behavior

Keywords: (1) History & future, (2) Basic VAWT aerodynamics , (3) Rotor representation, (4) VAWT modelling, (5) Wake dynamics, (6) Unsteady aerodynamics

HISTORY & FUTURE

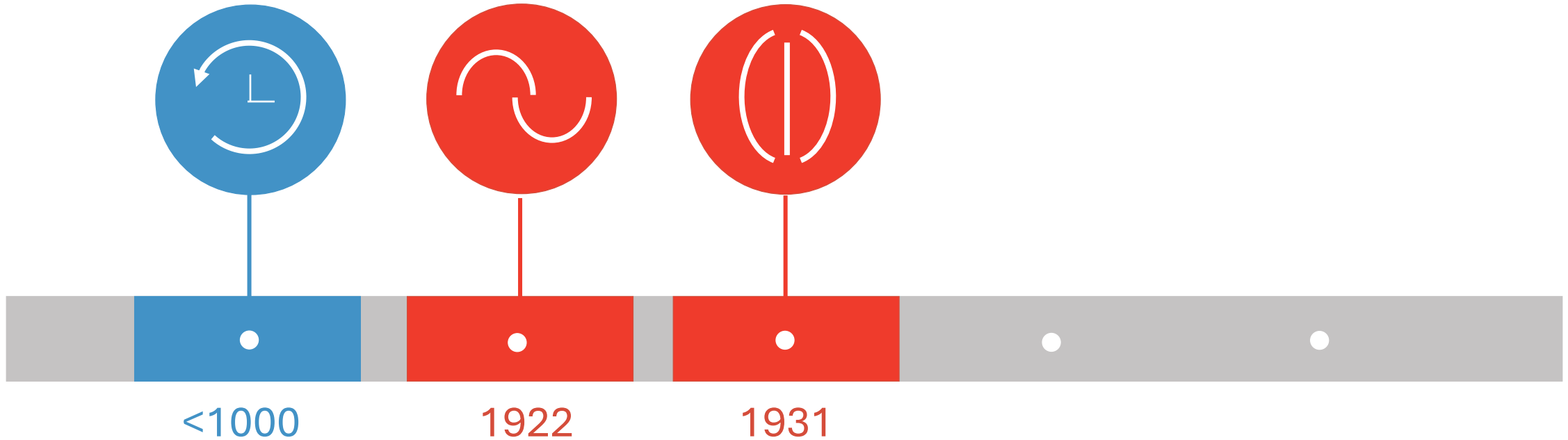
HISTORY



Ancient wind mills



HISTORY



Ancient wind mills Drag-driven VAWT Lift- driven VAWT

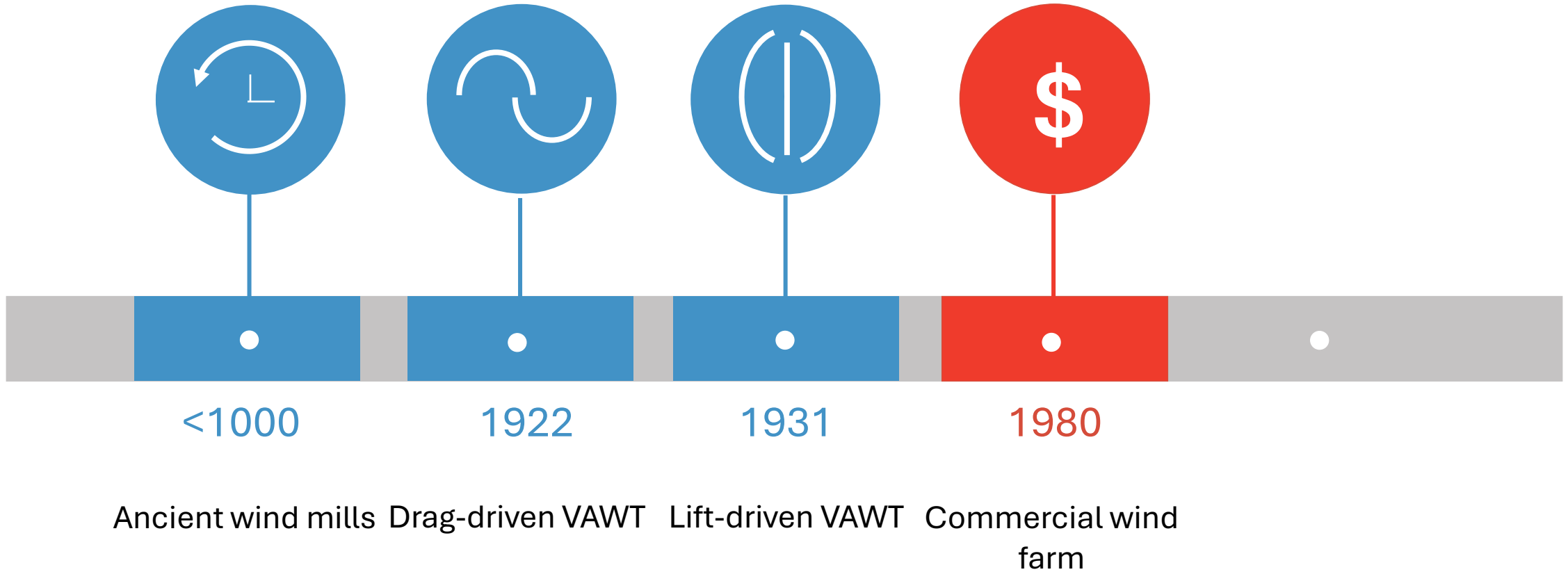
DRAG-DRIVEN



LIFT-DRIVEN

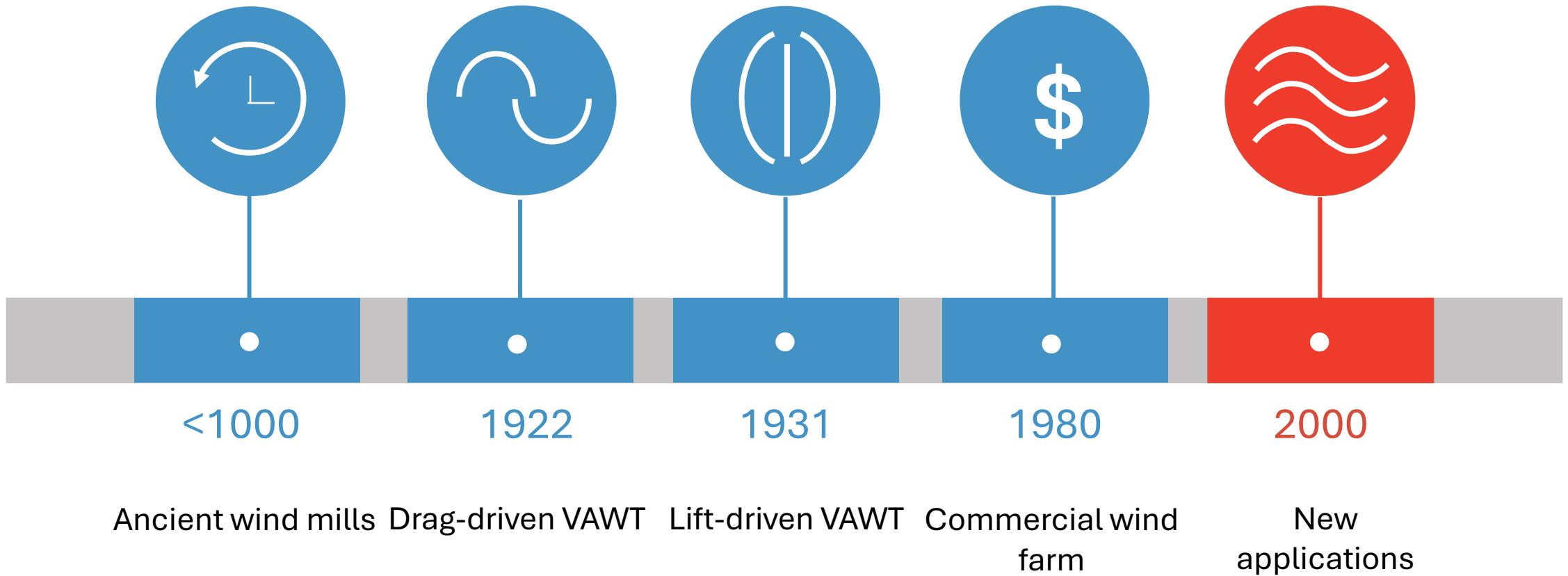


HISTORY

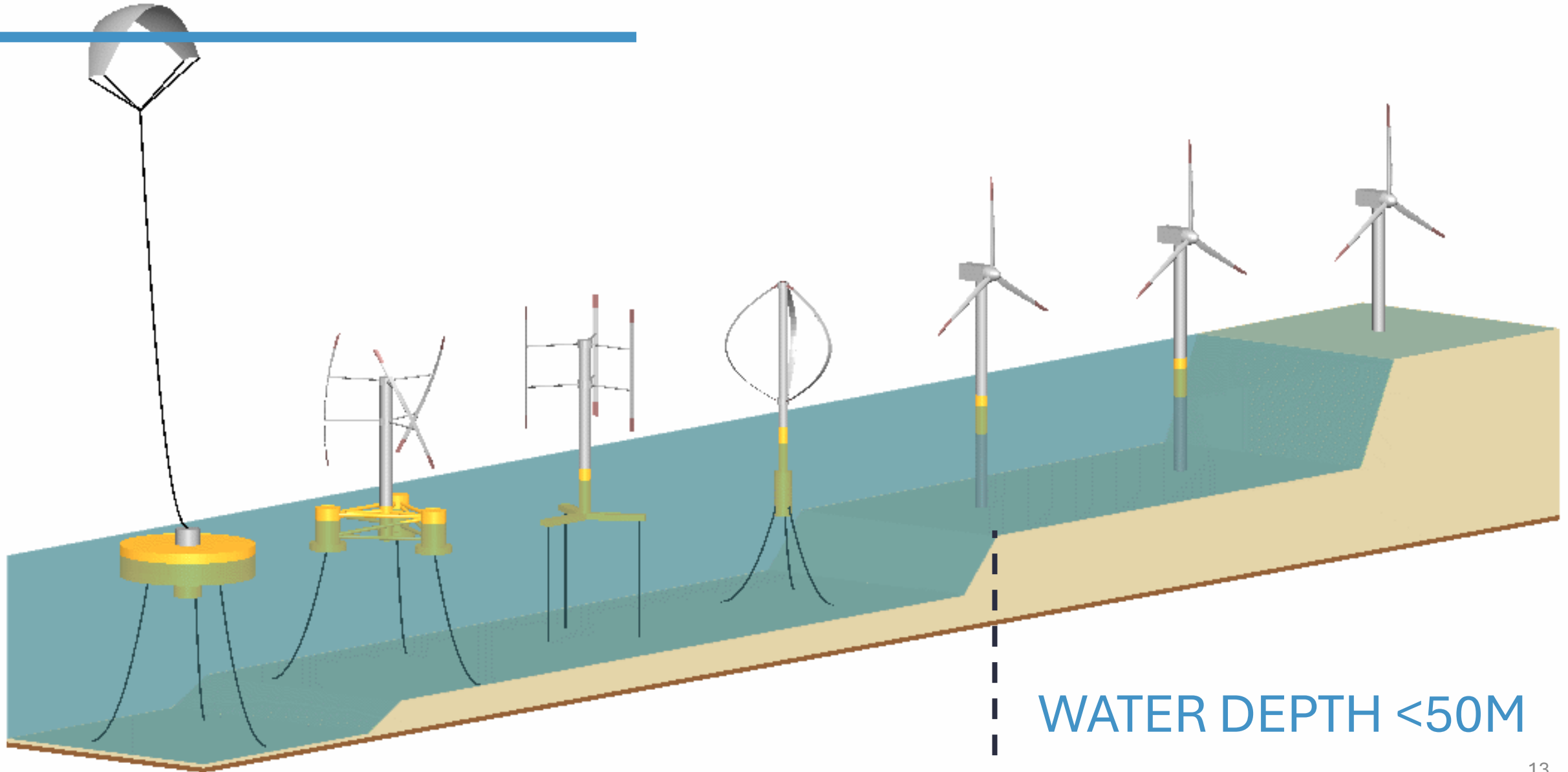




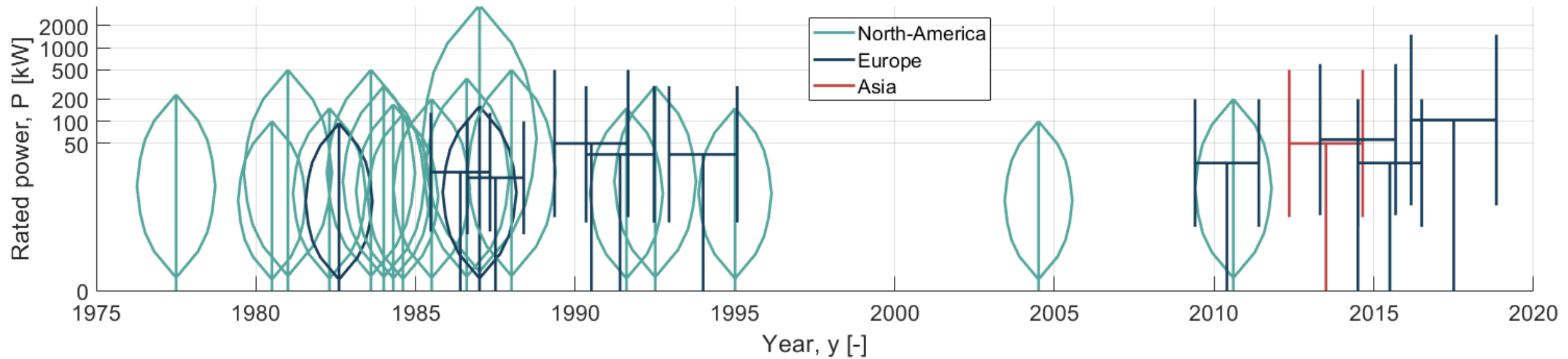
HISTORY & FUTURE

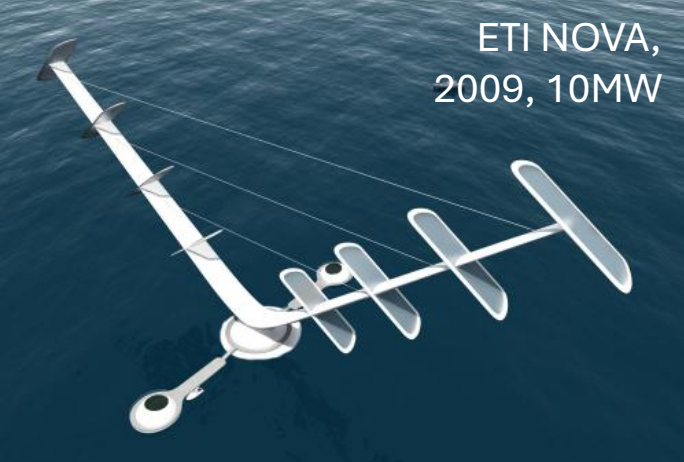


HISTORY & FUTURE



HISTORY & FUTURE





ETI NOVA,
2009, 10MW



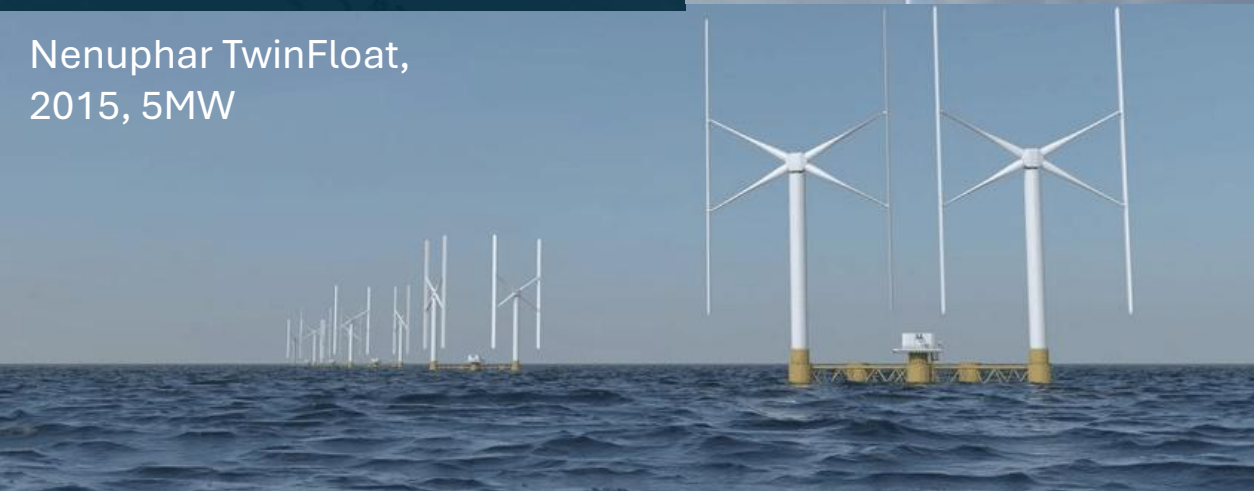
Arkom Tulyp,
2017, 1MW



Modec SKWID,
2010, 1MW



Eolfi Spinfloater,
2015, 5MW



Nenuphar TwinFloat,
2015, 5MW

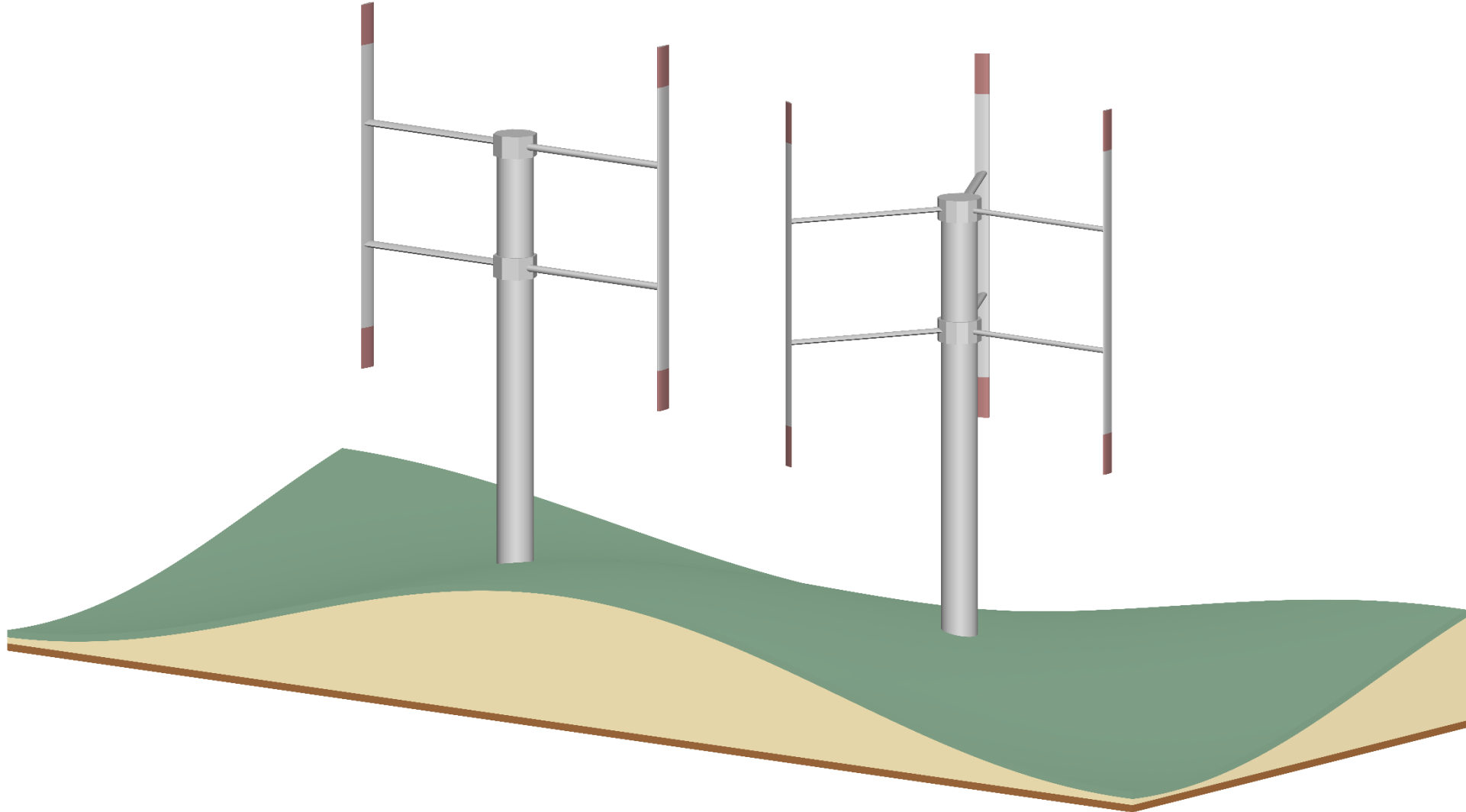


SeaTwirl S2,
2017, 1MW

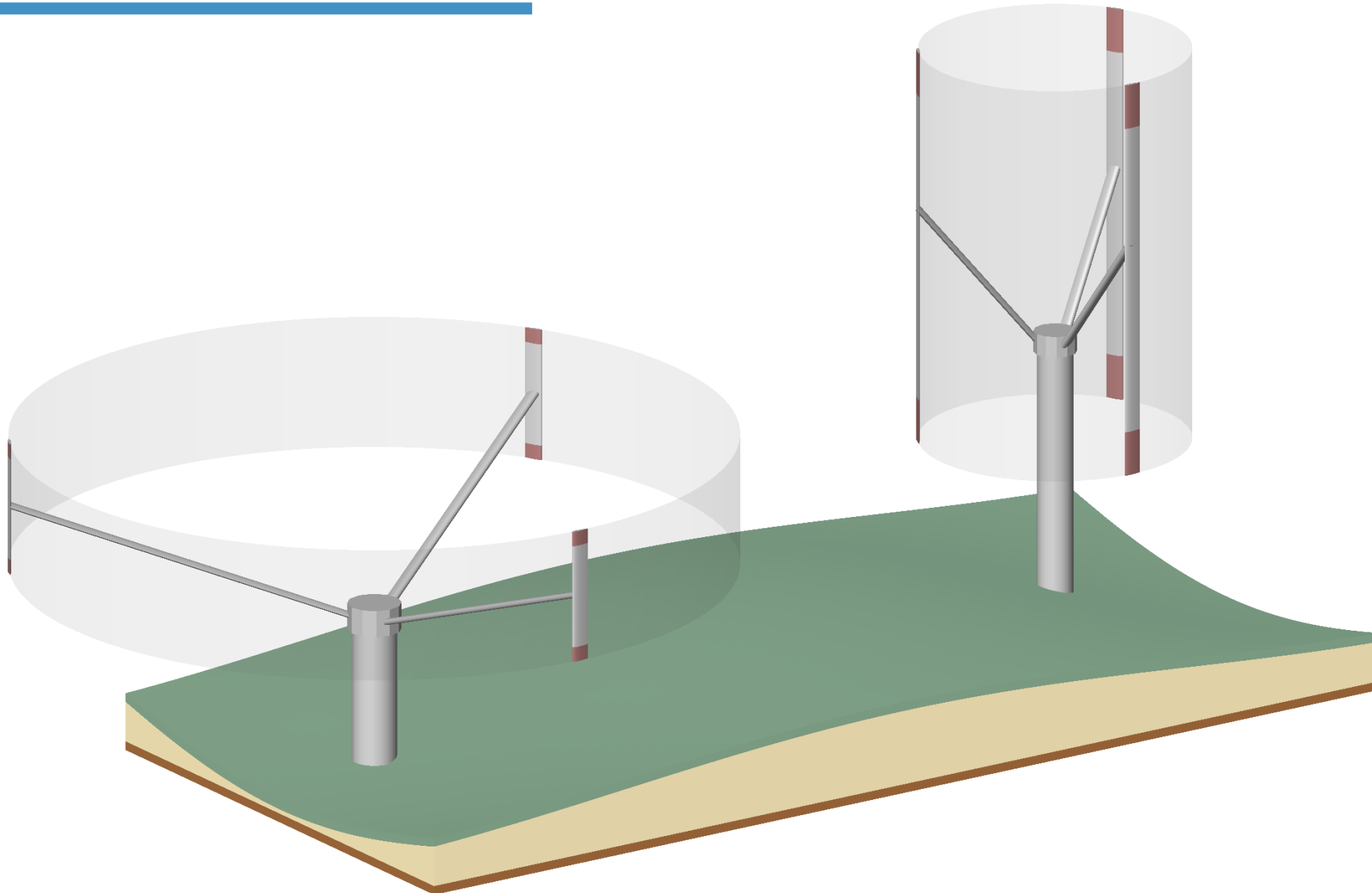


Wind Power Limited AerogeneratorX,
2009, 10MW

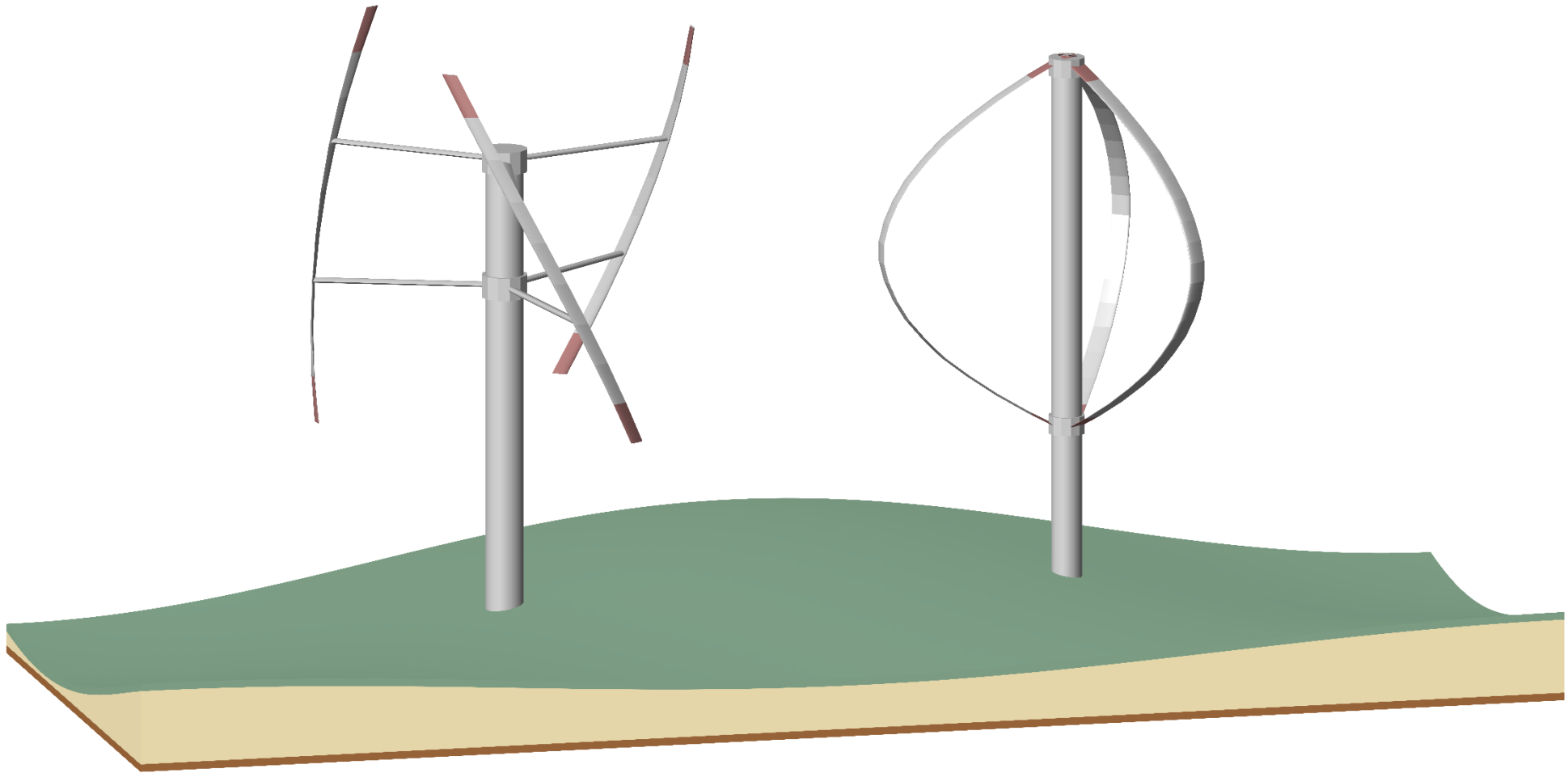
VAWTS IN DIFFERENT SHAPES



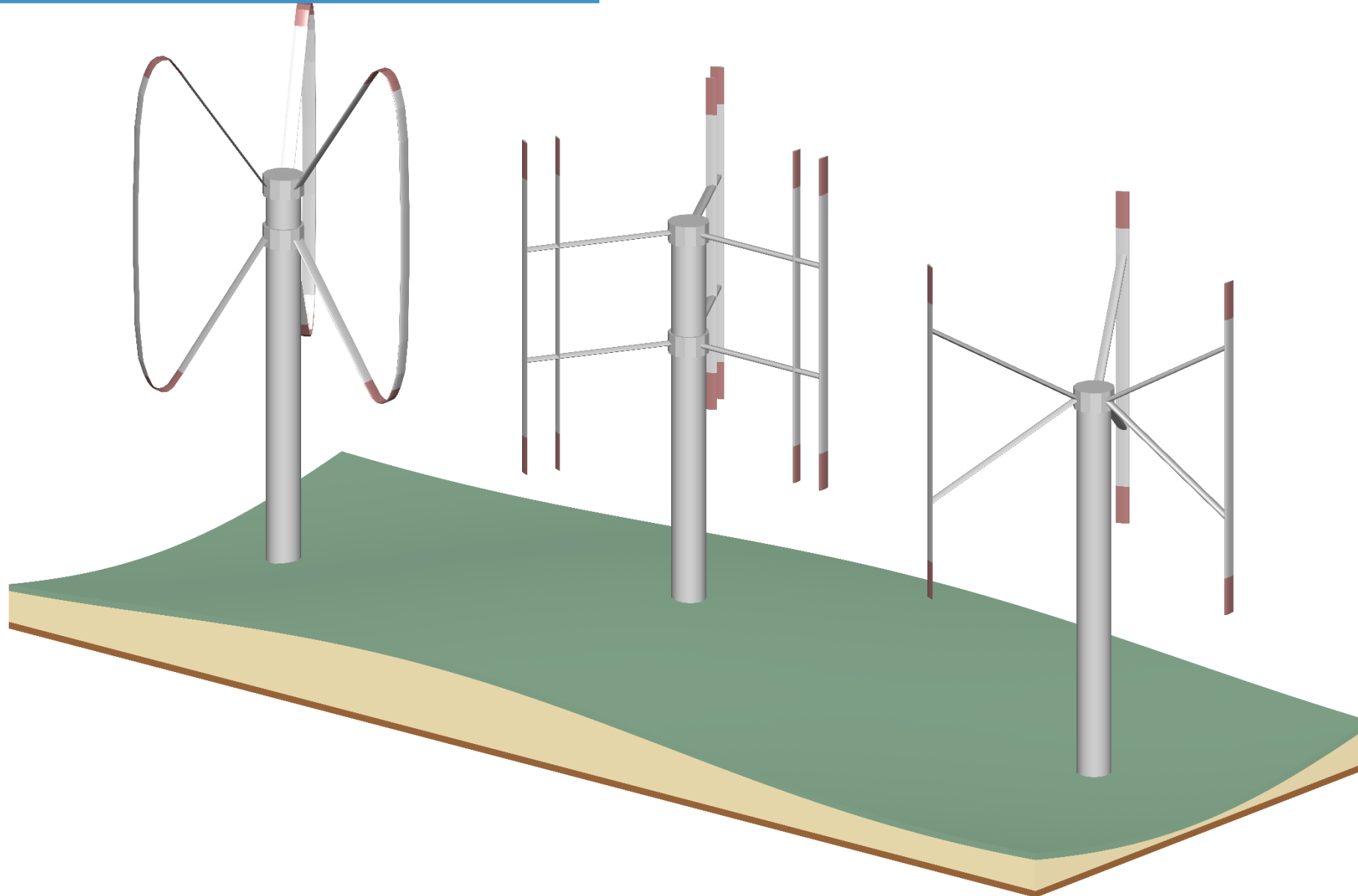
VAWTS IN DIFFERENT SHAPES



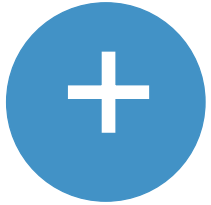
VAWTS IN DIFFERENT SHAPES



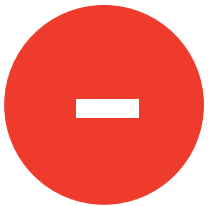
VAWTS IN DIFFERENT SHAPES



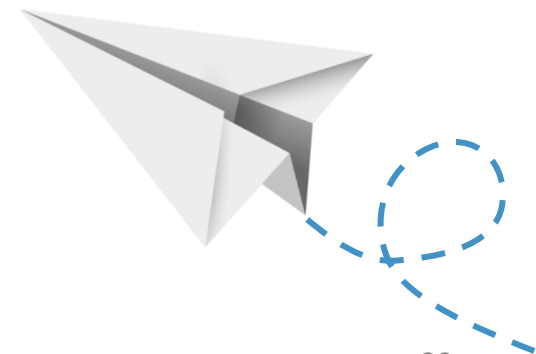
ADVANTAGES VS DISADVANTAGES



- Lower center of gravity
- Omni-directional (no yaw control required)
- Easy scalability
- Fast wake recovery
- Simple blade design (easy manufacturing)



- Cyclic loading (fatigue issues)
- Unsteady aerodynamics (dynamic stall, blade-vortex interaction)
- lower efficiency (in practice)
- Lower rotational speed (+ noise, - expensive generator)
- Vibrations

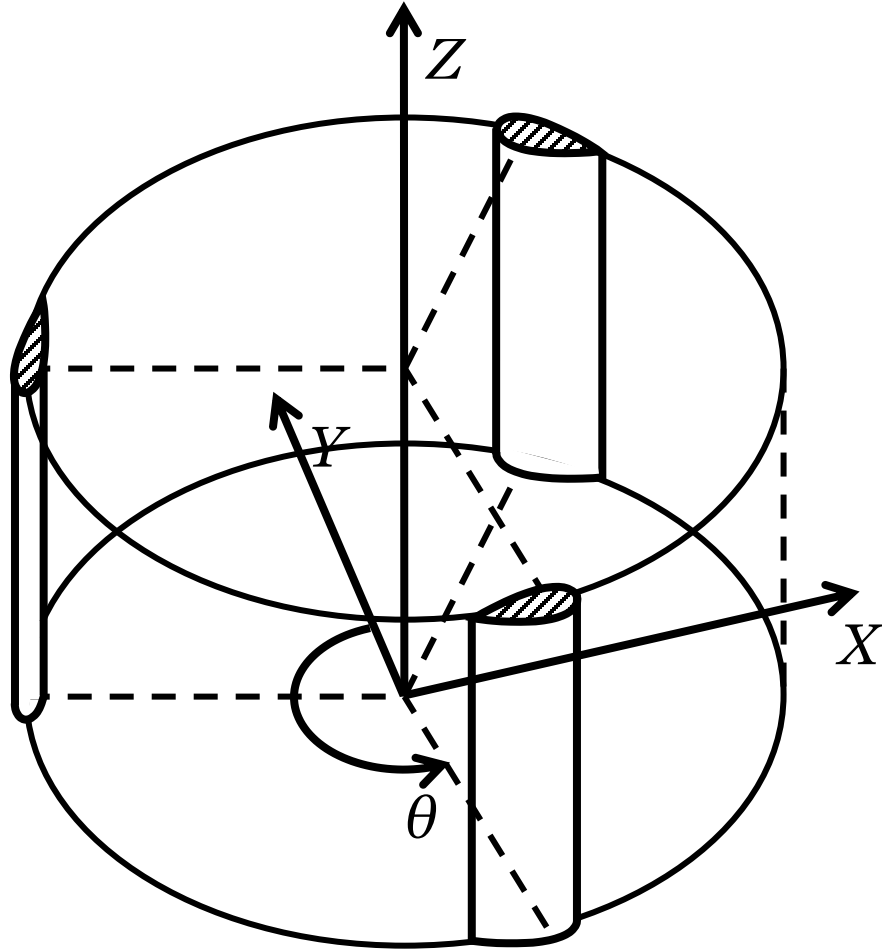


ANY QUESTIONS

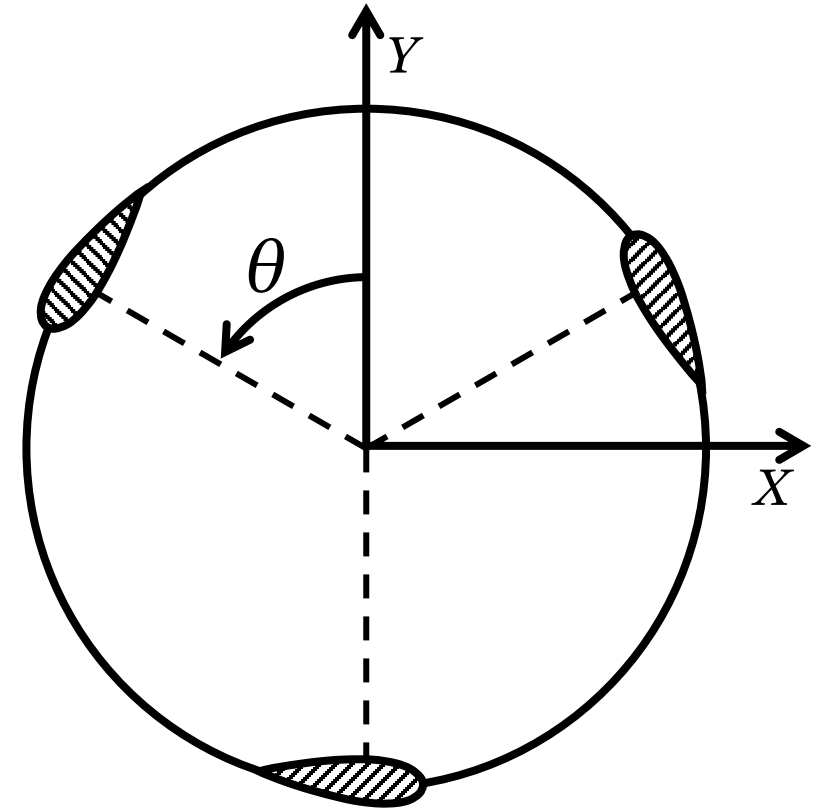


BASIC VAWT AERODYNAMICS

3D TO 2D VAWT

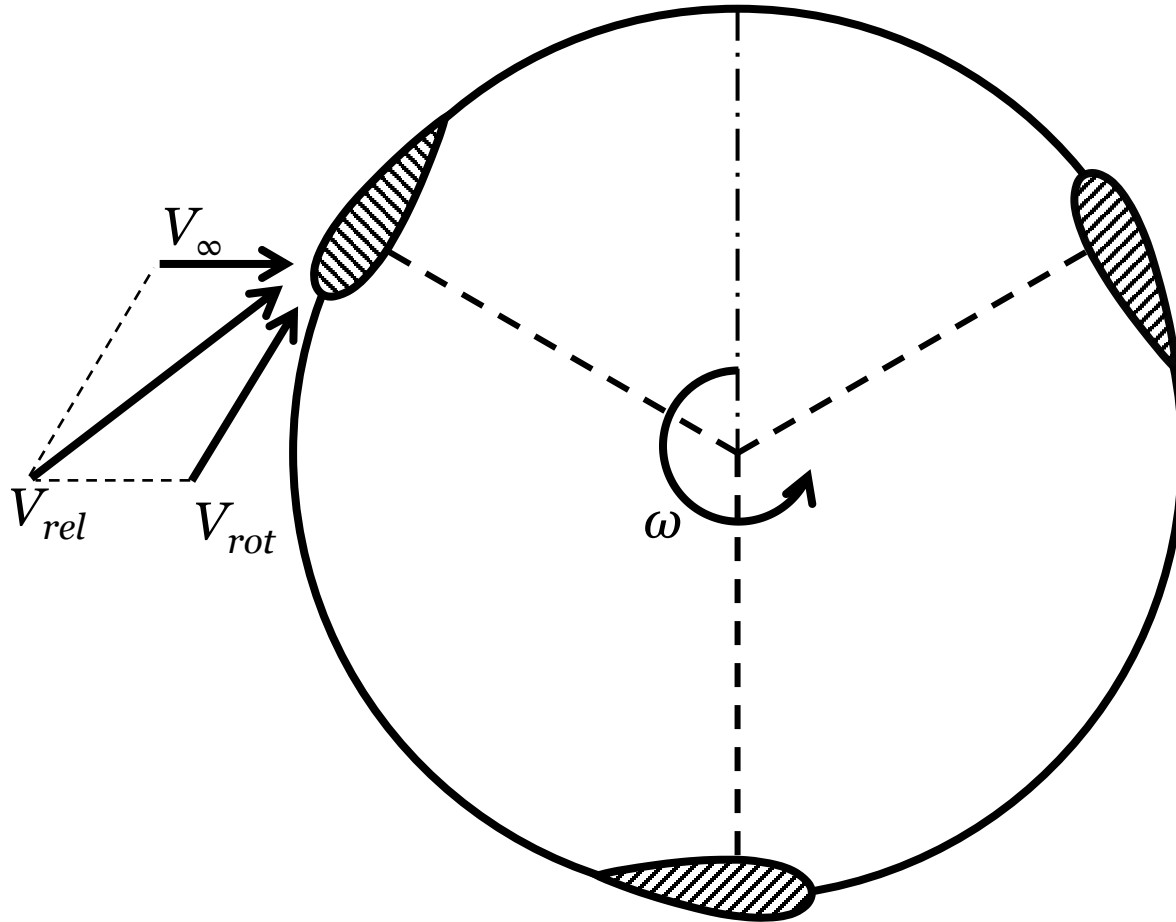


3D VAWT



2D VAWT

WIND TRIANGLE



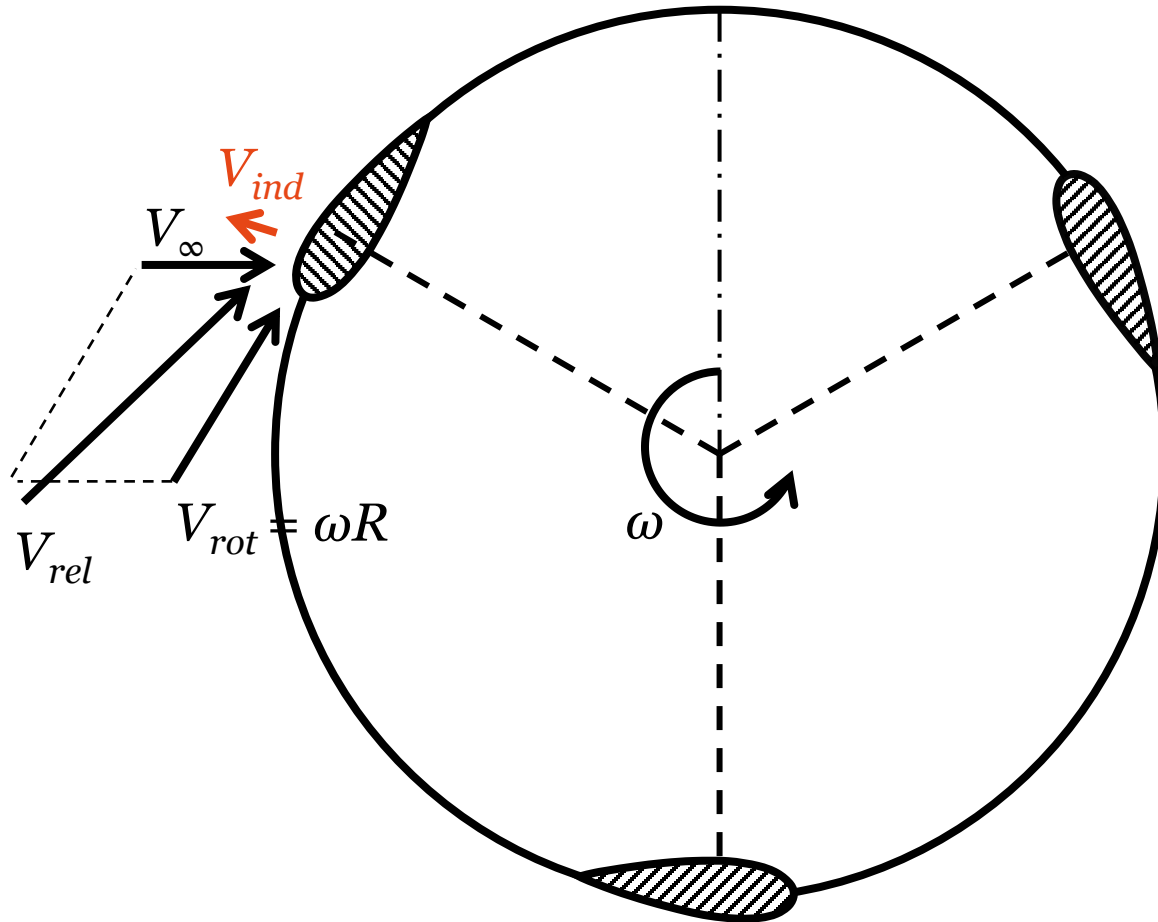
$$\vec{V}_{rel} = \vec{V}_\infty + \vec{V}_{rot}$$

V_{rot} = Rotational velocity

V_∞ = Incoming velocity

V_{rel} = Relative velocity

WIND TRIANGLE



$$\vec{V}_{rel} = \vec{V}_\infty + \vec{V}_{rot} + \vec{V}_{ind}$$

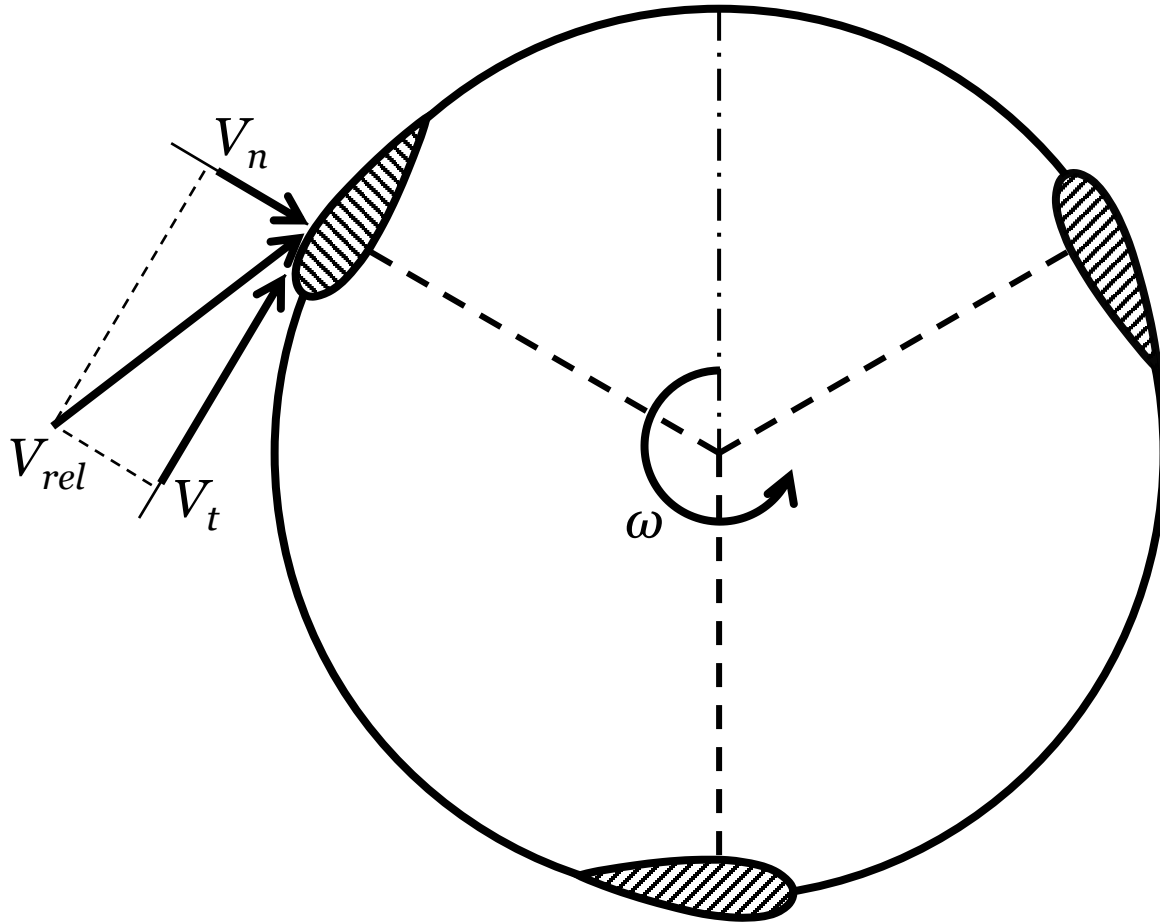
V_{rot} = Rotational velocity

V_∞ = Incoming velocity

V_{rel} = Relative velocity

V_{ind} = Induced velocity

WIND TRIANGLE



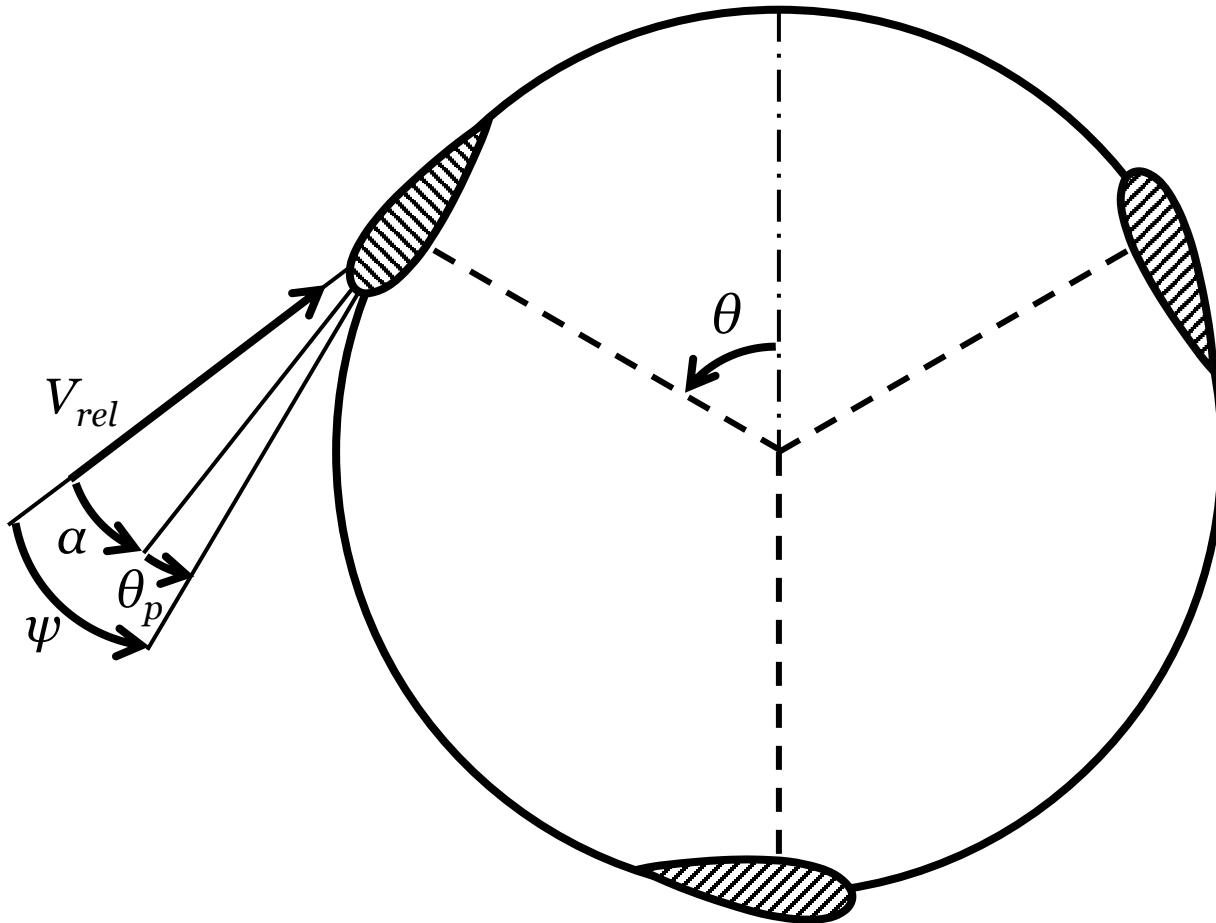
$$\vec{V}_{rel} = \vec{V}_n + \vec{V}_t$$

V_n = Normal velocity

V_t = Tangential velocity

ω = Rotational speed

WIND TRIANGLE



$$\alpha = \tan^{-1} \left(\frac{V_n}{V_t} \right)$$

V_n = Normal velocity

V_t = Tangential velocity

Ψ = Inflow angle

θ_p = Pitch angle

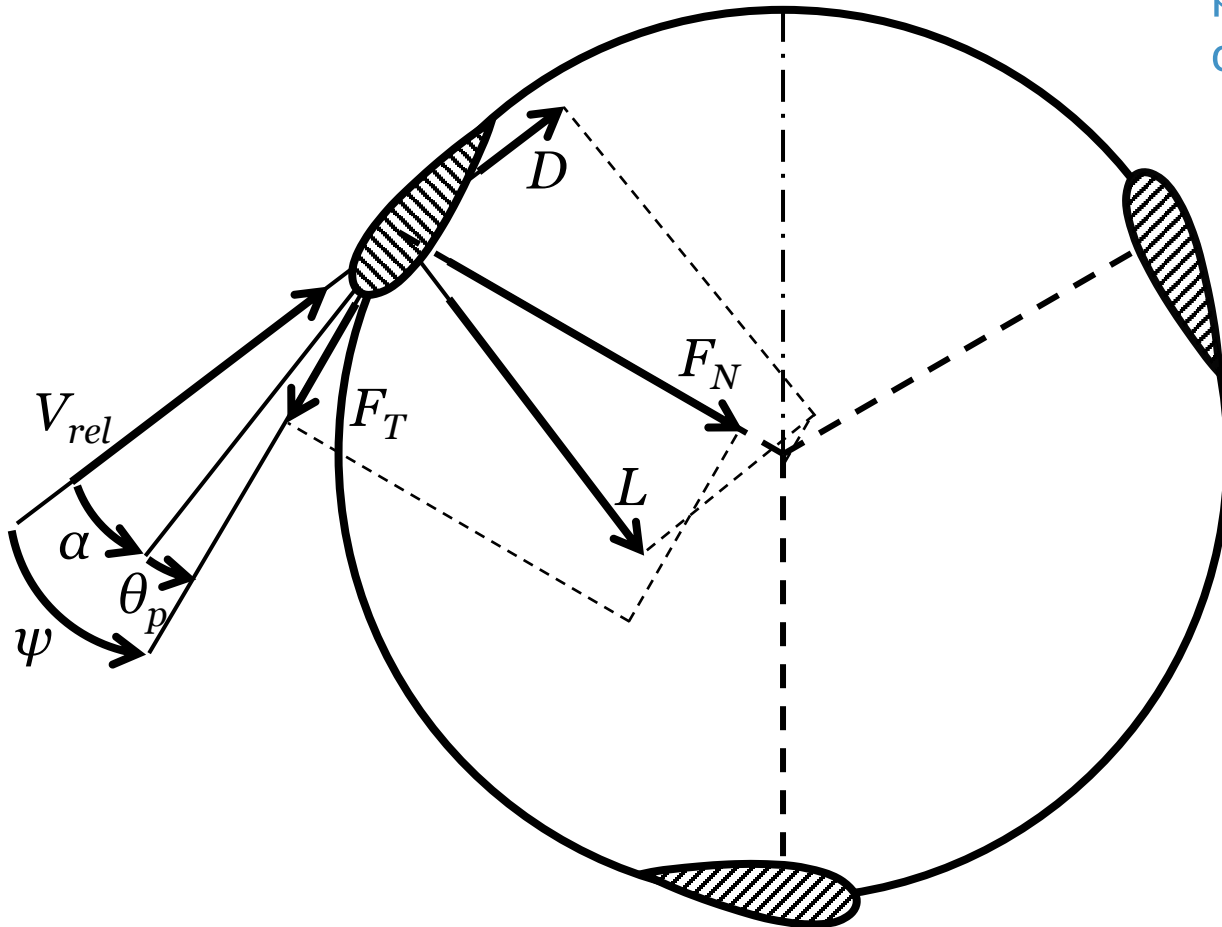
α = Angle of attack

θ = Azimuth angle

ANGLE DEFINITIONS

- **Azimuth angle, θ :** angle between y-axis and blade location
- **Angle of attack, α :** angle between relative velocity and airfoil chordline
- **Inflow angle, Ψ :** angle between relative velocity and tangent to rotating plane
- **Pitch angle, θ_p :** angle between airfoil chordline and tangent to rotating plane

WIND TRIANGLE



⚠ F_N and F_T are normal and perpendicular to the rotor plane. Note that in this figure they are indicating the sign convention and not drawn in scale!

$$F_T = L \sin(\Psi) - D \cos(\psi)$$

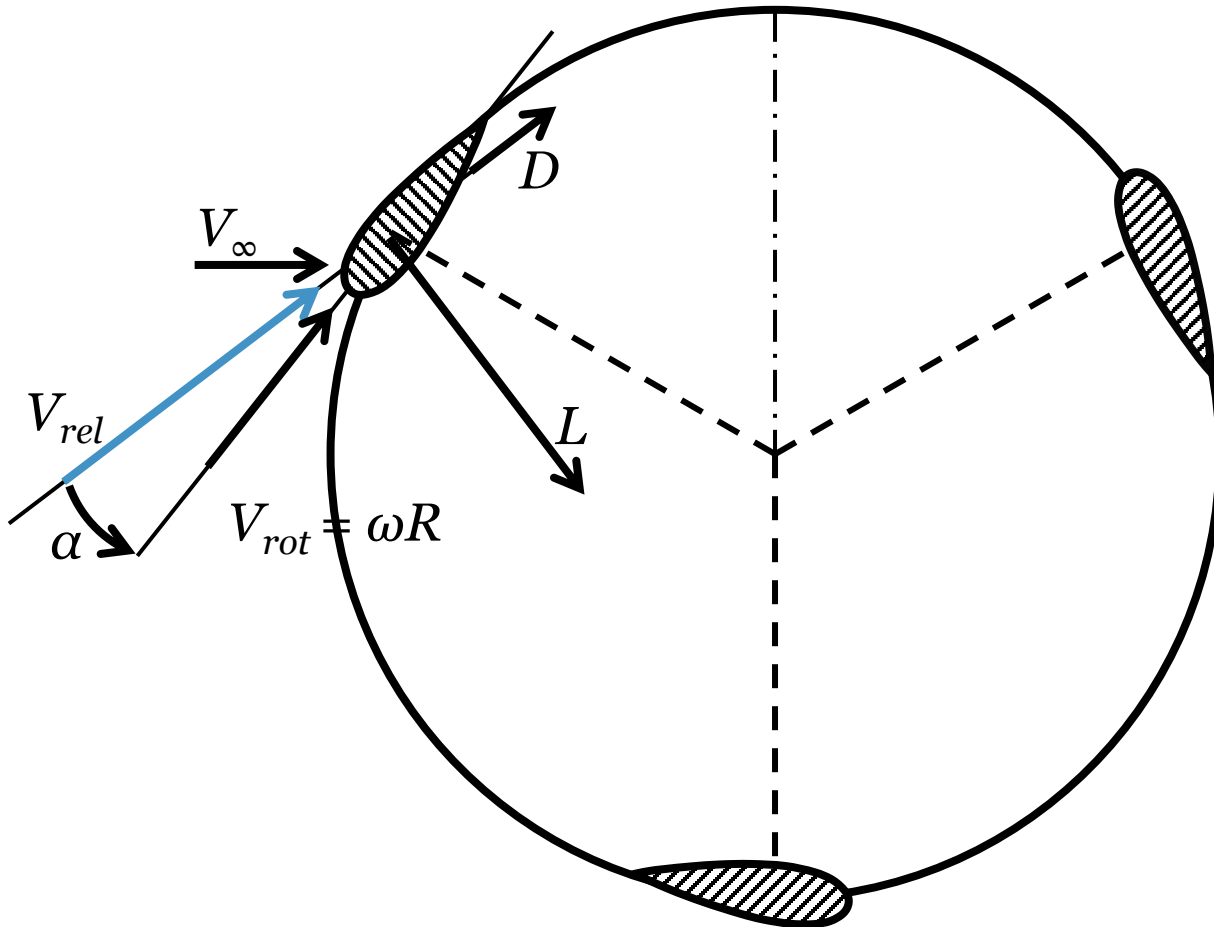
$$F_N = L \cos(\Psi) + D \sin(\Psi)$$

and

$$L = C_l(\alpha) \frac{1}{2} \rho V_{rel}^2 c$$

$$D = C_d(\alpha) \frac{1}{2} \rho V_{rel}^2 c$$

POWER AND THRUST



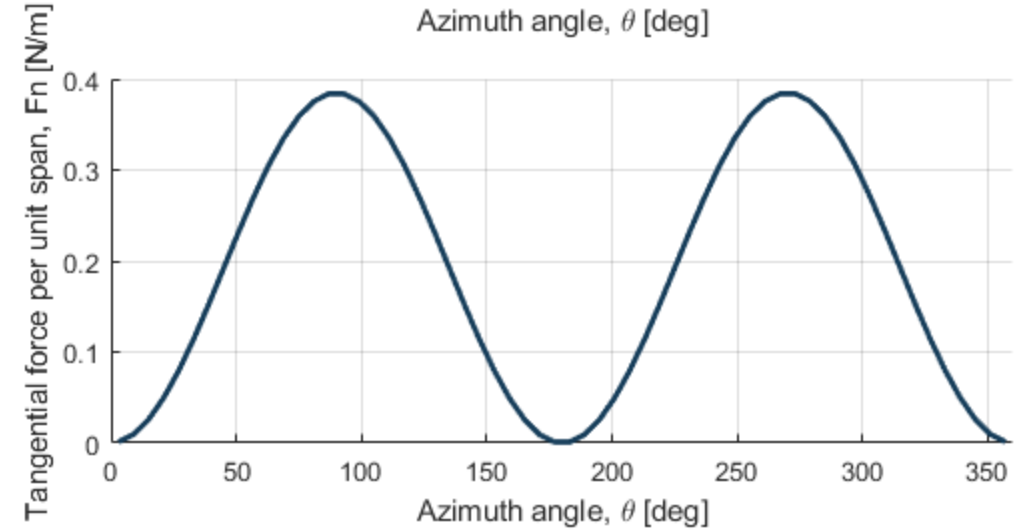
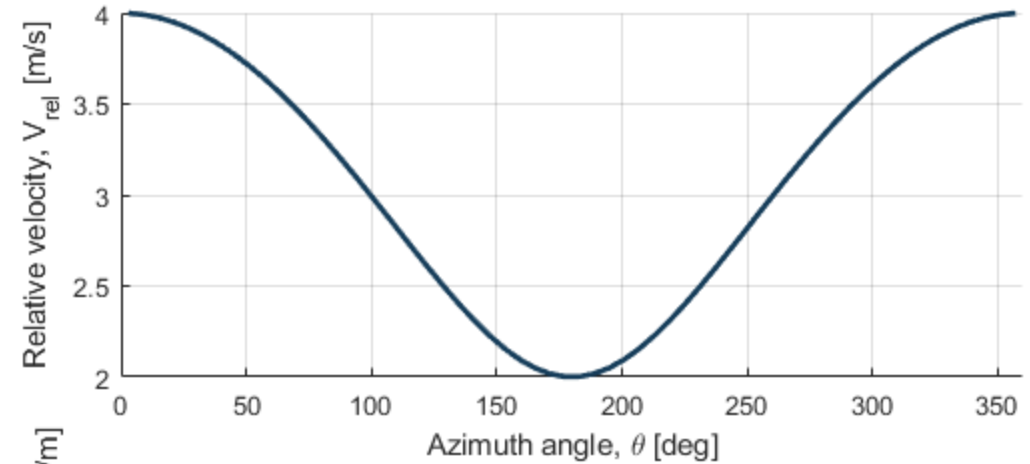
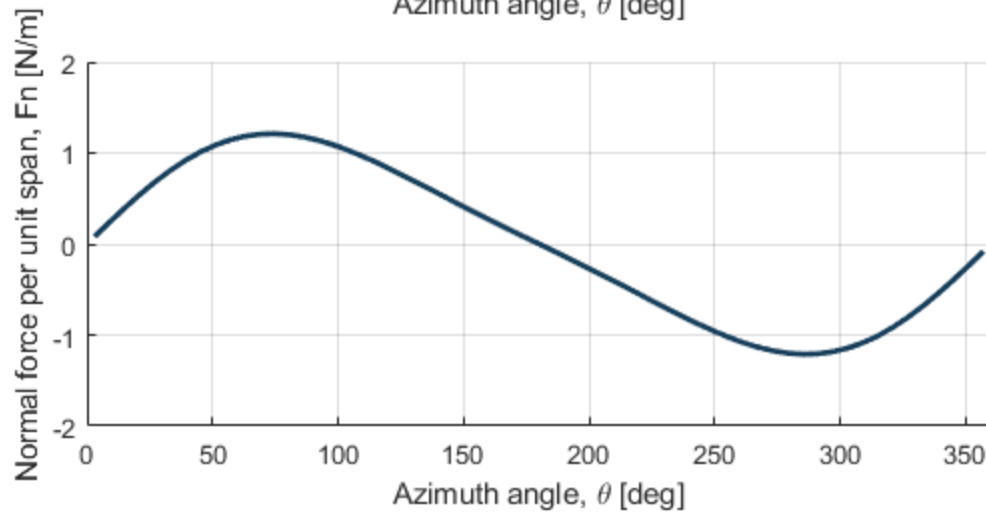
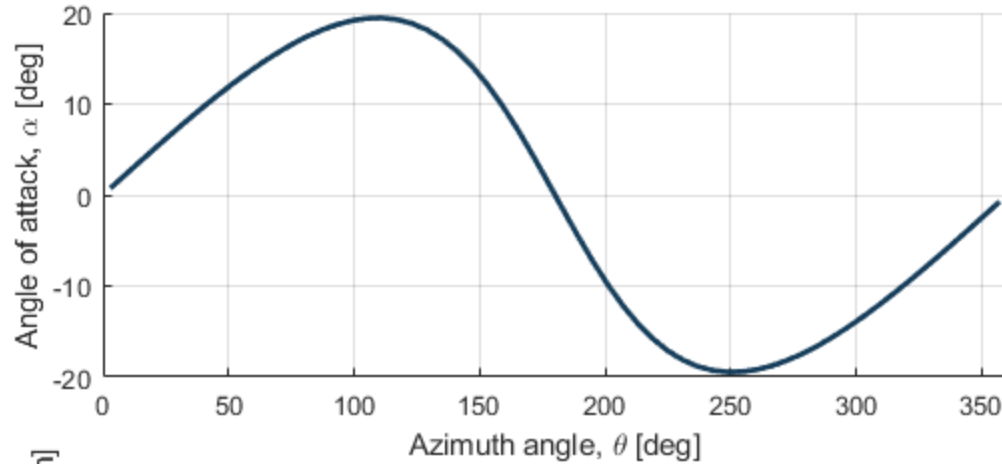
Can you draw

- The relative wind speed
- Angle of attack
- Lift and drag

for blade 2 and 3?

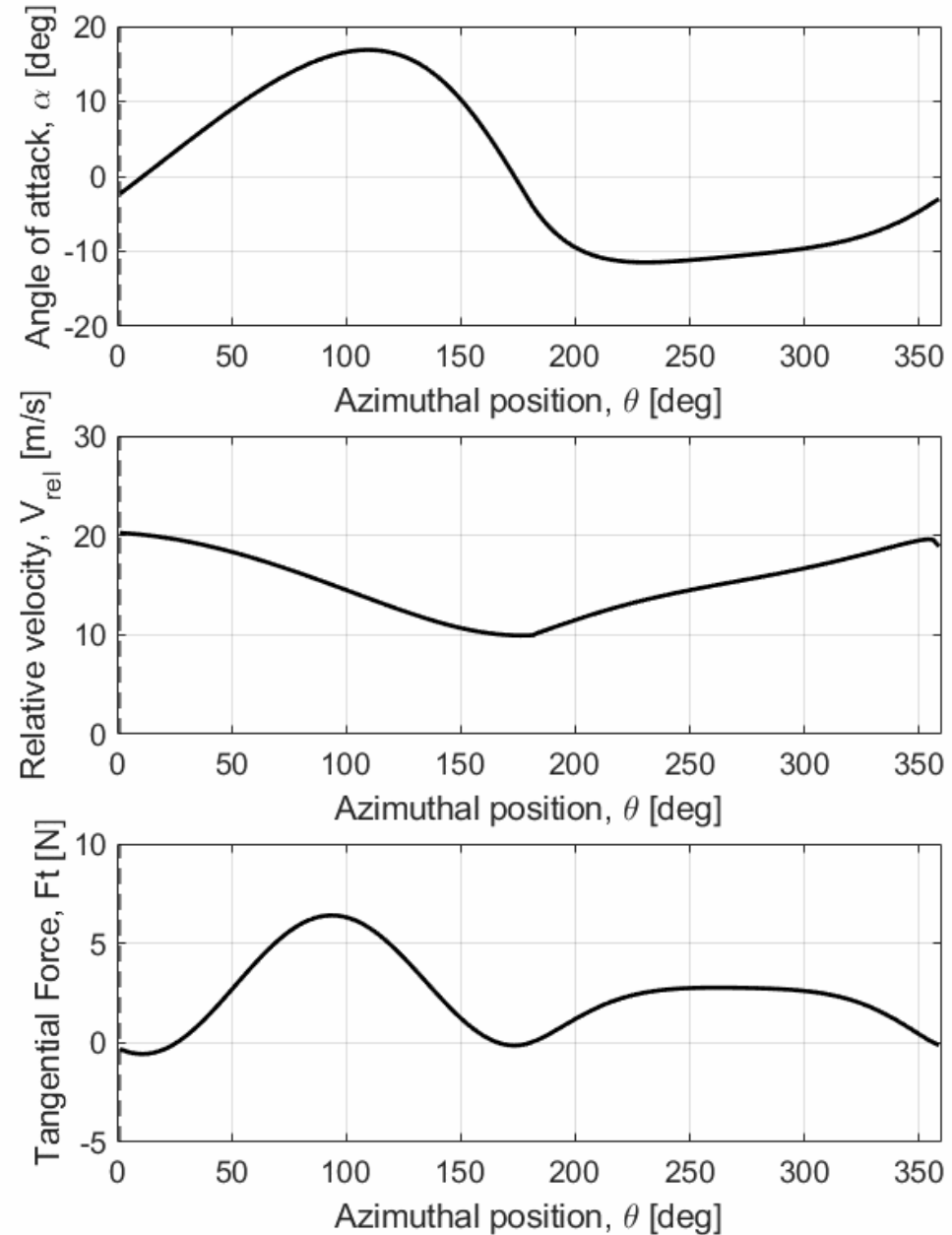
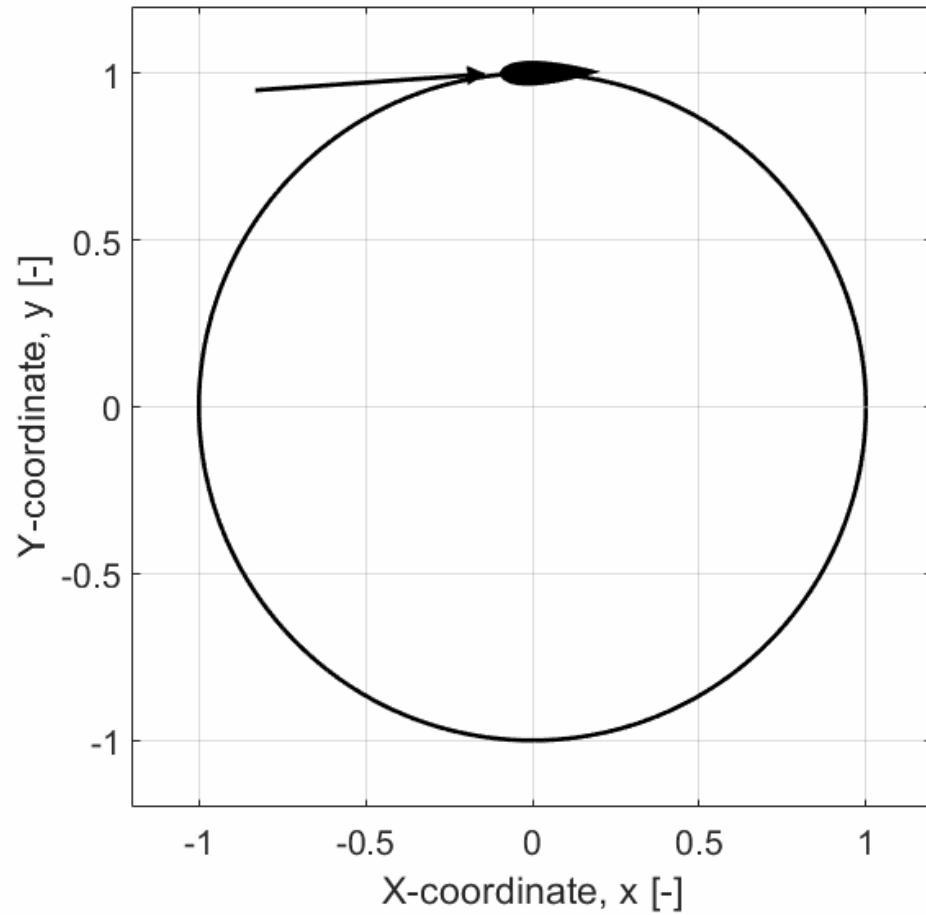
Assuming zero induction, how would the angle of attack vs azimuth angle look like?

GEOMETRICAL PERFORMANCE

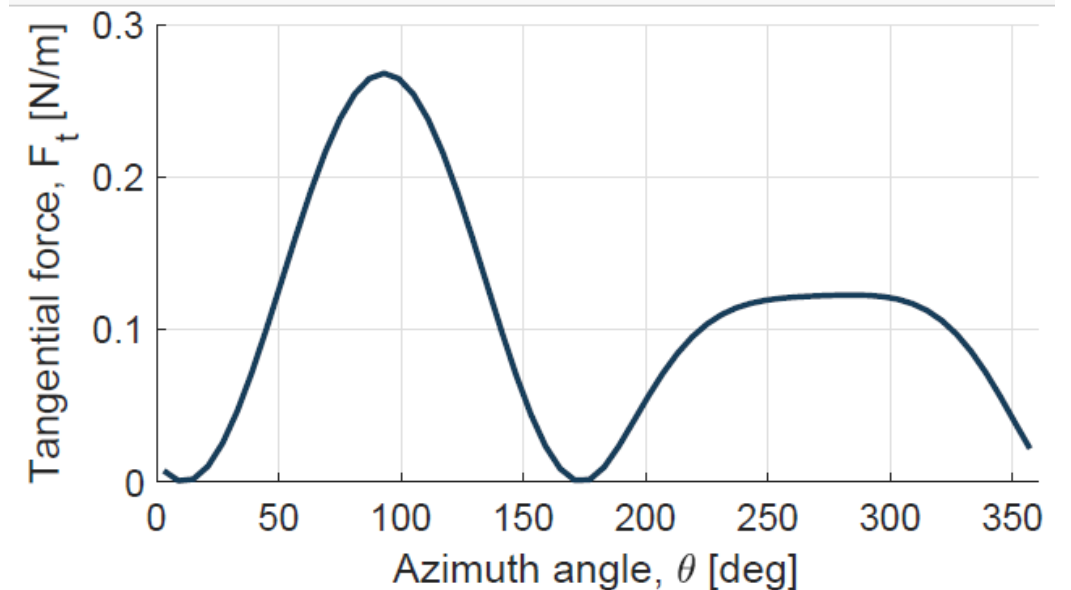
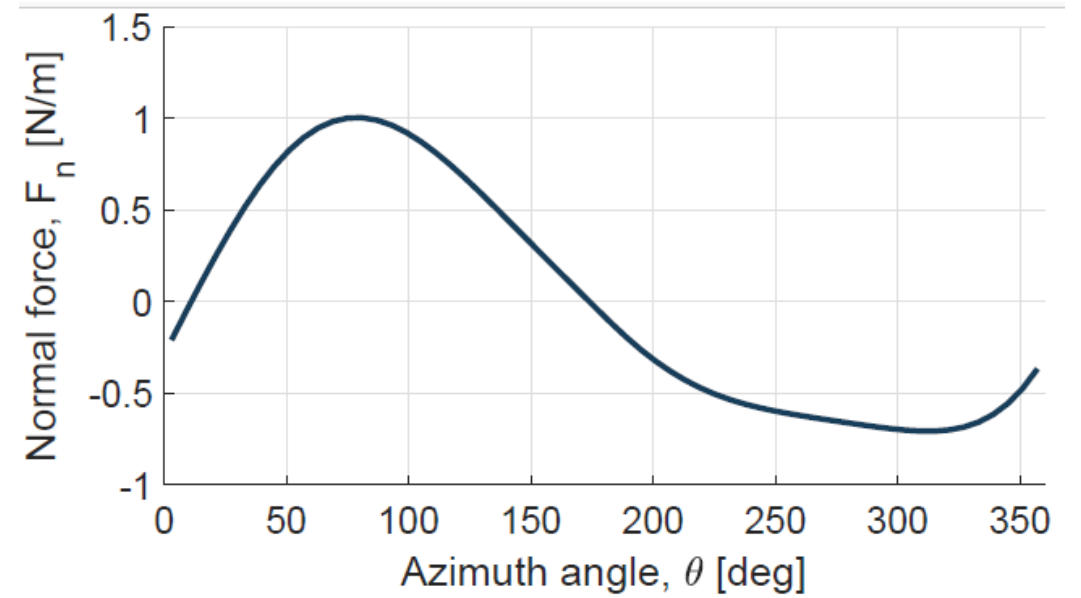
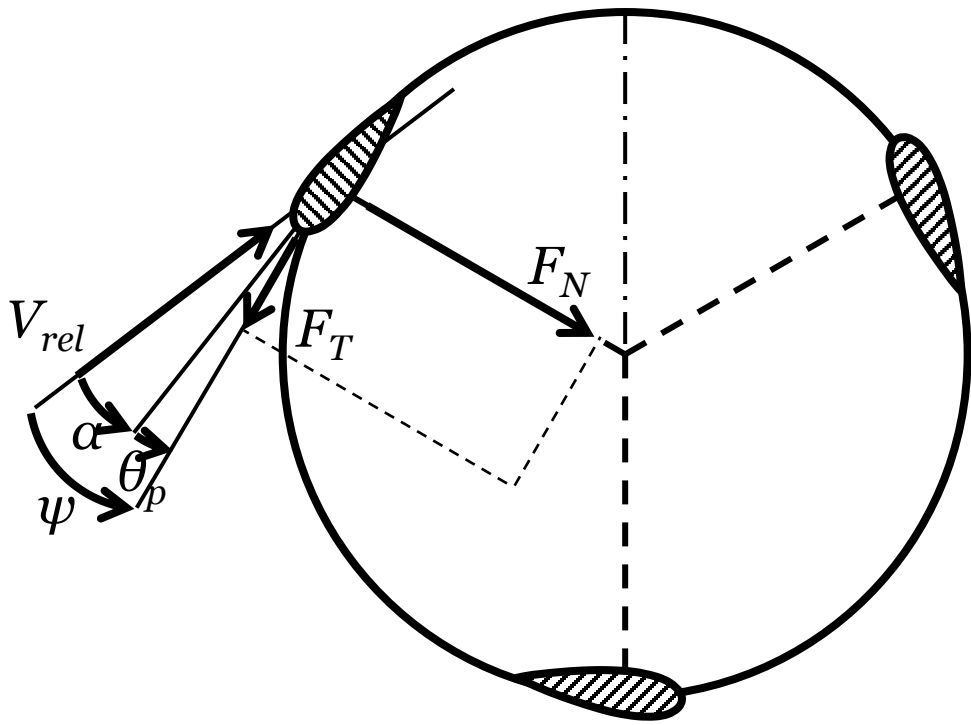


⚠ No induction

REAL PERFORMANCE



REAL PERFORMANCE



NON-DIMENSIONAL PROPERTIES

- **Tip speed ratio, λ** : ratio between blade velocity to the incoming wind speed

$$(\lambda = \frac{\omega R}{V_{\infty}})$$

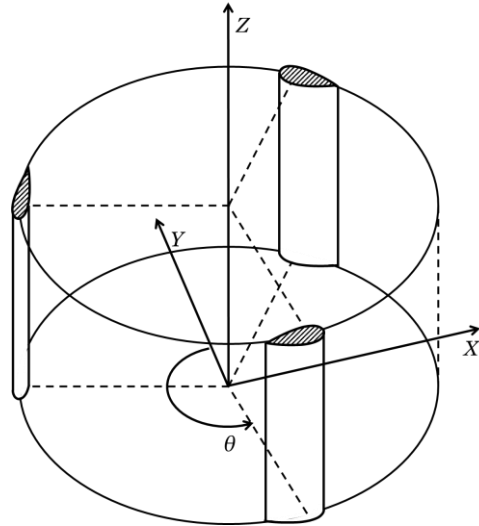
- **Rotor solidity, σ** : ratio of the total area of the rotor blades to the swept area of the rotor ($\sigma = \frac{Bc}{2R}$)

- **Aspect ratio, AR** : ratio between the blade length and rotor diameter ($AR = \frac{H}{2R}$)

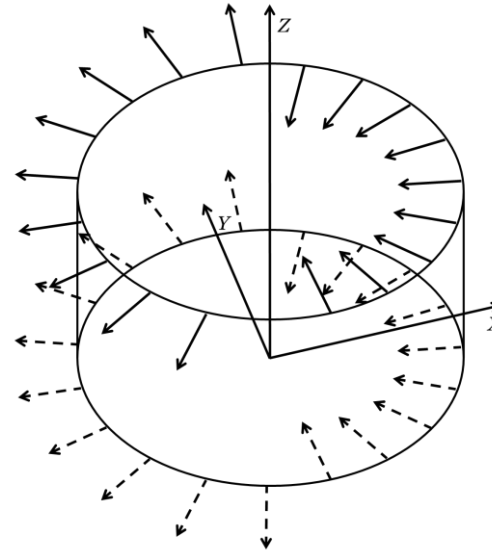
ROTOR REPRESENTATION

ROTOR REPRESENTATION

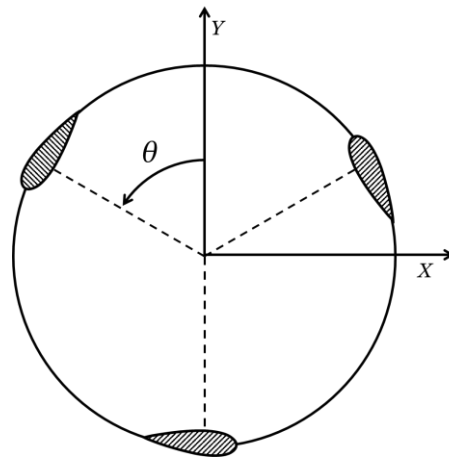
3D rotor



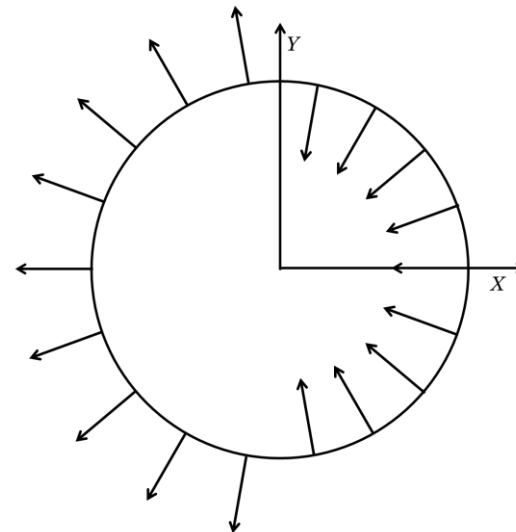
3D actuator disk



2D rotor

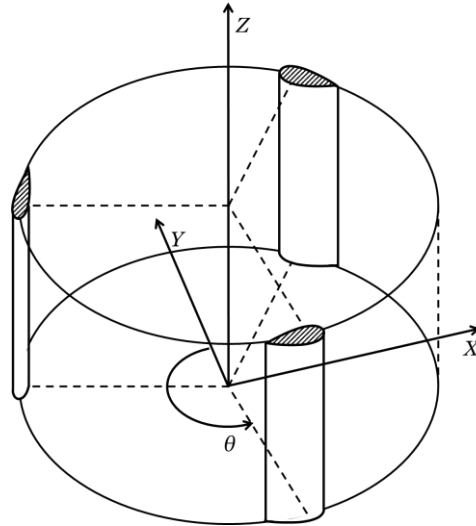


2D actuator disk



ROTOR REPRESENTATION

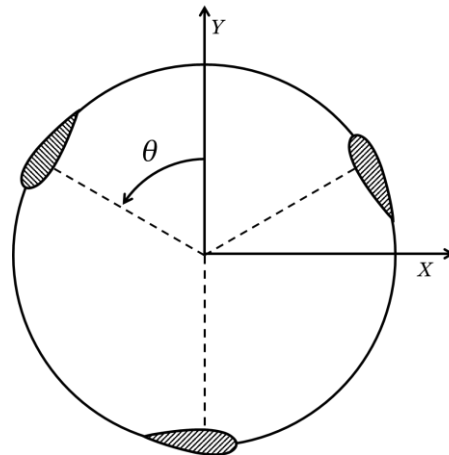
3D rotor



Characteristics:

- Finite number of blades
- Finite blade length, tip effects
- Time-marching

2D rotor



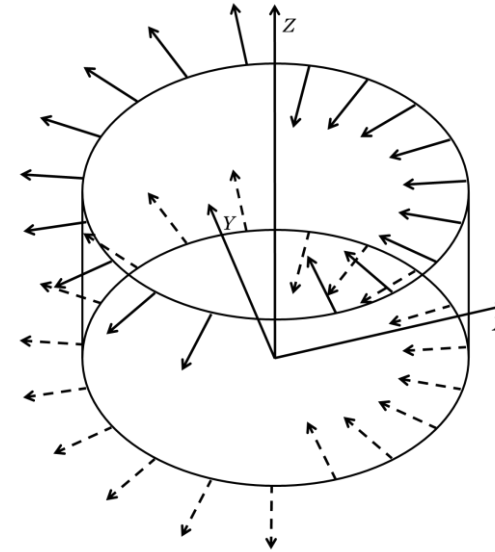
Characteristics:

- Finite number of blades
- Infinite blade length, no tip effects
- Time-marching

ROTOR REPRESENTATION

Characteristics:

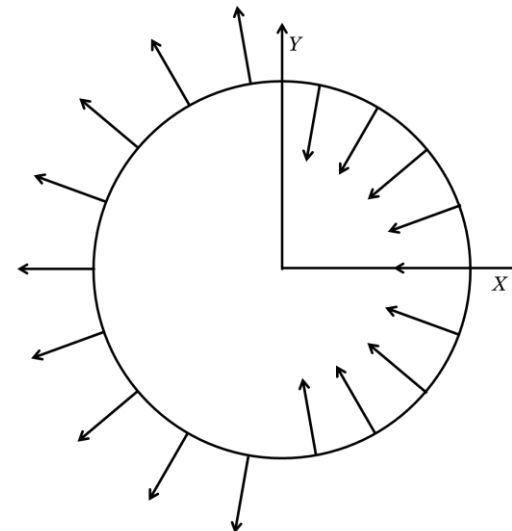
- Infinite number of blades
- Finite blade length, tip effects
- Average loading on actuator
- Steady force field and velocity field



3D actuator disk

Characteristics:

- Infinite number of blades
- Infinite blade length, tip effects
- Average loading on actuator
- Steady force field and velocity field



2D actuator disk

BREAK

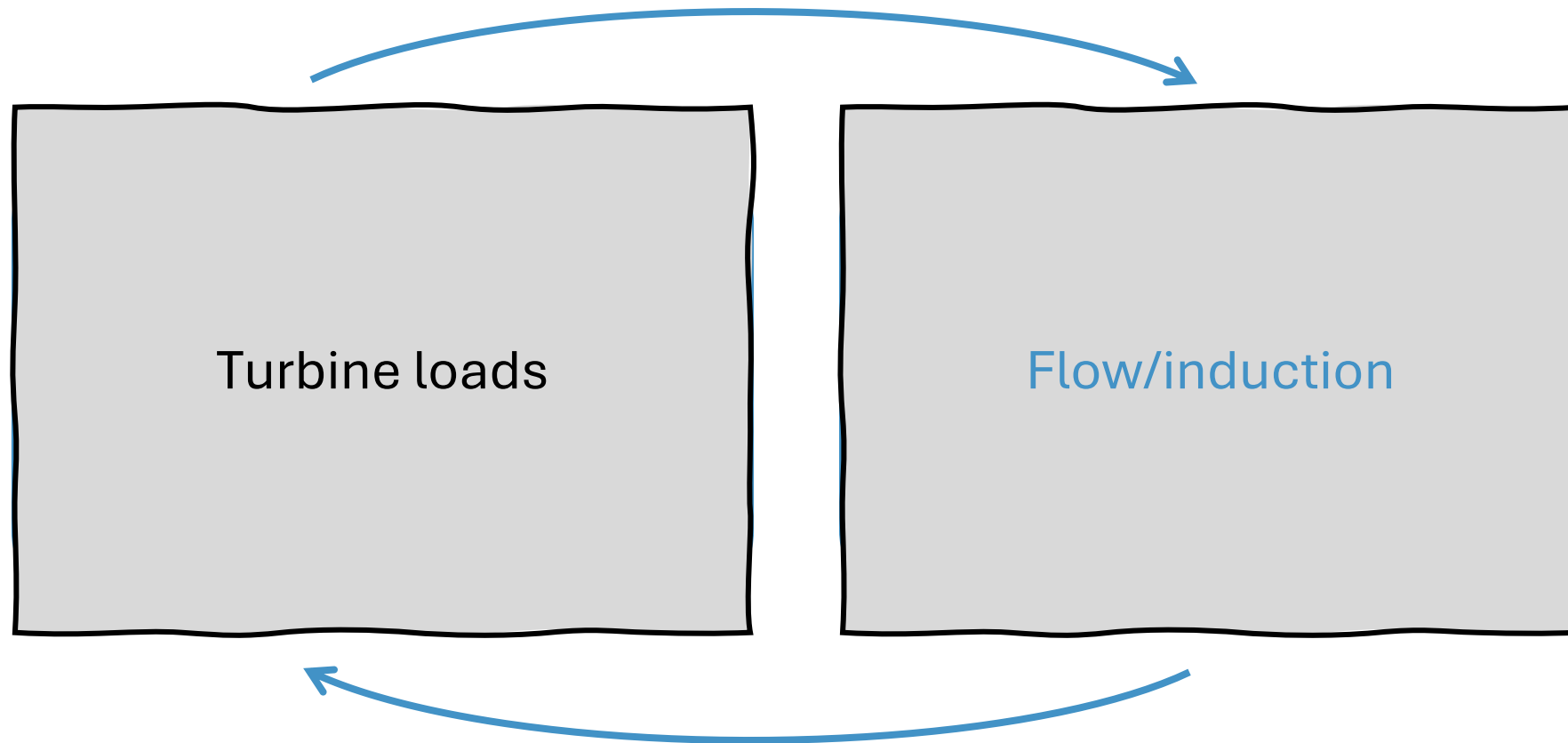


VAWT MODELLING

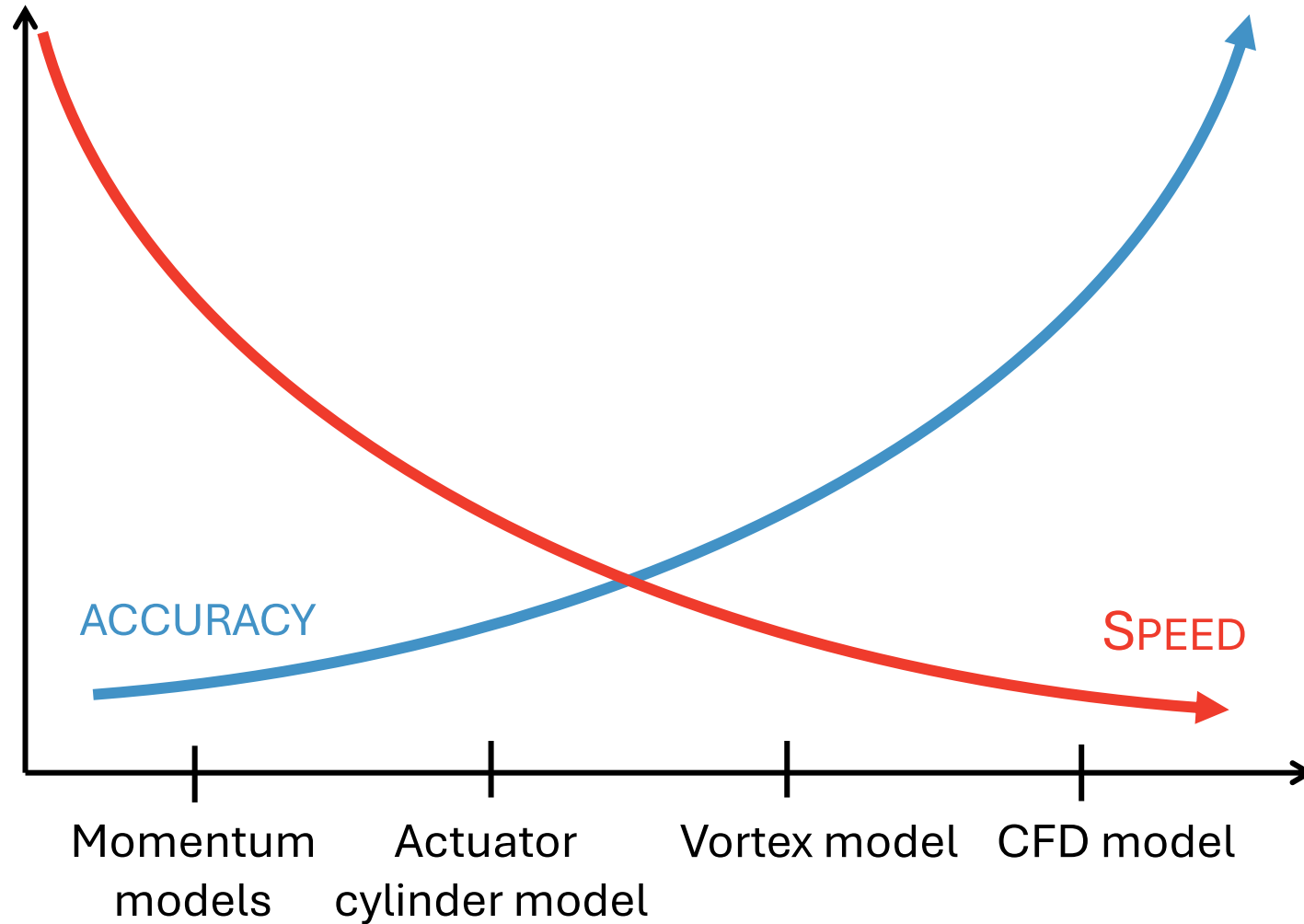
VAWT MODELLING

To compute the **turbine loads**, we need to know the local **flow velocity**, and to compute the local **flow velocity**, we need to know the **turbine loads**

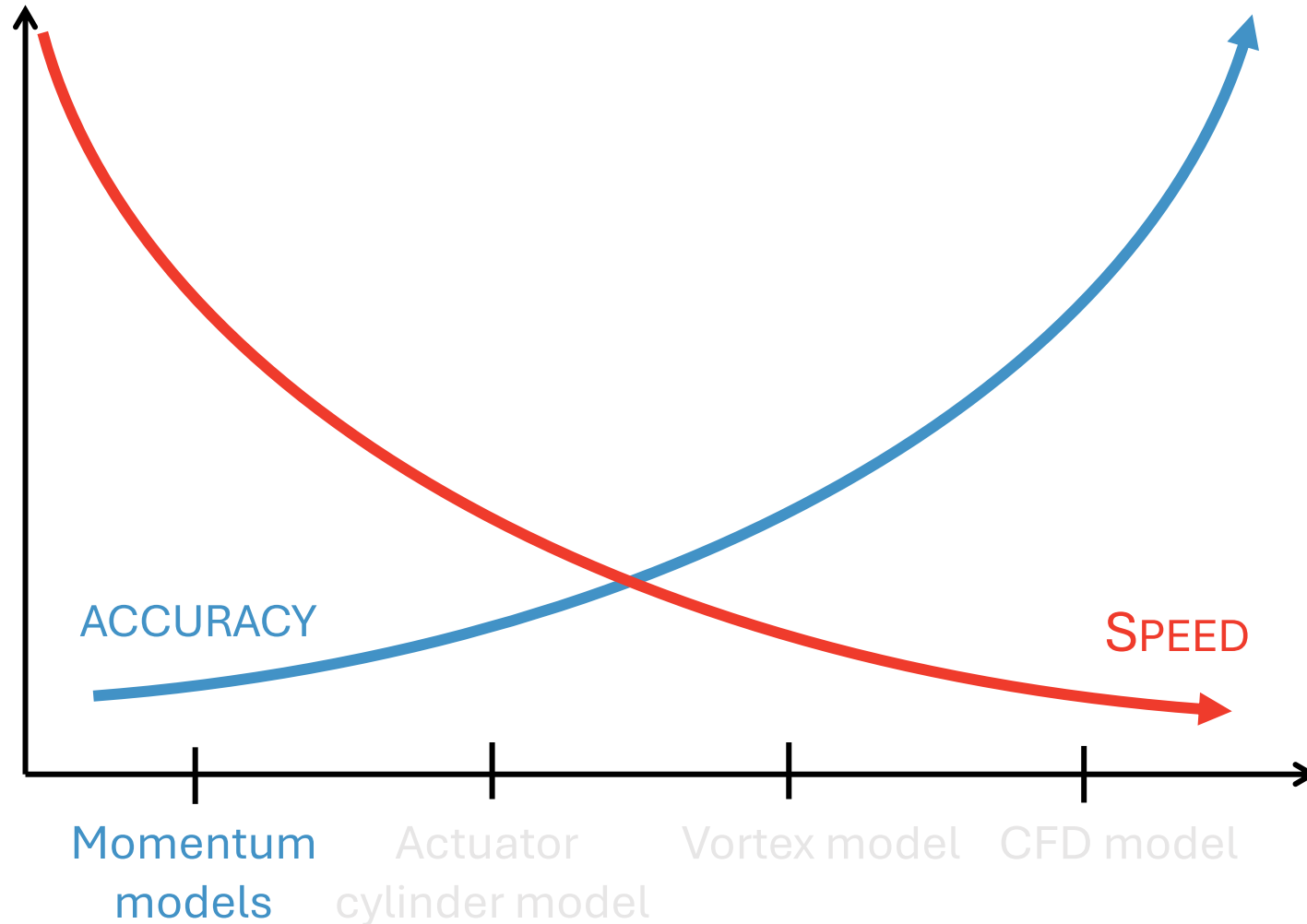
VAWT MODELLING



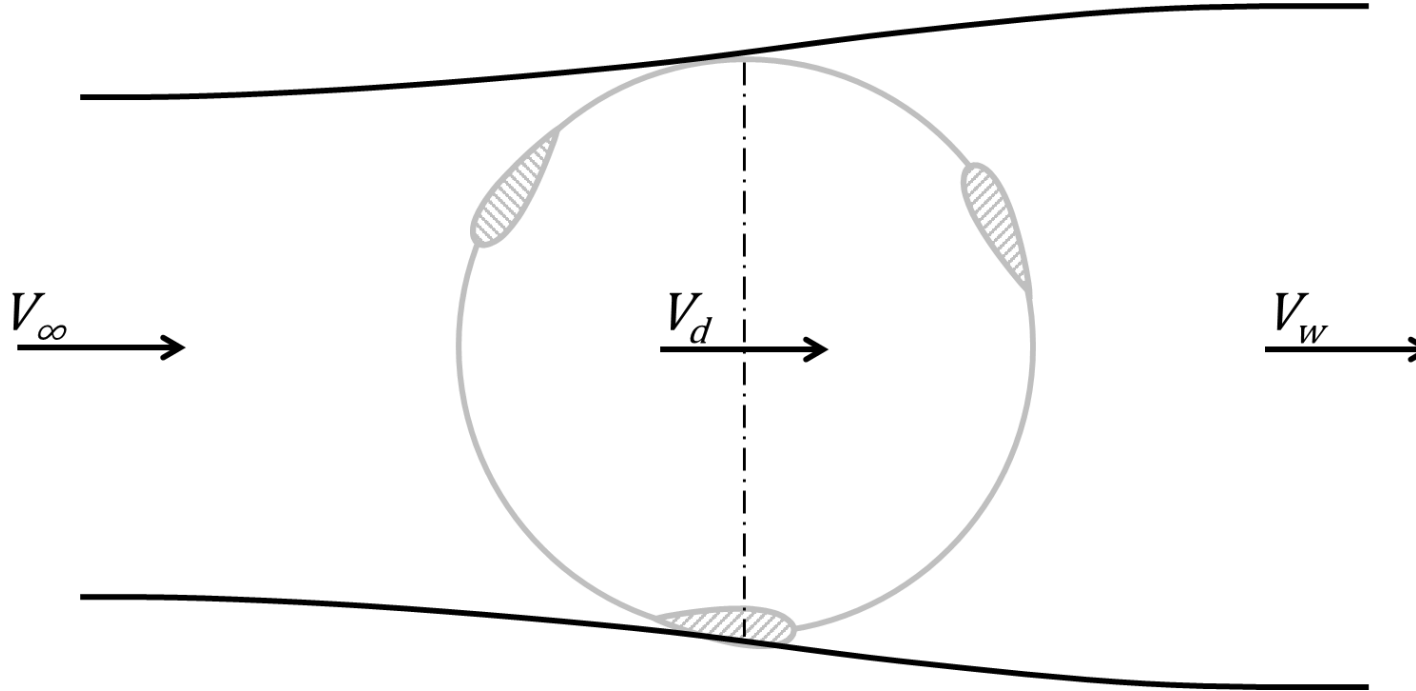
MODEL OVERVIEW



MODEL OVERVIEW



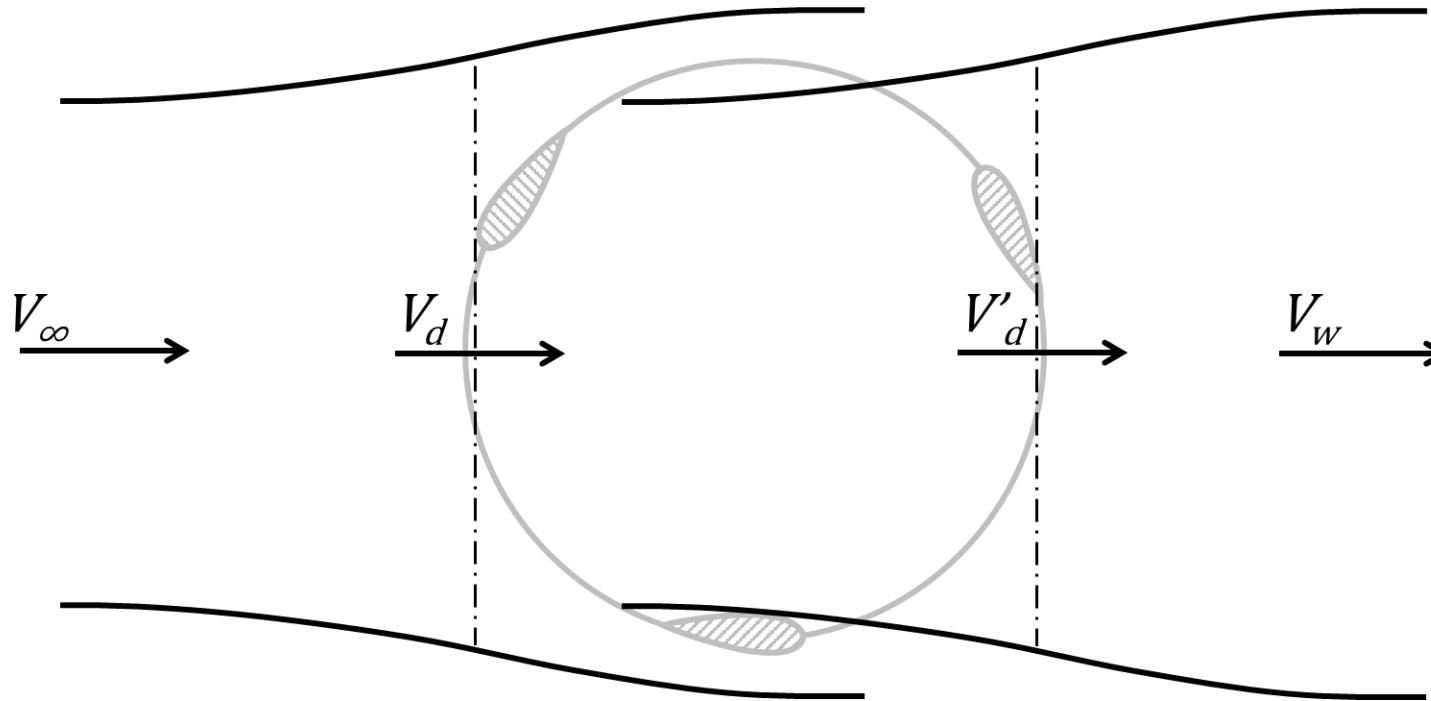
1 - MOMENTUM MODELS



1A – Single streamtube:

- R. Templin
- Replace entire rotor by equivalent actuator disk
- Constant induction for entire rotor

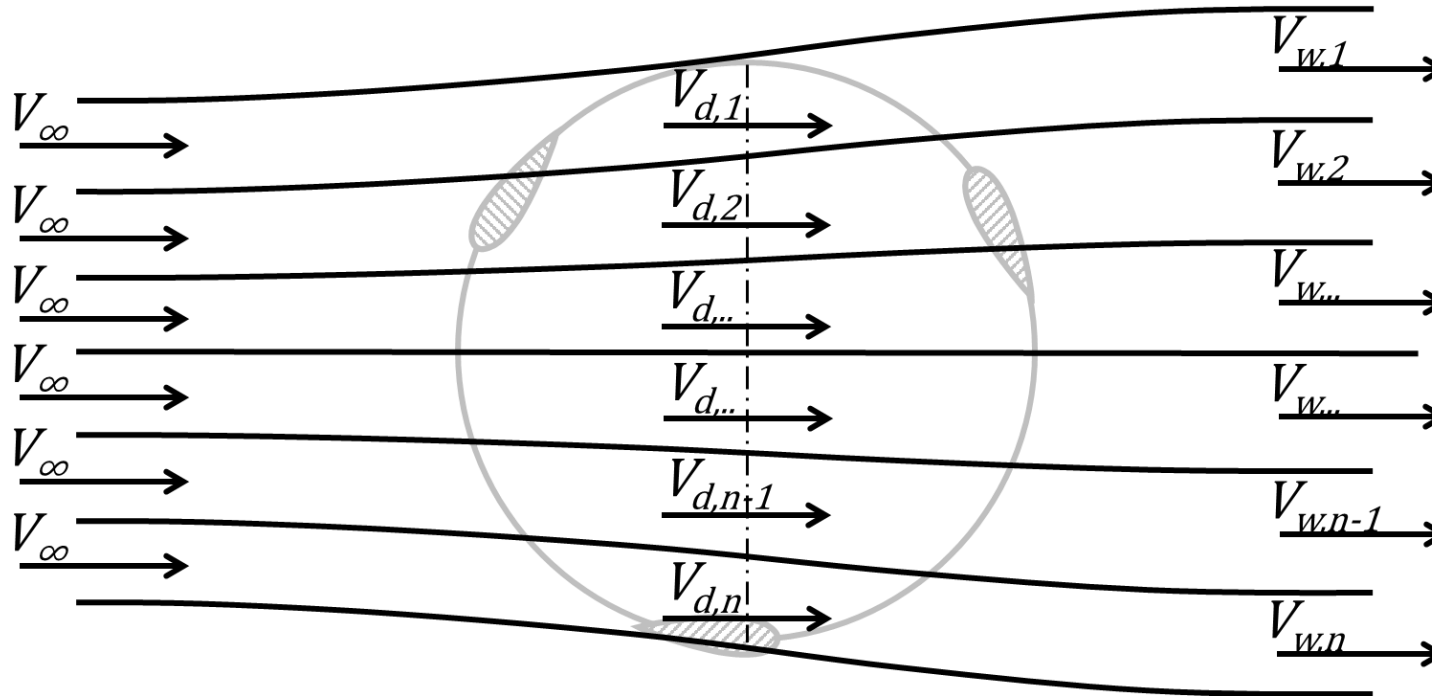
1 - MOMENTUM MODELS



1B – Double streamtube:

- Replace upwind and downwind part of rotor by 2 equivalent actuator disks
- Constant induction for upwind and downwind part of the rotor

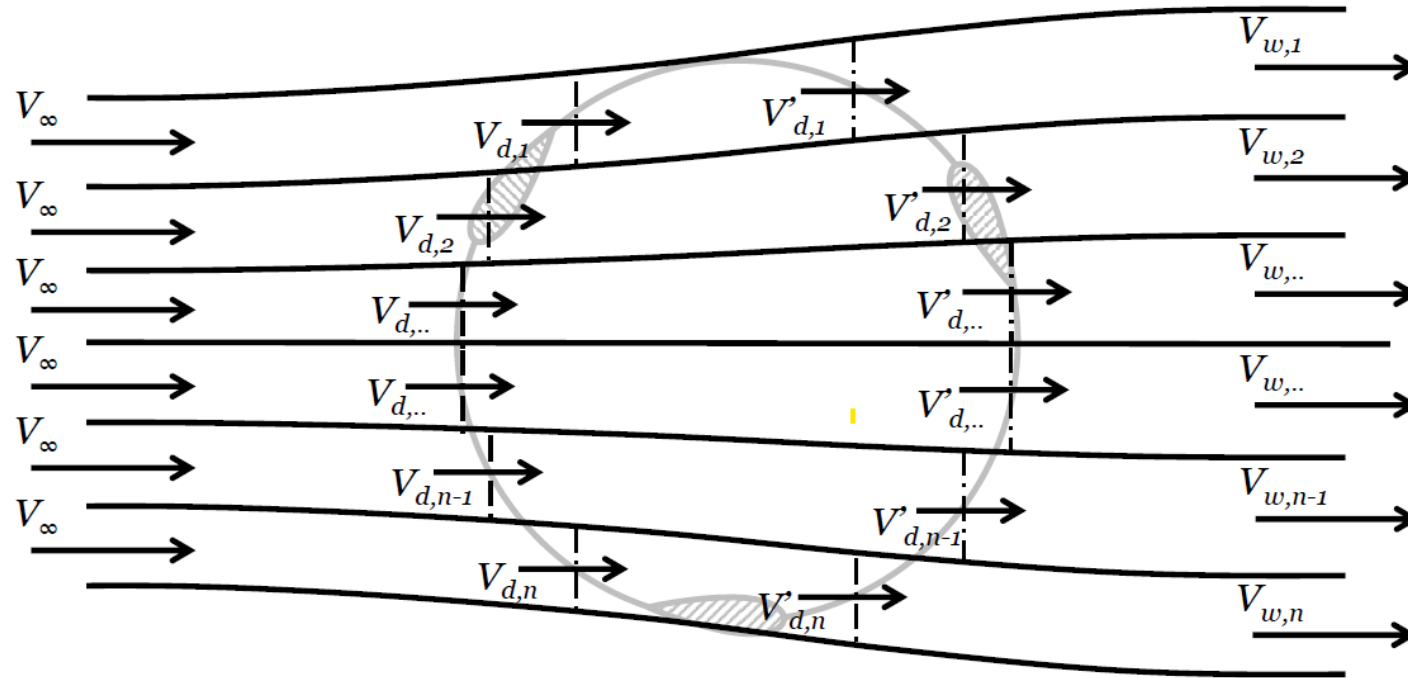
1 - MOMENTUM MODELS



1C – Multiple streamtube:

- Wilson and Lissaman, and Stickland
- Series of adjacent aerodynamically independent streamtubes.
- Replace rotor by one equivalent actuator disks per streamtube
- Different induction for each streamtube, but same upwind and downwind

1 - MOMENTUM MODELS



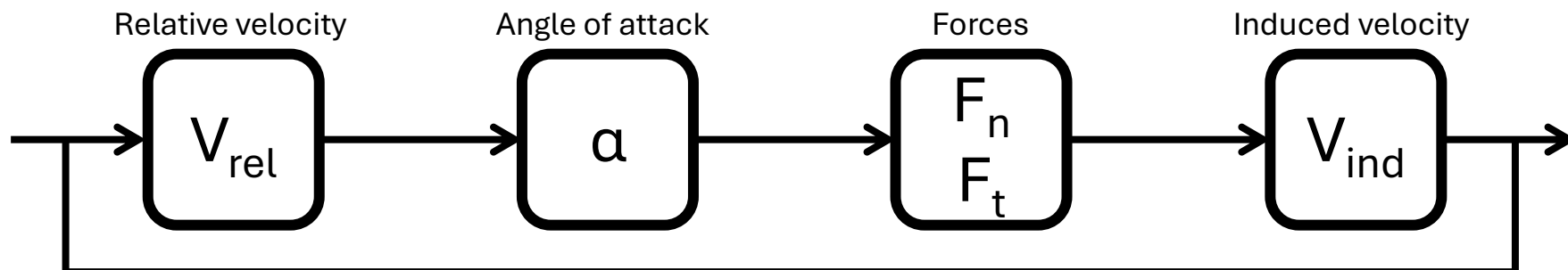
1D – Double multiple streamtube:

- I. Paraschivoiu
- Series of adjacent aerodynamically independent streamtubes.
- Replace upwind and downwind part of rotor by 2 equivalent actuator disks per streamtube
- Different induction for each streamtube, different upwind and downwind

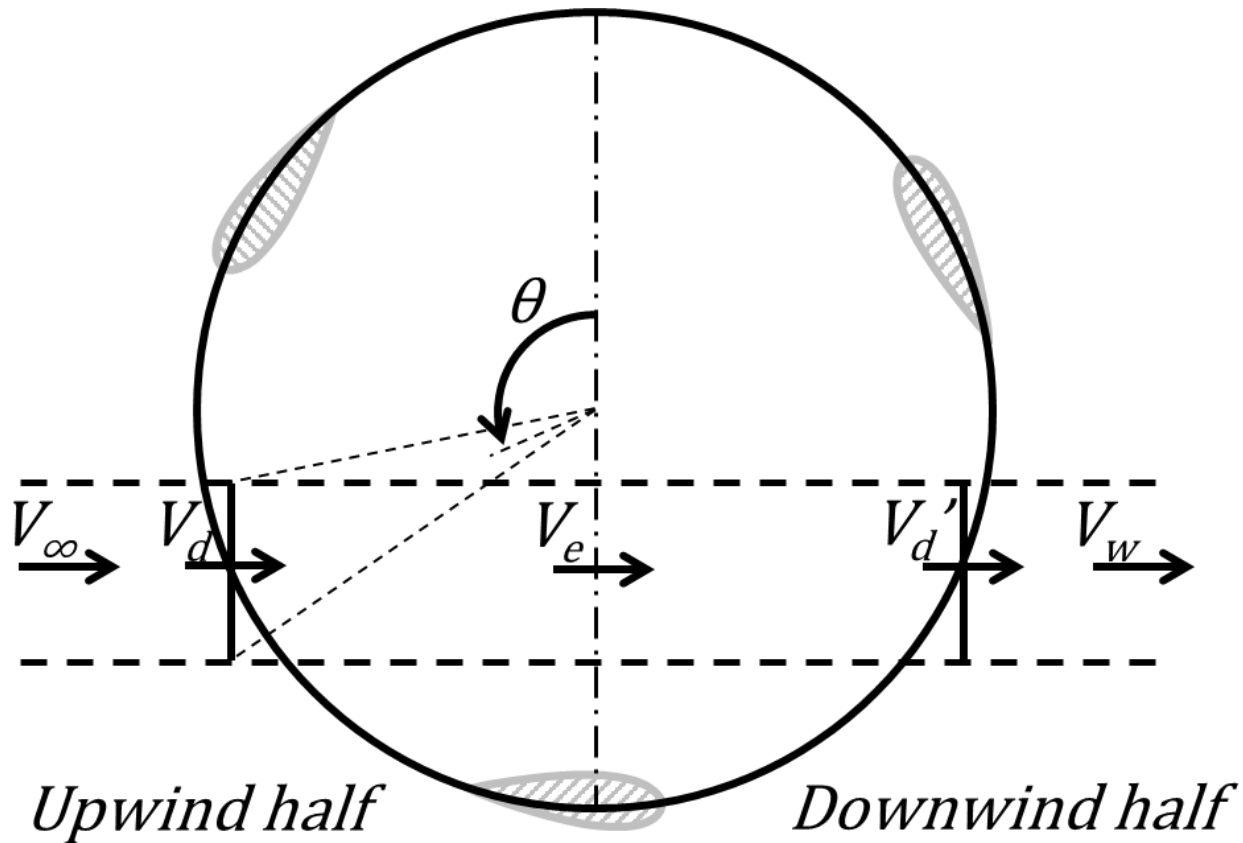
1 - MOMENTUM MODELS

Principle of double multiple streamtube model:

The main principle is to use the **blade-element theory** to calculate the forces of the blades from the local flow velocity. The flow velocity is obtained using the **momentum theory**. The solution can be found iteratively.



1 - MOMENTUM MODELS



$$V_d = \frac{V_\infty + V_e}{2}$$

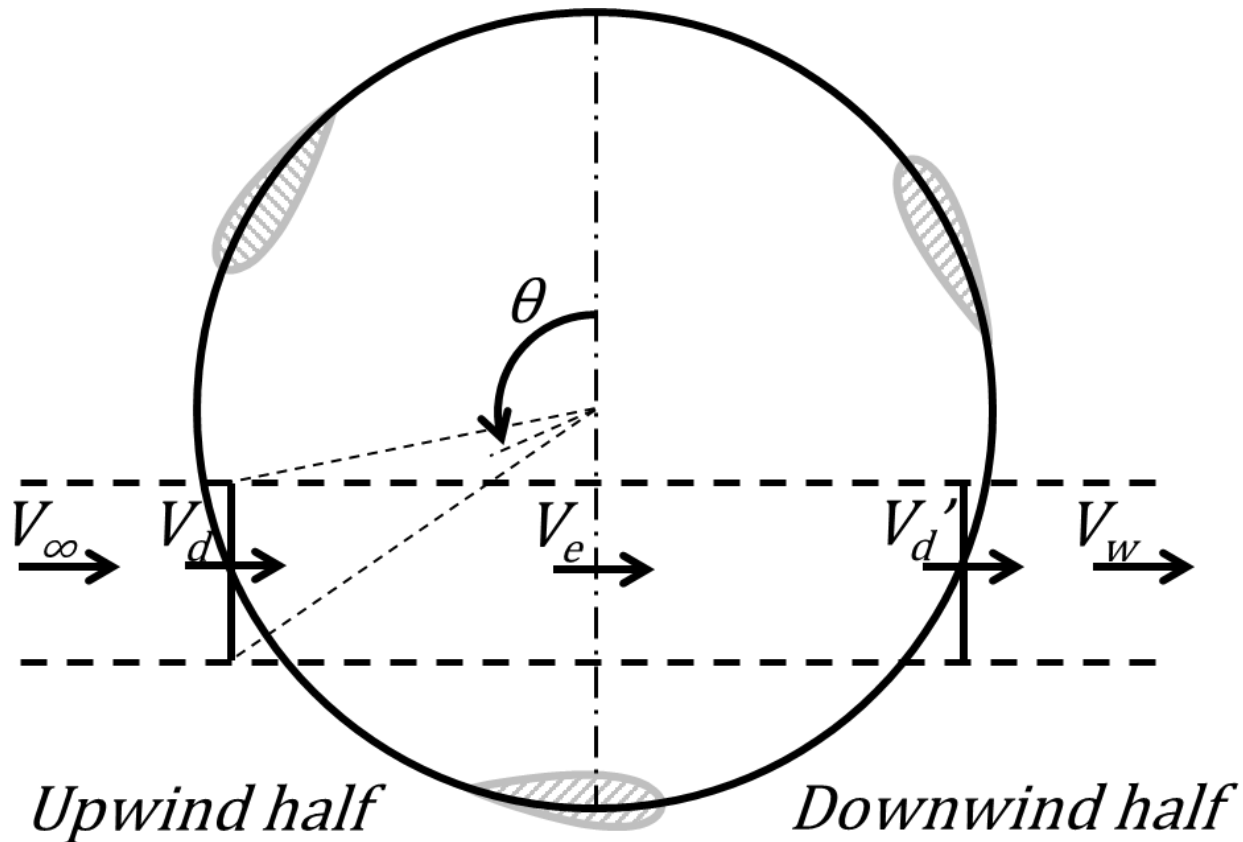
$$= (1 - a)V_\infty$$

$$V_d' = \frac{V_e + V_w}{2}$$

$$= (1 - a')(1 - 2a)V_\infty$$

You assume that the wake of the upwind half is fully developed before reaching the downwind half

1 - MOMENTUM MODELS



$$C_{T,BEM} = \frac{T_{avg}}{\frac{1}{2} \rho V_\infty^2 A_{st}}$$

where

Time-average thrust force:

$$T_{avg} = \frac{B \Delta \theta}{2\pi} T_{inst}$$

Instantaneous thrust force:

$$T_{inst} = \frac{1}{2} \rho V_{rel}^2 c (-C_t \cos(\theta) + C_n \sin(\theta))$$

Swept area of streamtube:

$$A_{st} = R \Delta \theta \sin(\theta)$$

1 - MOMENTUM MODELS

Upwind:

$$C_{T,MOM} = 4a(1 - a)$$

$$C_{T,BEM} = \frac{Bc}{R} \frac{1}{2\pi} \left(\frac{V_{rel}}{V_{\infty}} \right)^2 \left(-C_t \frac{\cos(\theta)}{\sin(\theta)} + C_n \right)$$

Downwind:

$$C_{T,MOM} = 4a'(1 - a')$$

$$C_{T,BEM} = \frac{Bc}{R} \frac{1}{2\pi} \left(\frac{V'_{rel}}{V_e} \right)^2 \left(-C_t \frac{\cos(\theta)}{\sin(\theta)} + C_n \right)$$

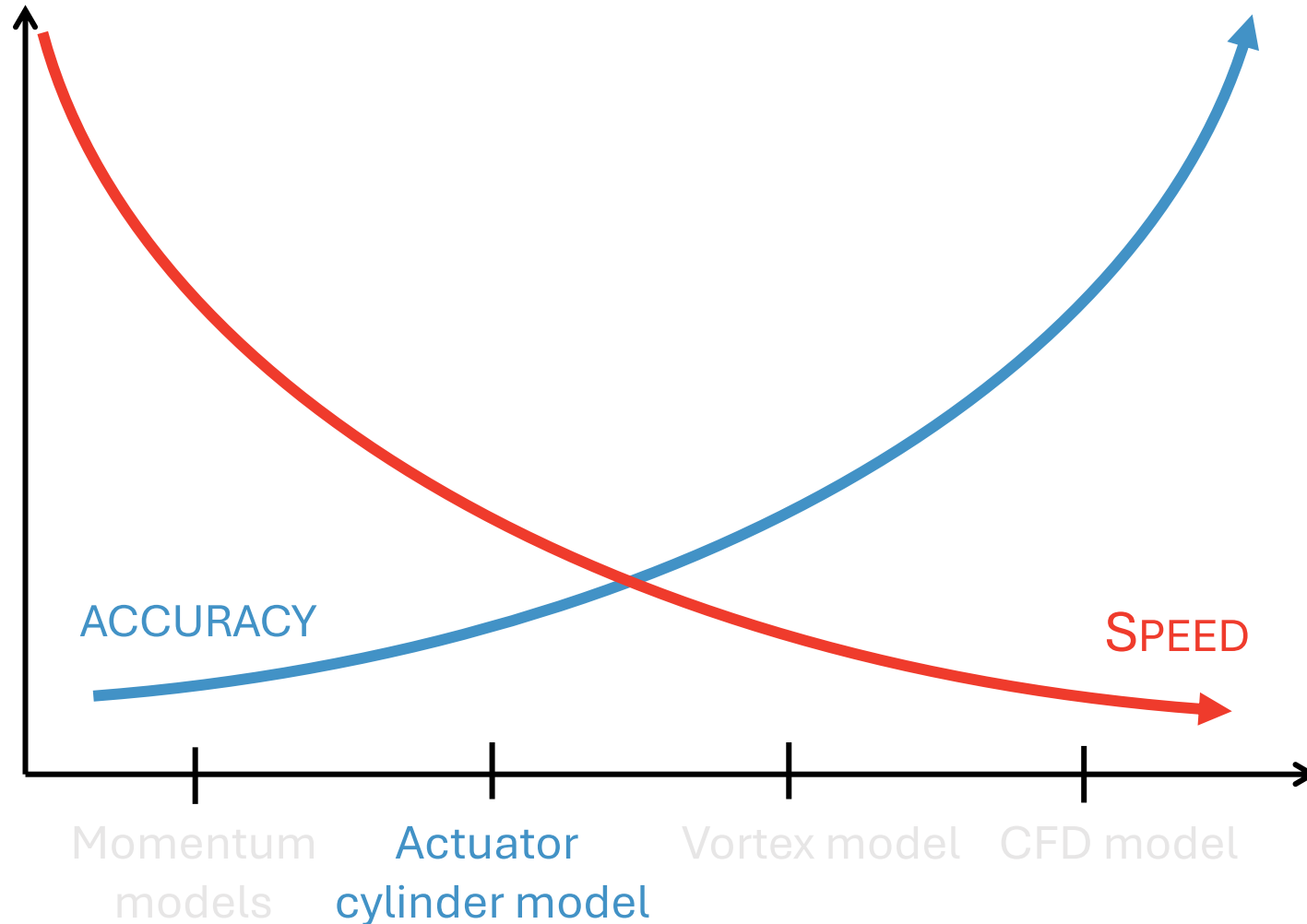
Equating both equations allows to determine the induction upwind and downwind, for each streamtube

1 - MOMENTUM MODELS

Assumptions of double multiple streamtube model:

- The wake of the upwind half is fully developed before reaching the downwind half and as such that you have atmospheric pressure in the centre of the turbine.
- The original DMST model by Paraschivoiu uses straight streamtubes. Read and Sharpe include flow expansion.
- The streamtubes are strictly independent
- The induction of the downwind actuator disk on the upwind actuator disk is not taken into account

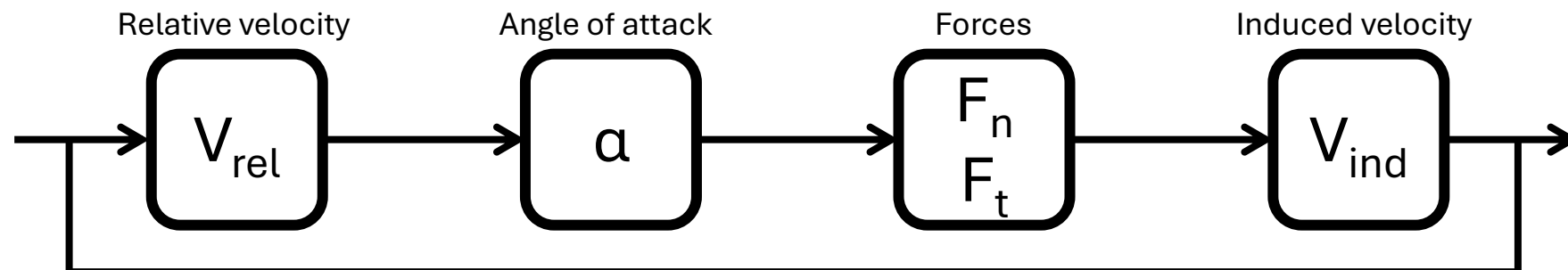
MODEL OVERVIEW



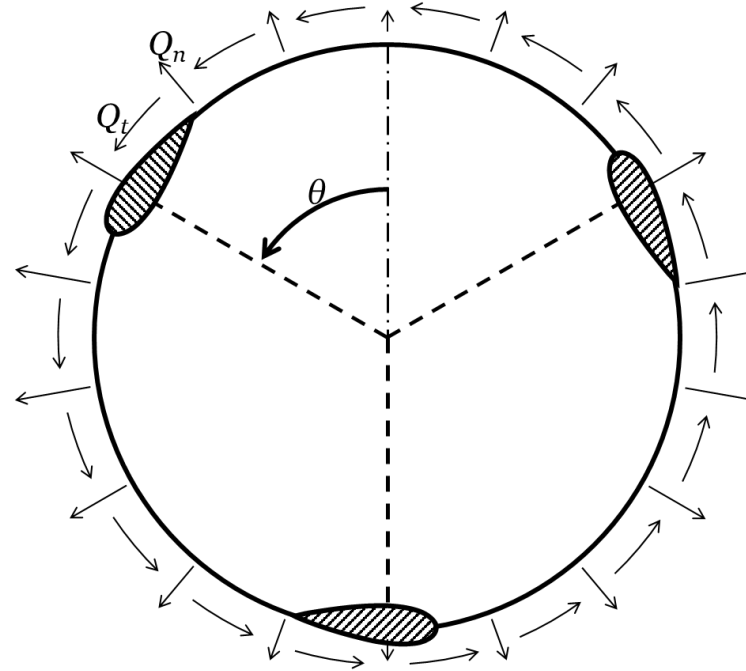
2 – ACTUATOR CYLINDER MODEL

Principle of actuator cylinder model:

The main principle is to use the **blade-element theory** to calculate the forces of the blades from the local flow velocity. The flow velocity is obtained using the **2D, steady, incompressible Euler momentum equations and the equation of continuity**. The solution can be found iteratively.



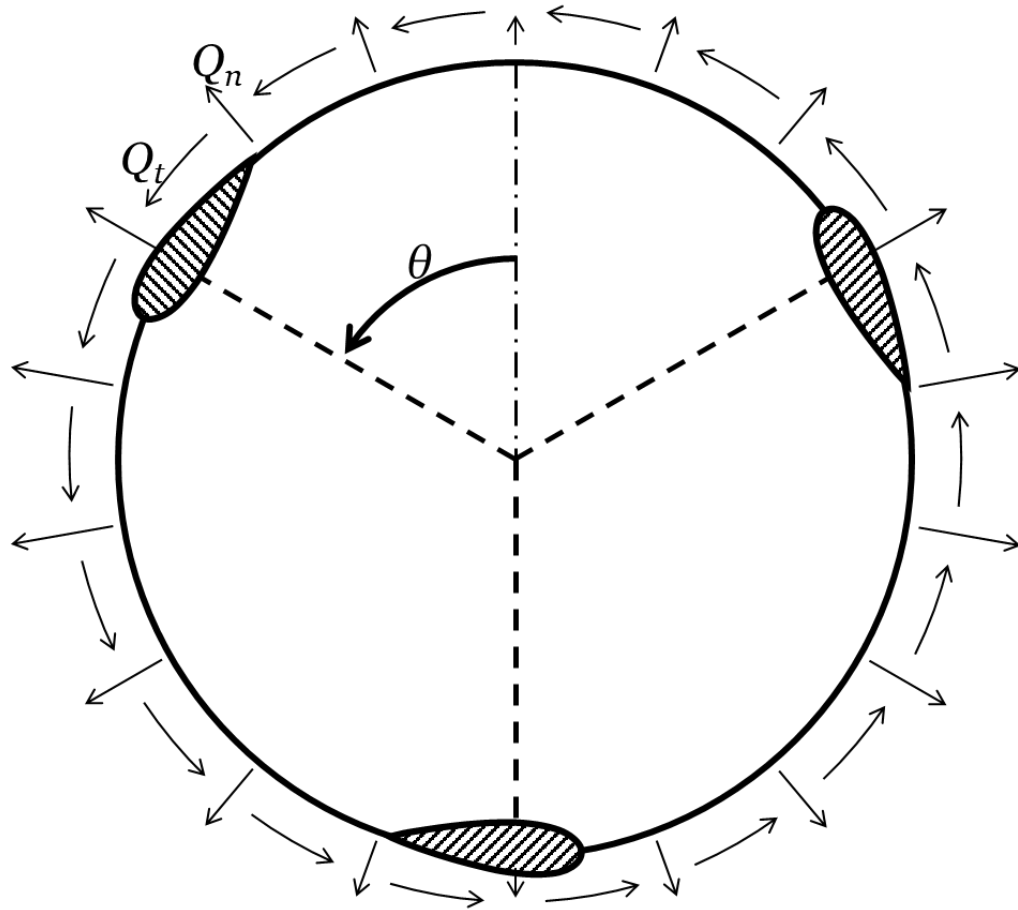
2 – ACTUATOR CYLINDER MODEL



2 – Actuator cylinder model

- H. Madsen
- 2D model but 3D equations can be derived in similar way
- The reaction of the blade forces are applied in the flow as distributed body forces in the normal and tangential direction

2 – ACTUATOR CYLINDER MODEL

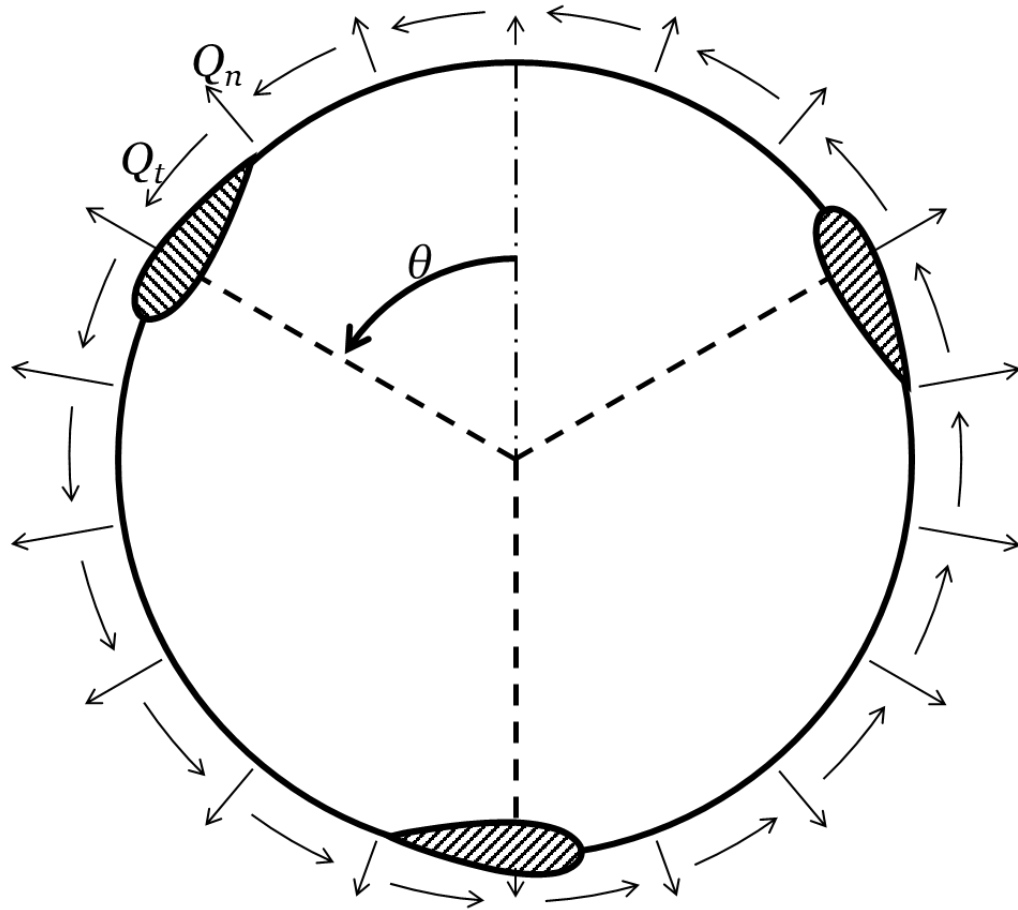


$$Q_n(\theta) = \frac{BF_n(\theta)}{2\pi R\rho V_\infty^2}$$

$$Q_t(\theta) = \frac{-BF_t(\theta)}{2\pi R\rho V_\infty^2}$$

⚠ Be careful with the sign convention wrt F_n and F_t (slide 29)

2 – ACTUATOR CYLINDER MODEL

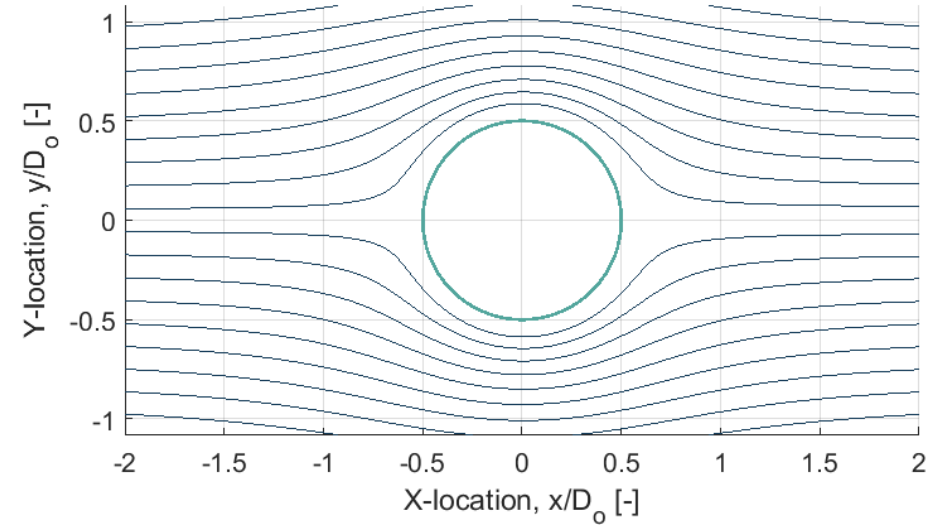
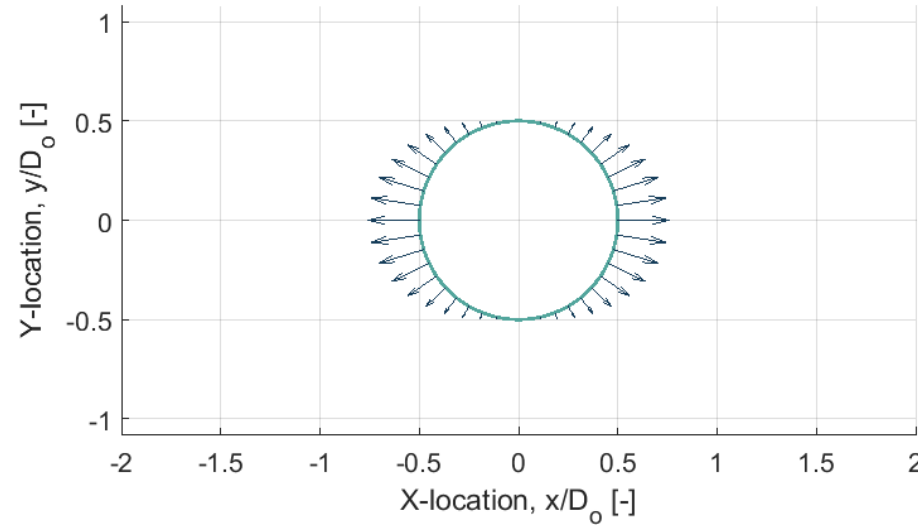


Using the Euler momentum equation in x and y, and the continuity equation, a solution can be found for the induction:

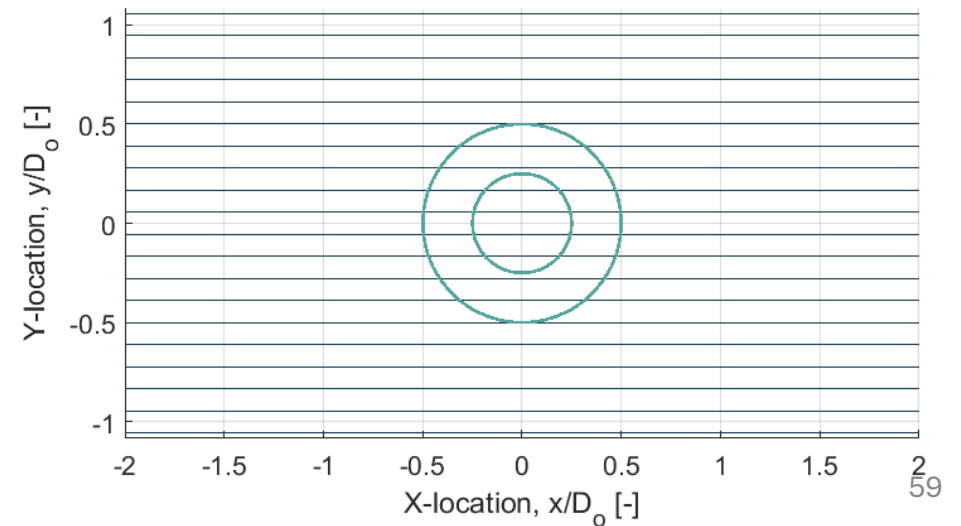
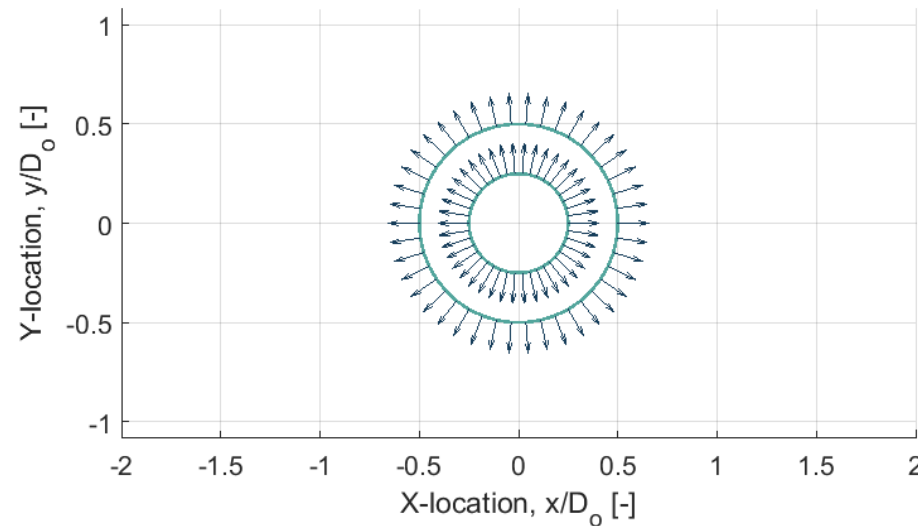
$$\begin{aligned}
 u(x, y) = & \frac{1}{2\pi} \int_0^{2\pi} Q_n(\theta) \frac{(x + \sin(\theta))\sin(\theta) - (y - \cos(\theta))\cos(\theta)}{(x + \sin(\theta))^2 + (y - \cos(\theta))^2} d\theta \\
 & + \frac{1}{2\pi} \int_0^{2\pi} Q_t(\theta) \frac{(x + \sin(\theta))\cos(\theta) + (y - \cos(\theta))\sin(\theta)}{(x + \sin(\theta))^2 + (y - \cos(\theta))^2} d\theta \\
 & - Q_n(\cos^{-1}(y)) - Q_t(\cos^{-1}(y)) \frac{y}{\sqrt{1-y^2}} \quad (\text{in the cylinder and wake}) \\
 & + Q_n(-\cos^{-1}(y)) - Q_t(-\cos^{-1}(y)) \frac{y}{\sqrt{1-y^2}} \quad (\text{in the wake}) \\
 v(x, y) = & \frac{1}{2\pi} \int_0^{2\pi} Q_n(\theta) \frac{(x + \sin(\theta))\cos(\theta) + (y - \cos(\theta))\sin(\theta)}{(x + \sin(\theta))^2 + (y - \cos(\theta))^2} d\theta \\
 & - \frac{1}{2\pi} \int_0^{2\pi} Q_t(\theta) \frac{(x + \sin(\theta))\sin(\theta) - (y - \cos(\theta))\cos(\theta)}{(x + \sin(\theta))^2 + (y - \cos(\theta))^2} d\theta
 \end{aligned}$$

2 – ACTUATOR CYLINDER MODEL

Flow around cylinder

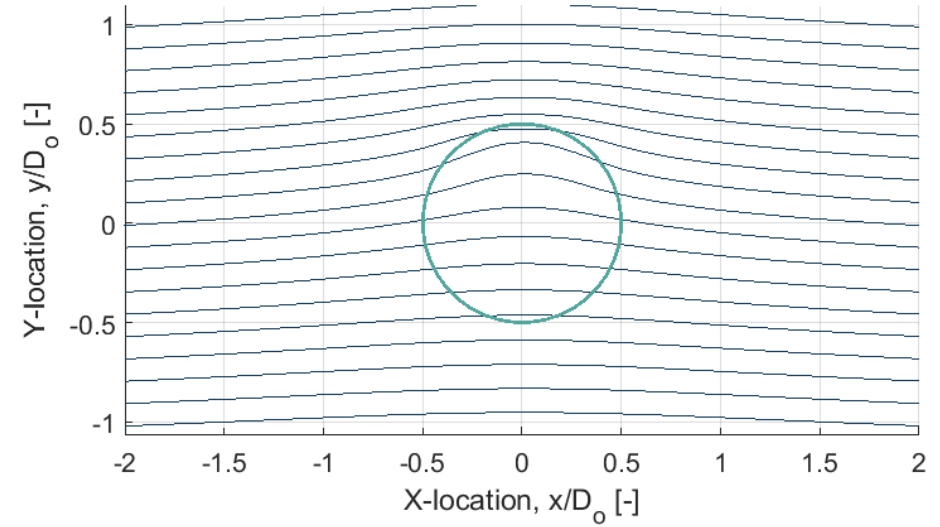
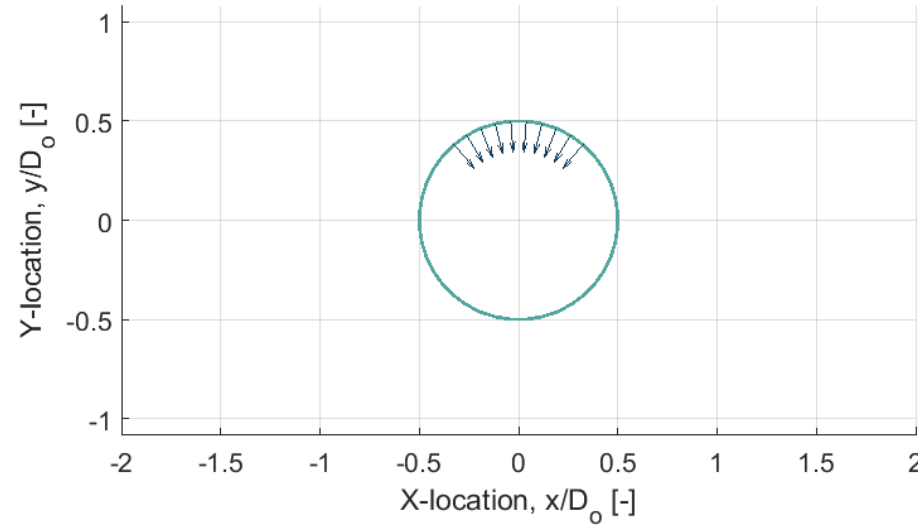


Conservative forcefield

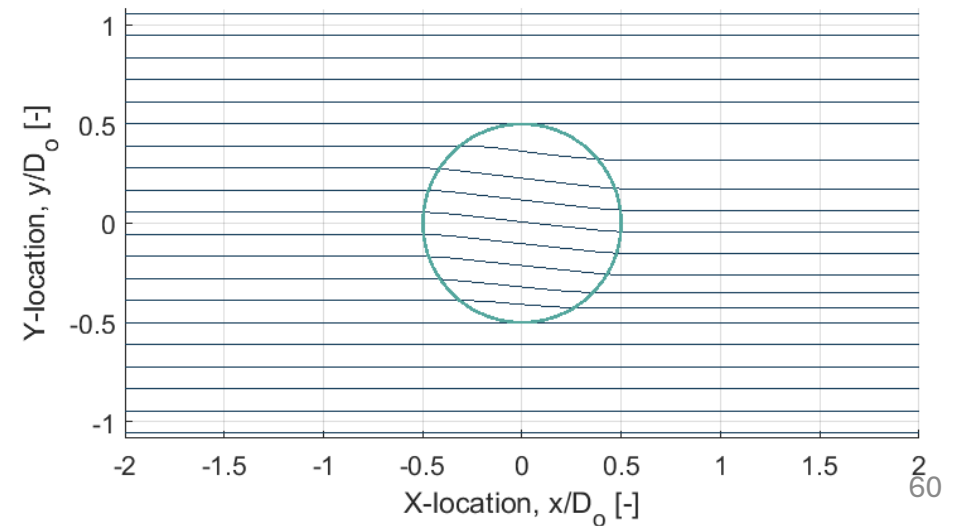
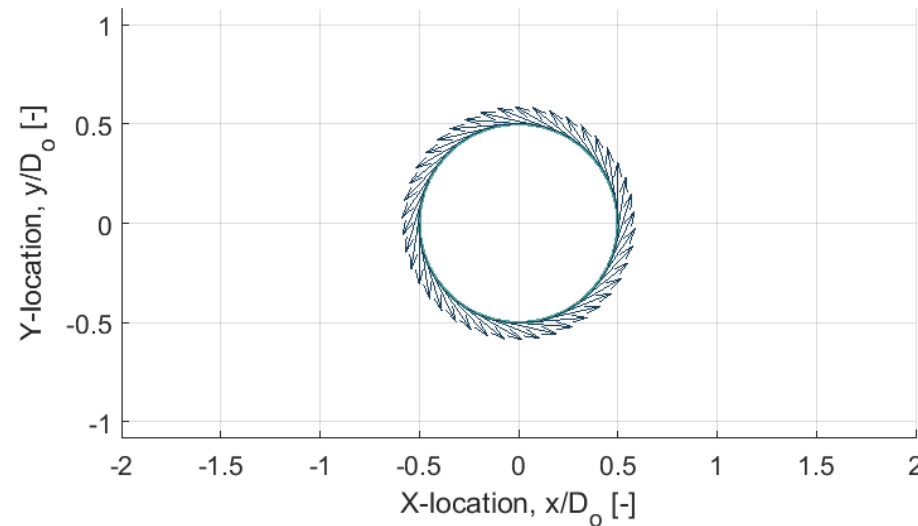


2 – ACTUATOR CYLINDER MODEL

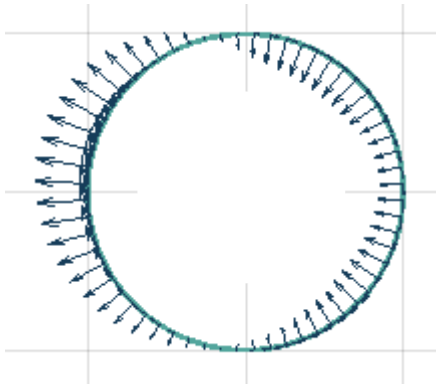
Airfoil flow



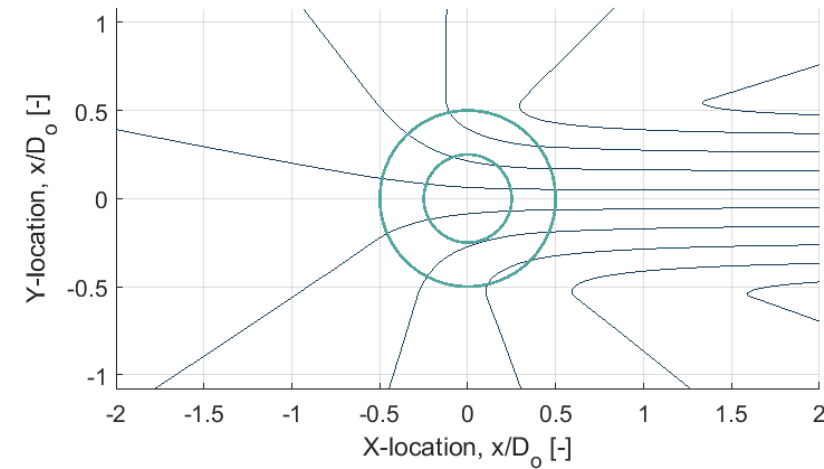
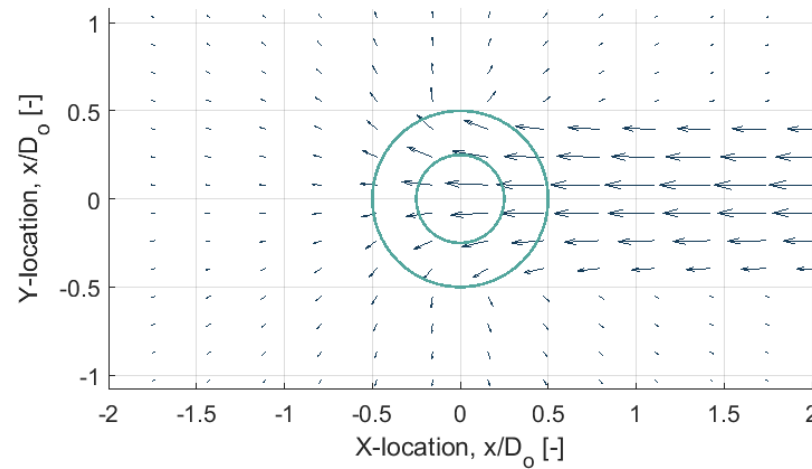
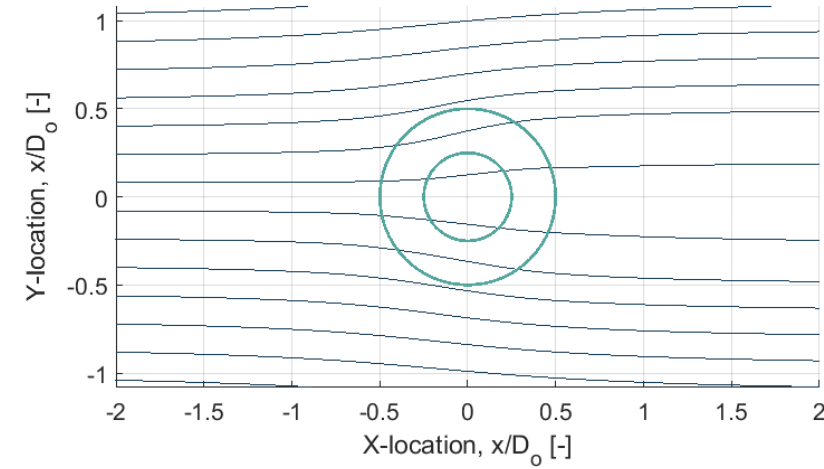
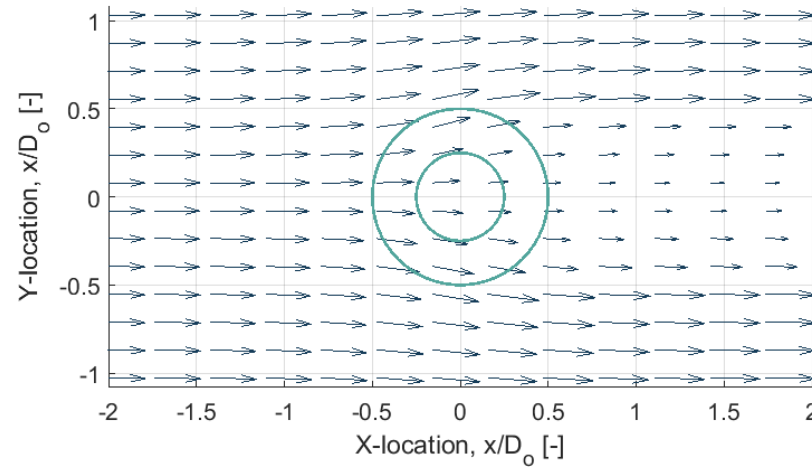
Non-conservative
forcefield



2 – ACTUATOR CYLINDER MODEL



Typical VAWT forcefield

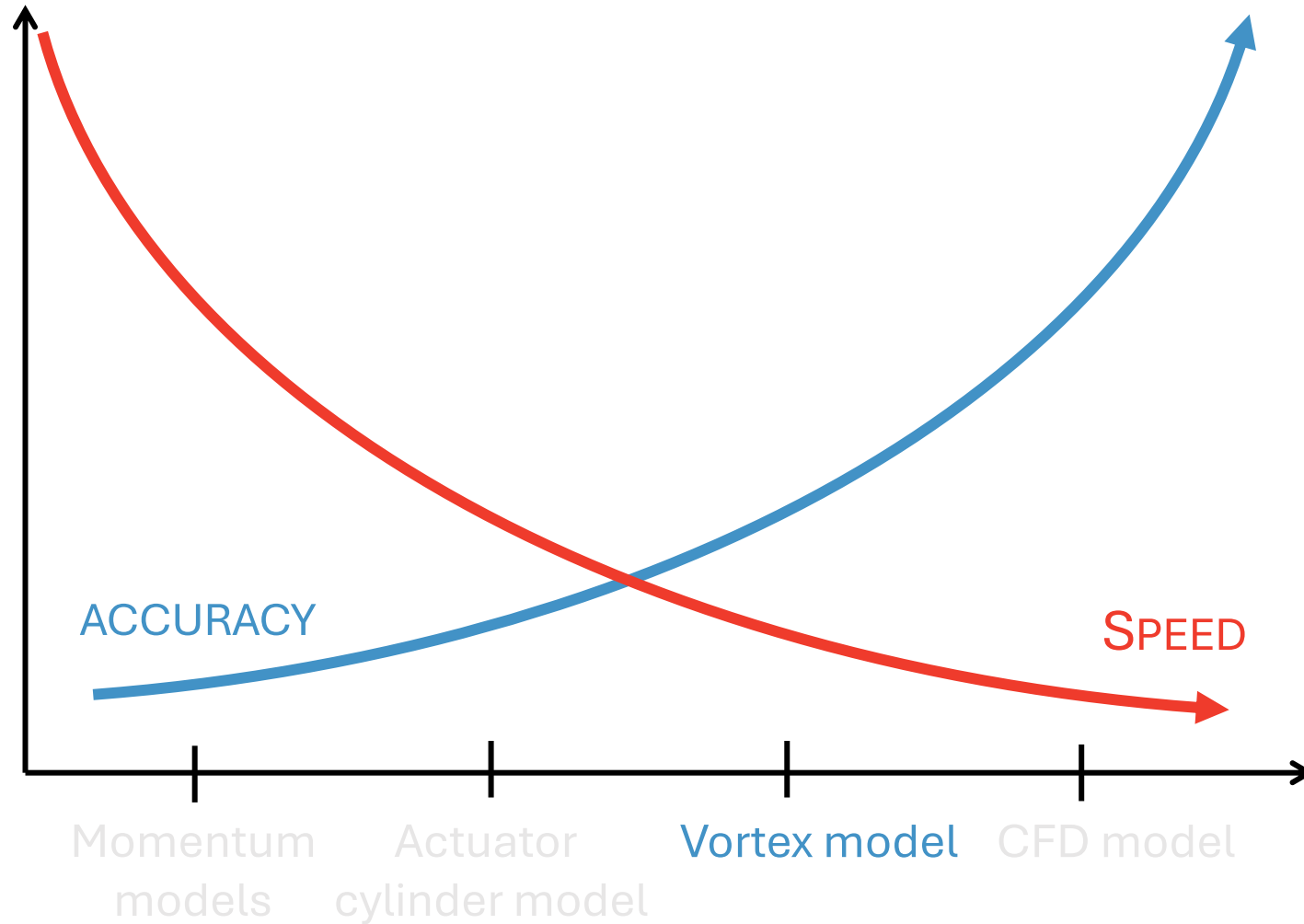


2 – ACTUATOR CYLINDER MODEL

Assumptions of actuator cylinder model:

- The elements/streamtubes are no longer independent
- The induction of the downwind part on the upwind part is taken into account
- Represent 3D rotor by stacking 2D independent slices (or derive the 3D equations)
- Steady solution

MODEL OVERVIEW

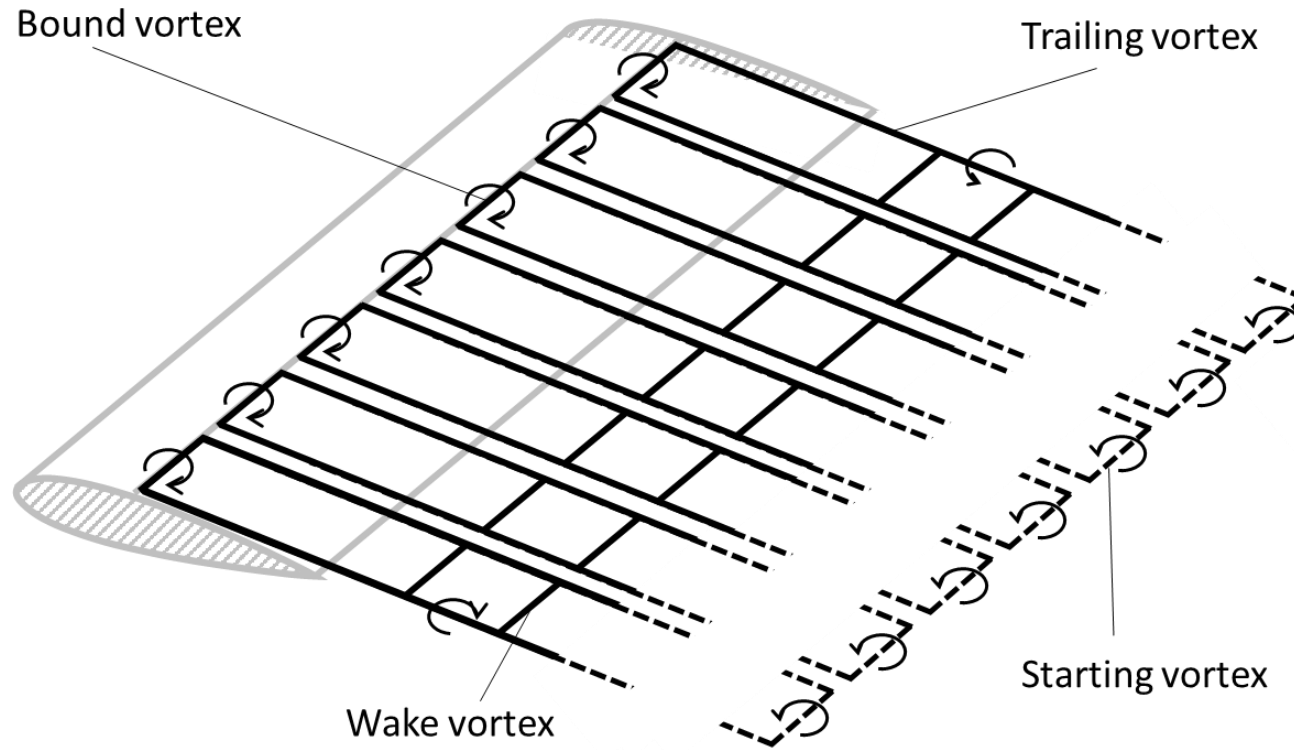


3 – VORTEX MODEL

Principle of vortex model:

Vortex models consult the **vorticity field** to determine the induced velocity and as such the relative velocity at the blades. The blade loads are either calculated using **lifting-line theory** or the principle of a **lifting surface**.

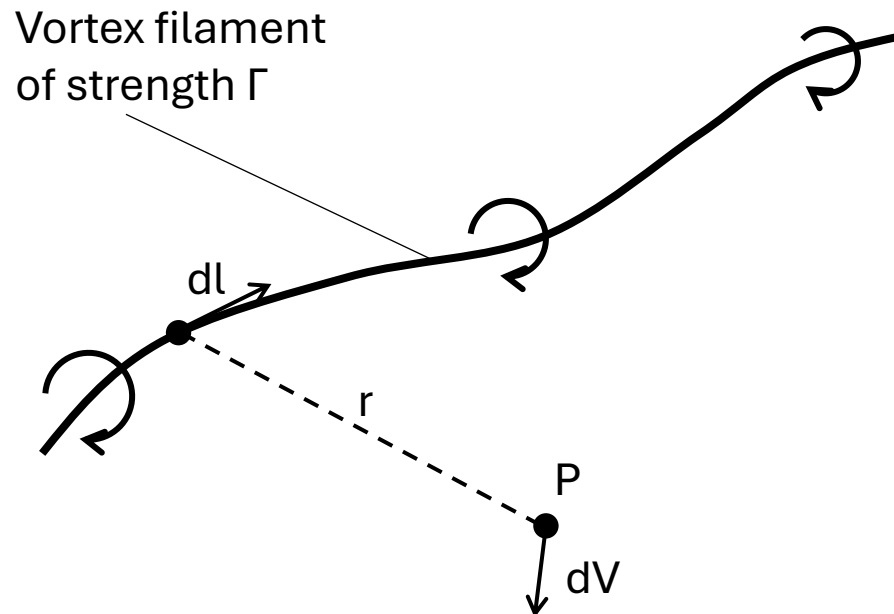
3 – VORTEX MODEL



3 – Vortex model

- Helmholtz' vortex theorem: circulation is constant along vortex line and cannot end
- Kelvin's circulation theorem: time rate of change of circulation in closed loop is zero

3 – VORTEX MODEL



Biot-Savart law:

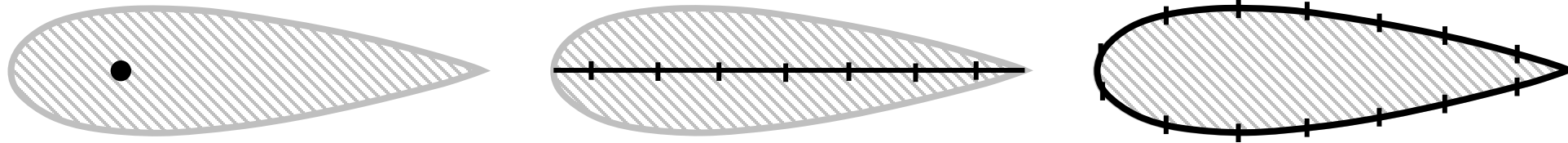
$$\overrightarrow{dV} = \frac{\gamma}{4\pi} \frac{\overrightarrow{dl} \times \vec{r}}{|\vec{r}|^3}$$

The wake will be represented by a vortex lattice, where at every time step each blade element is producing a new shed vortex line segment connected to the bound vortex by two trailing vortex line segments.

2 types:

- **Free wake method:** the wake convects by the local velocity at every wake point
- **Fixed wake method:** location of wake points is prescribed

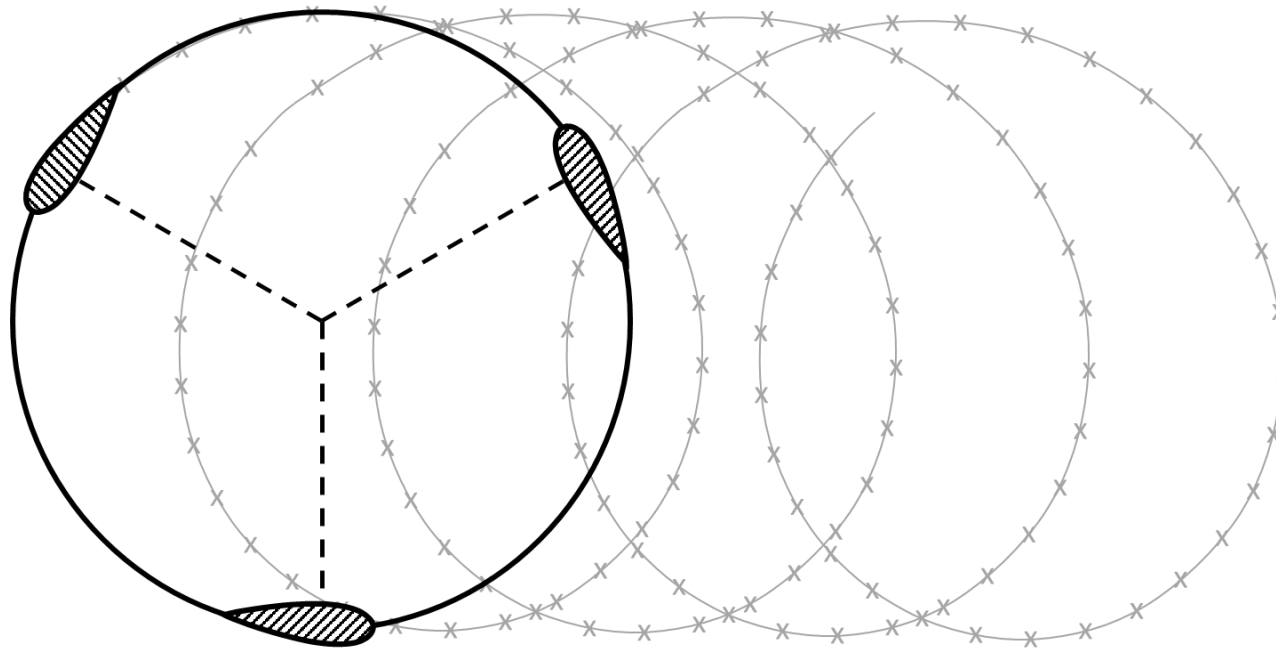
3 – VORTEX MODEL



The strength of the bound vortex can be prescribed in many ways.

- Blade is represented by **single lifting line** (often) at quarter-chord point. Circulation is concentrated along one line and calculated using Kutta-Joukowski theorem.
- Blade represented by **vortex sheet**. Straight vortex filaments side by side forming a lifting surface:
 - Surface coincide with airfoil mean camber line
 - Surface coincide with airfoil surface

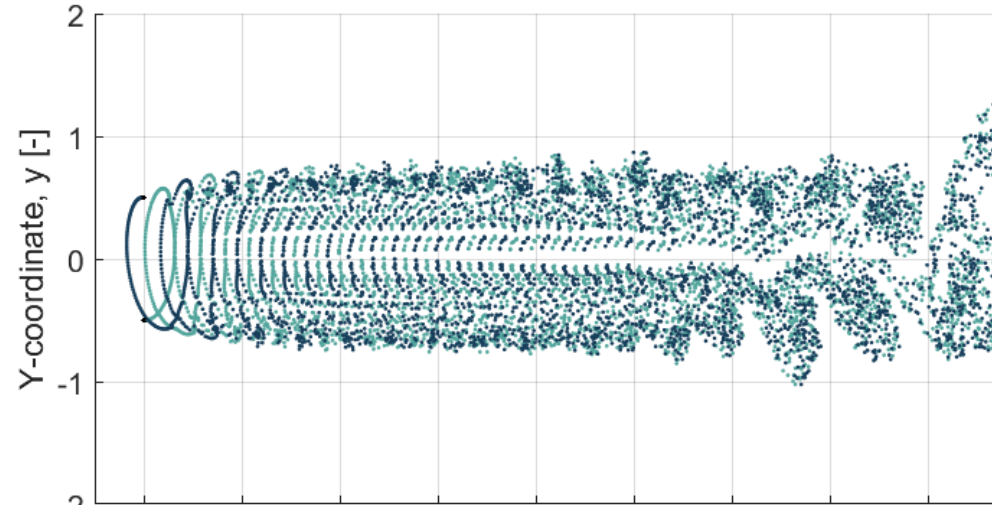
3 – VORTEX MODEL



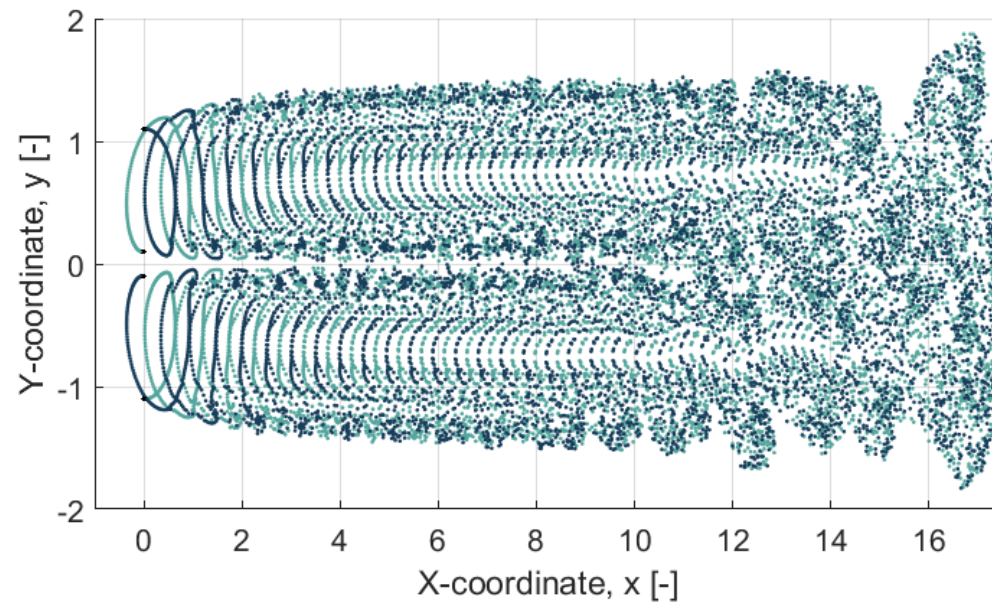
⚠ In 2D, the bound vortex is infinitely long and thus the trailing vortices are infinitely far. As such, the vortex lament can be replaced by a vortex point with strength Γ

3 – VORTEX MODEL

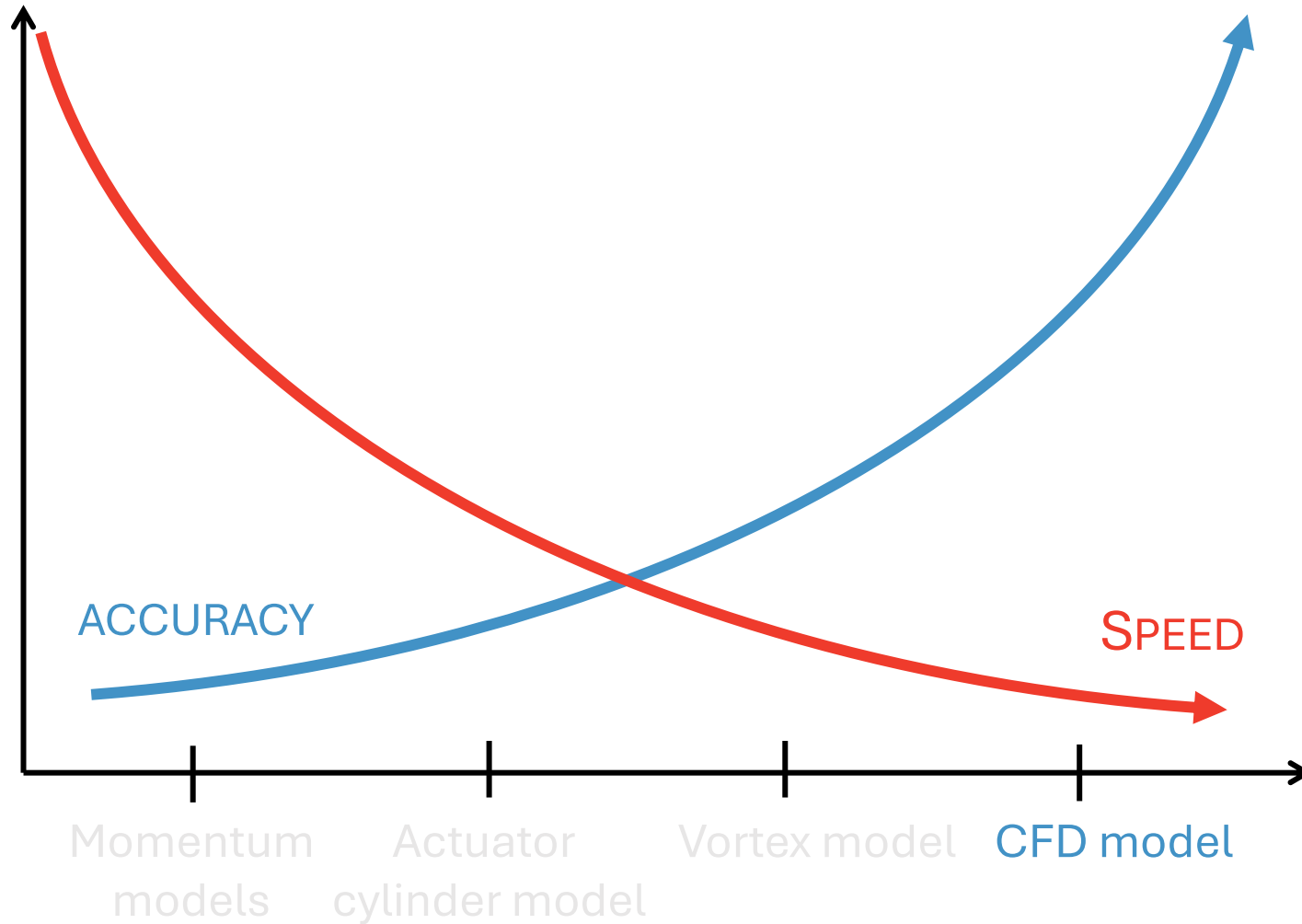
Single rotor



Double rotor



MODEL OVERVIEW



4 – CFD MODEL

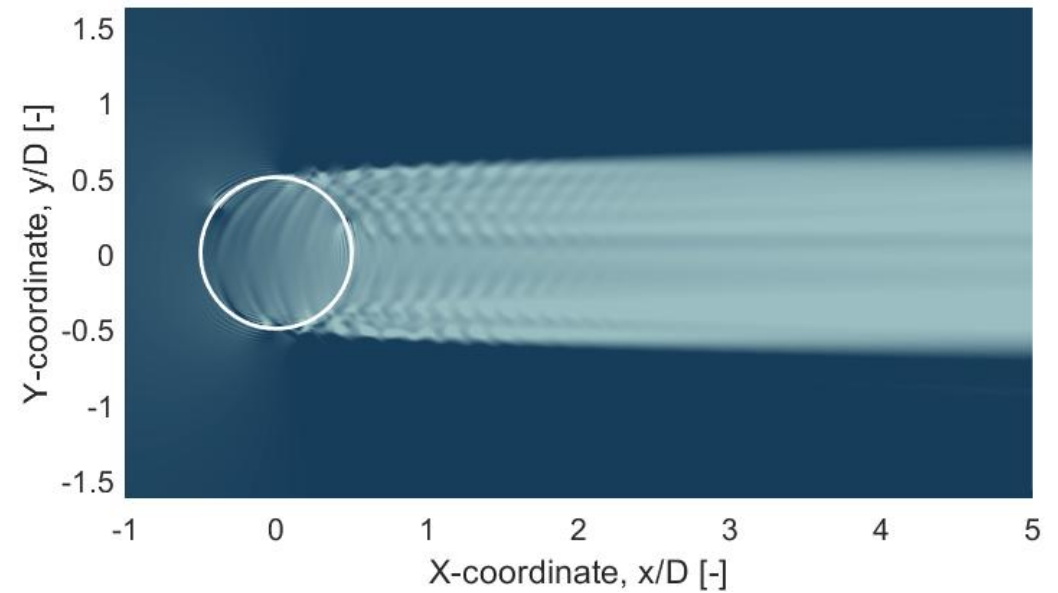
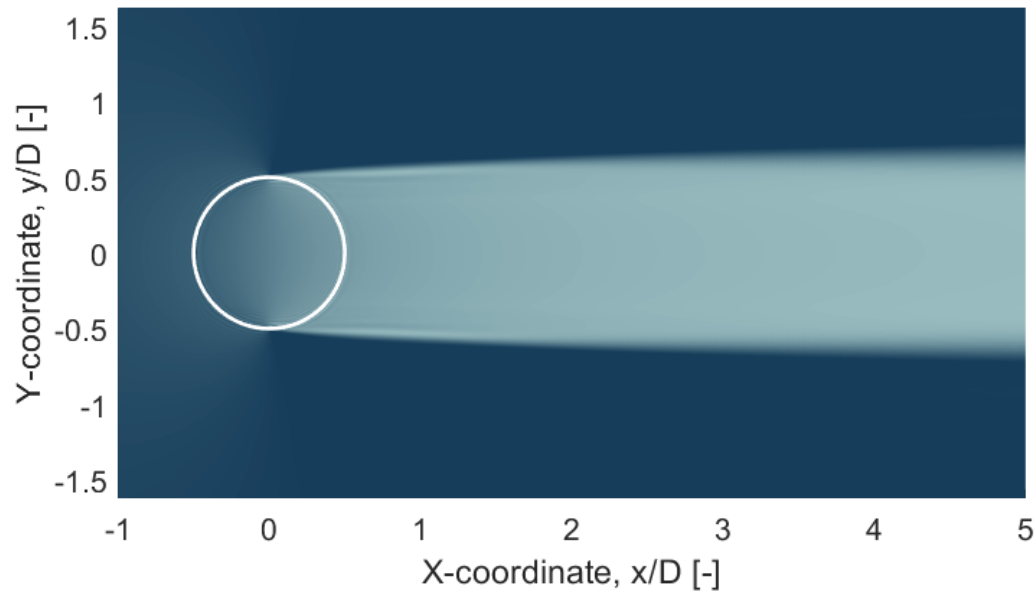
Principle of CFD model:

CFD models consult the Navier-Stokes equations to determine the pressure and velocity field. The blade are either included as geometrical surfaces or represented by volume forces.

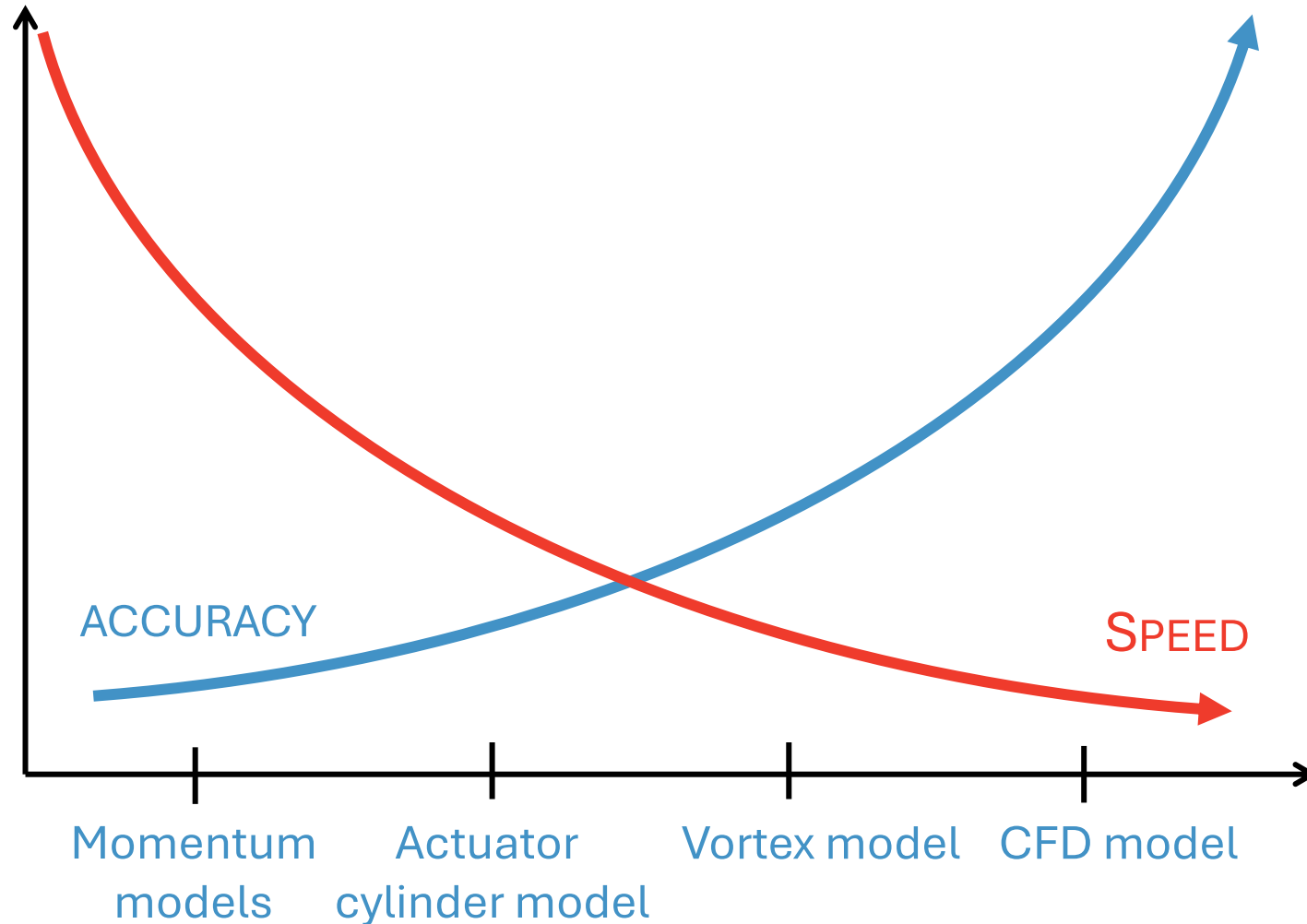
⚠ You could also apply the time-averaged volume forces to model the infinite bladed actuator cylinder

4 – CFD MODEL

Finite and infinite number of blades using CFD:



MODEL OVERVIEW



MODEL COMPARISON

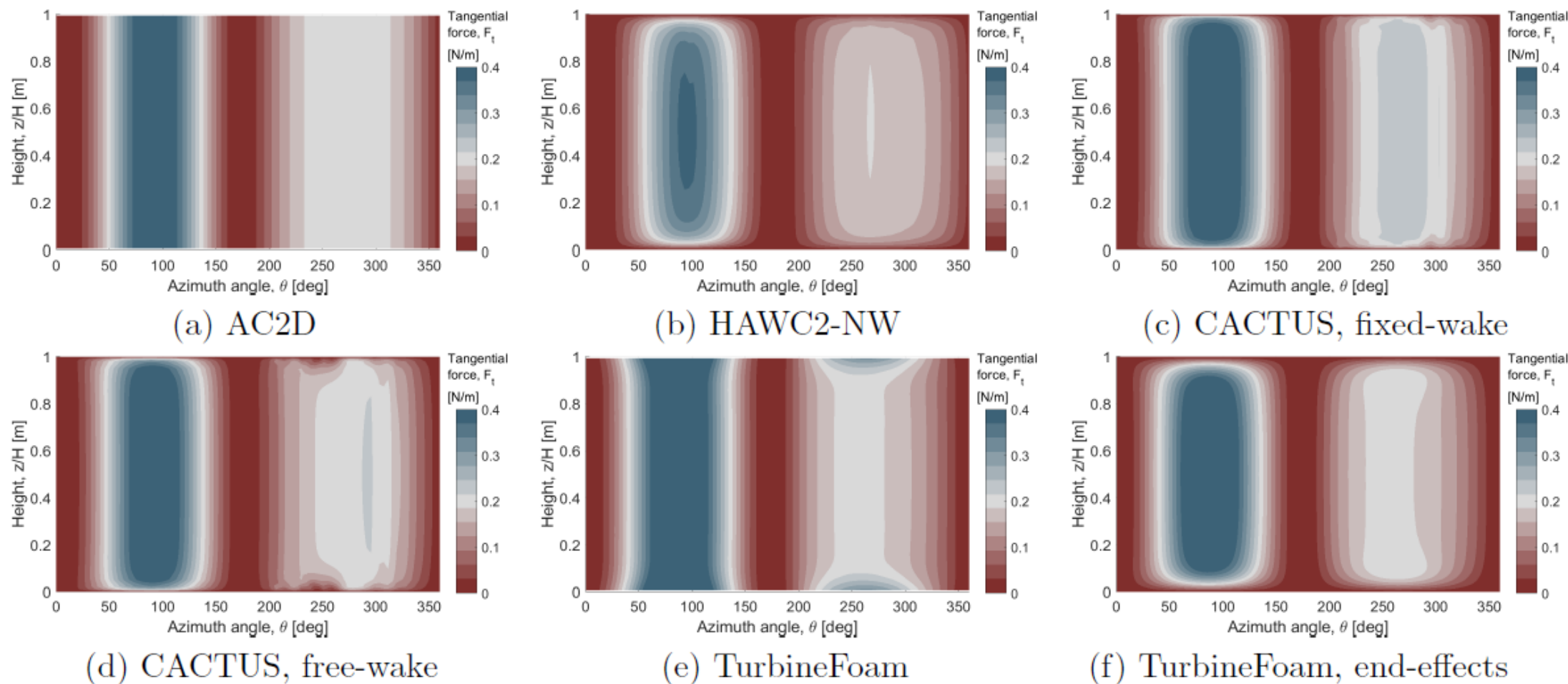


Figure 2: Tangential blade force vs azimuth angle and spanwise direction for the 3-bladed H -shaped baseline turbine using six different models. The rotor solidity is 0.085, the aspect ratio is 1 and the tip speed ratio is 3.

WAKE DYNAMICS

HAWT VORTEX SYSTEM

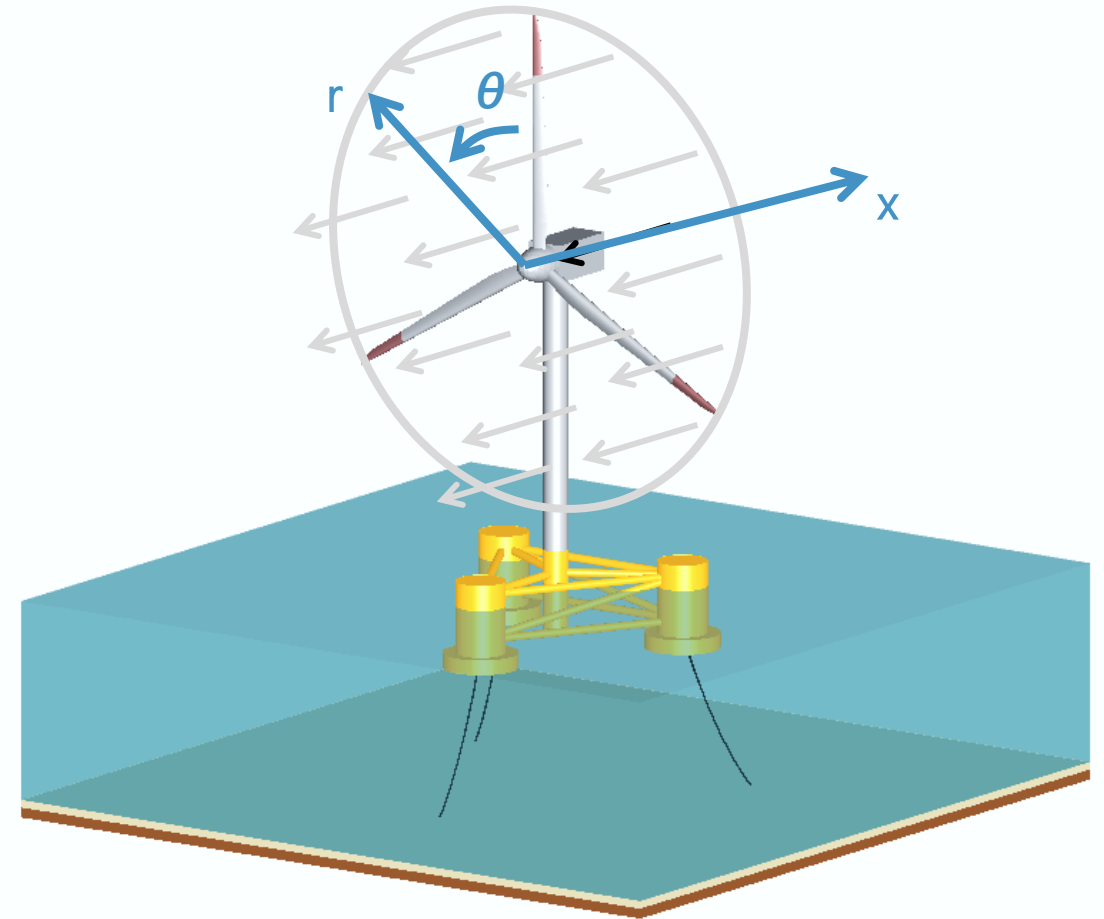
Where is vorticity created:

The vorticity equations for incompressible, inviscid flows states that vorticity can only be created by a change in the force field, i.e. at the locations where the curl of the force field is non-zero.

$$\frac{D\omega}{Dt} = \nabla \times f$$

In polar coordinates (r, θ, x) this becomes for a uniform 3D disk:

$$\frac{D\omega}{Dt} = \begin{bmatrix} \cancel{\frac{1}{r} \frac{df_x}{d\theta}} - \cancel{\frac{df_\theta}{dx}} \\ \cancel{\frac{df_x}{dr}} + \cancel{\frac{df_r}{dx}} \\ \cancel{\frac{df_\theta}{dr}} - \cancel{\frac{1}{r} \frac{df_r}{d\theta}} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{df_x}{dr} \\ 0 \end{bmatrix}$$



HAWT VORTEX SYSTEM

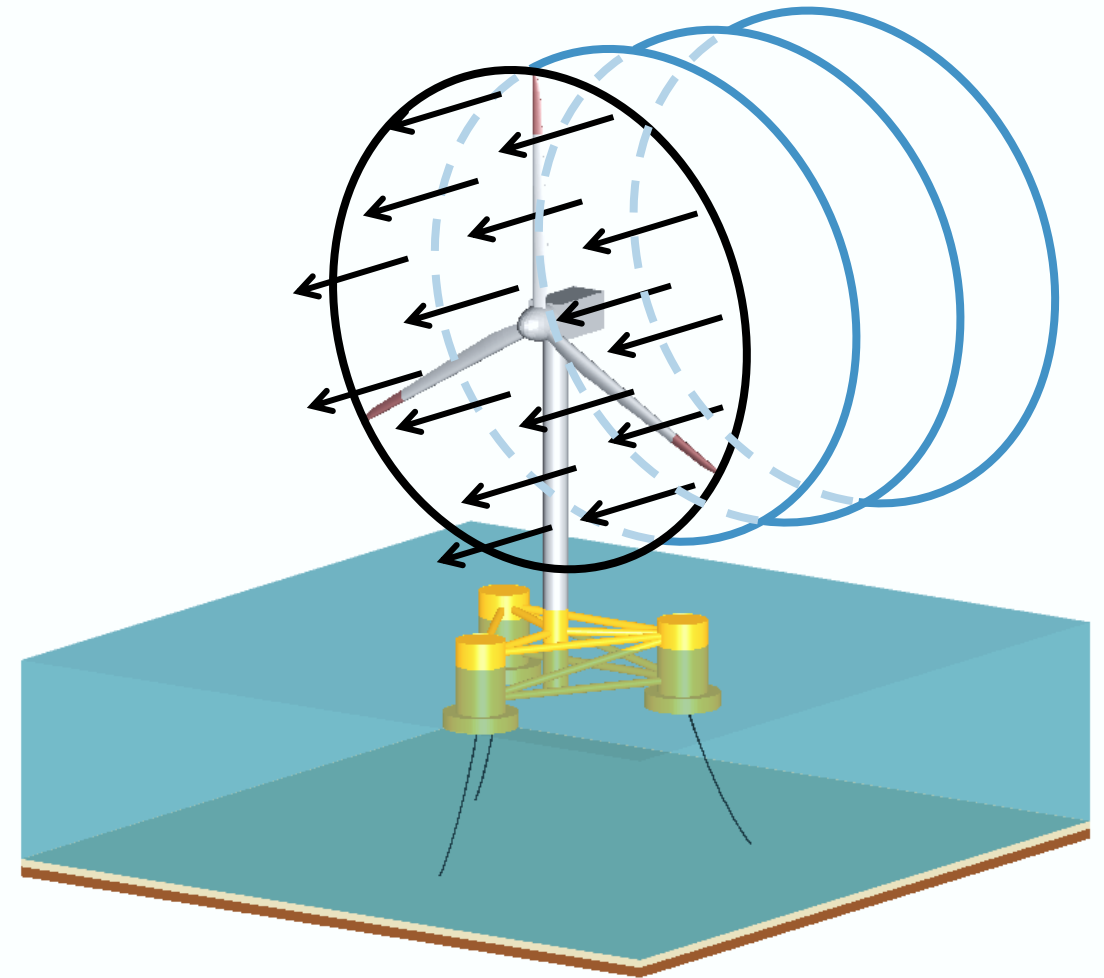
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VAWT VORTEX SYSTEM

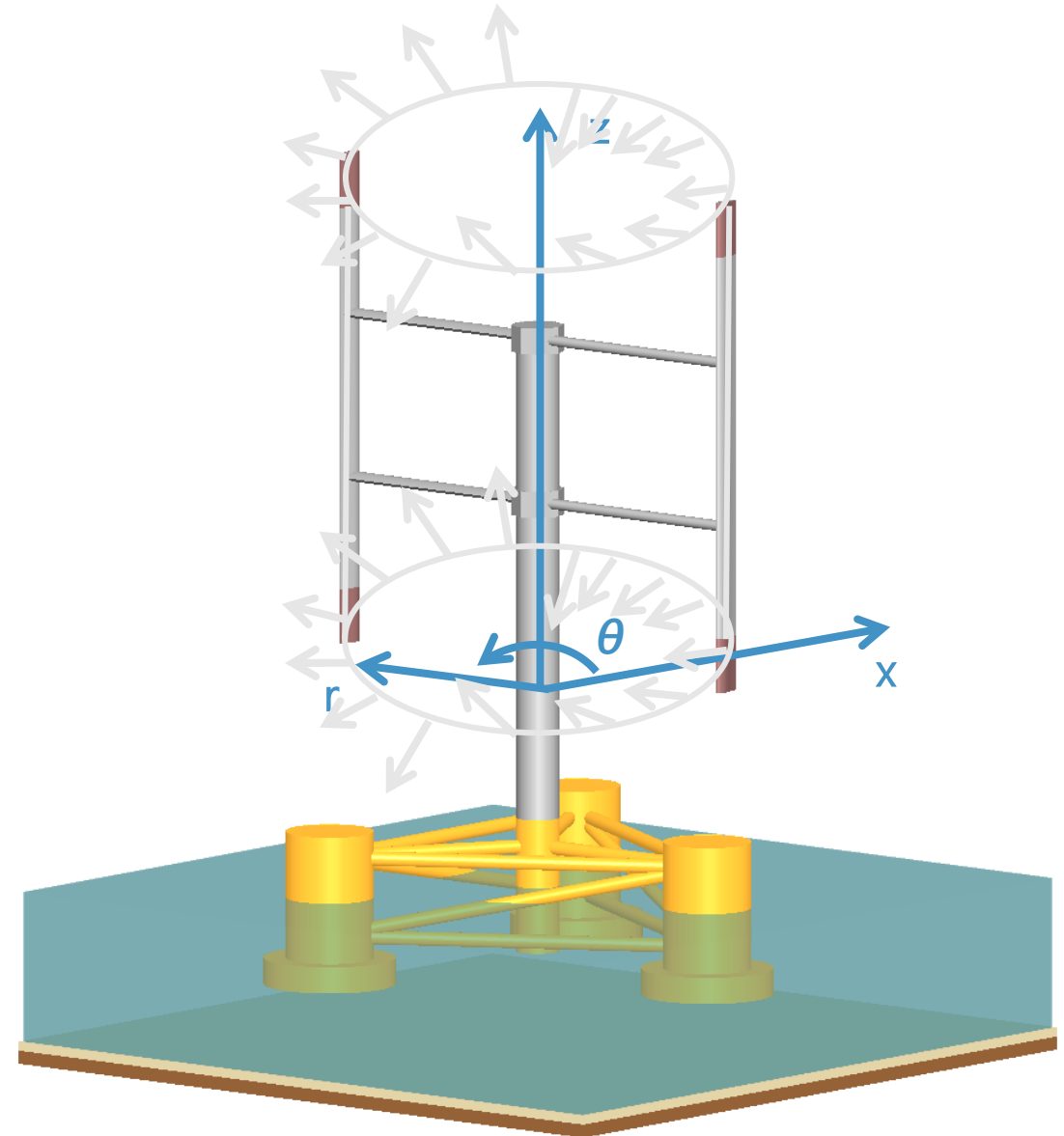
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$$\frac{D\omega}{Dt} = \nabla \times f$$

In polar coordinates (r, θ, z) this becomes for a uniform 3D cylinder:

$$\frac{D\omega}{Dt} = \begin{bmatrix} \cancel{\frac{1}{r} \frac{df_z}{d\theta}} - \cancel{\frac{df_\theta}{dz}} \\ \cancel{\frac{df_z}{dr}} + \frac{df_r}{dz} \\ \cancel{\frac{df_\theta}{dr}} - \cancel{\frac{1}{r} \frac{df_r}{d\theta}} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{df_r}{dz} \\ -\frac{1}{r} \frac{df_r}{d\theta} \end{bmatrix}$$



VAWT VORTEX SYSTEM

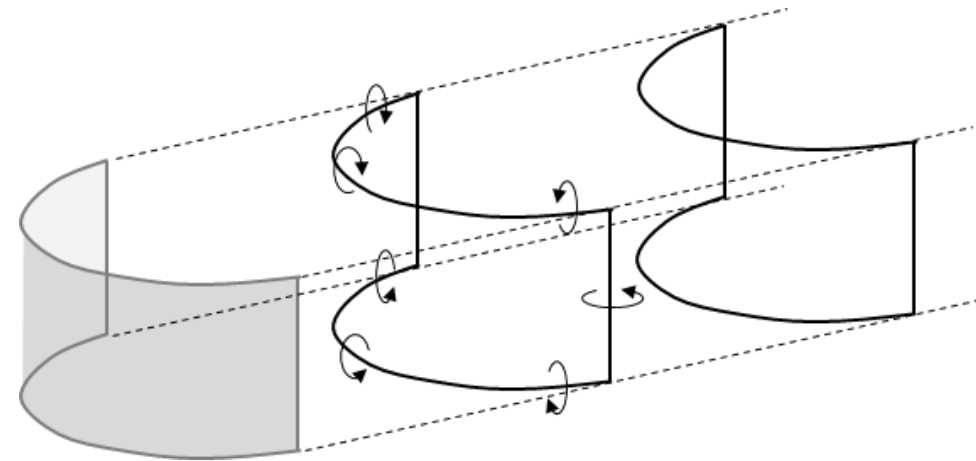
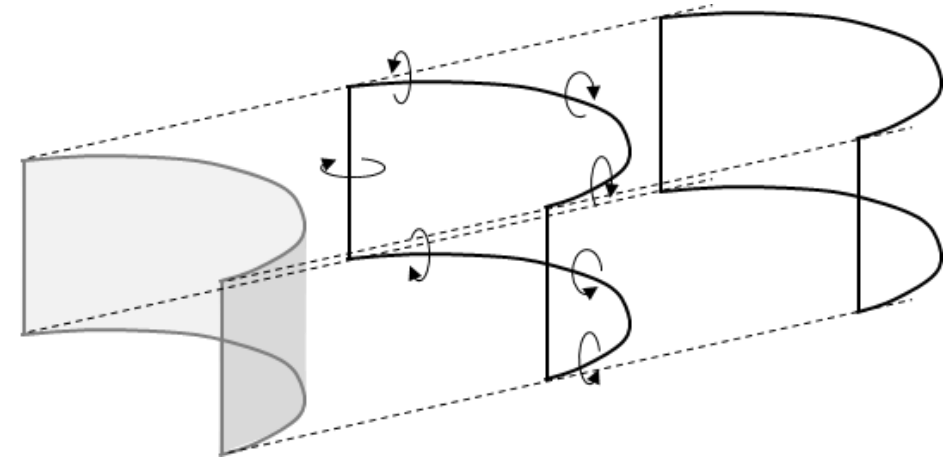
Where is vorticity created:

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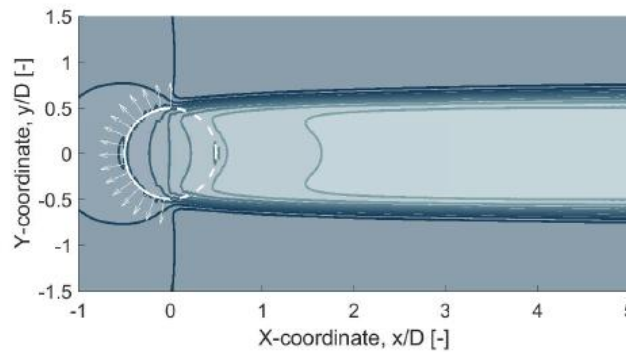
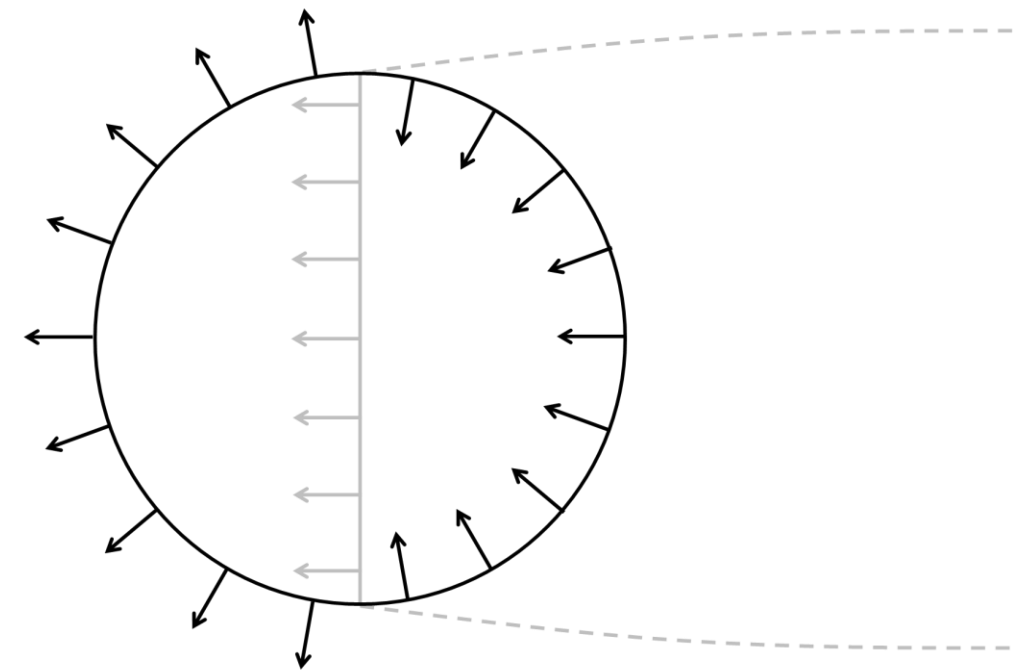
$$\frac{D\omega}{Dt} = \nabla \times f$$

In polar coordinates (r, θ, x) this becomes for a uniform 3D disk:

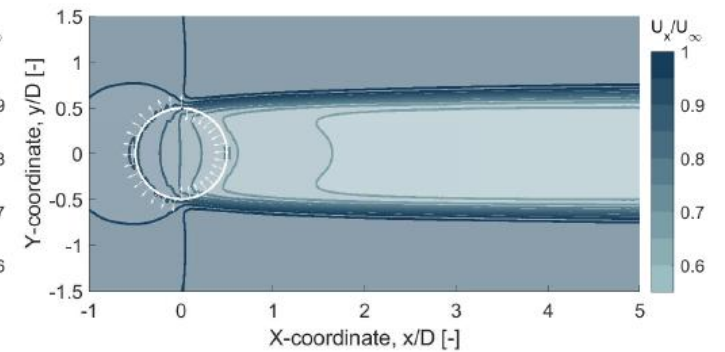
$$\frac{D\omega}{Dt} = \begin{bmatrix} \cancel{\frac{1}{r} \frac{df_z}{d\theta}} - \cancel{\frac{df_\theta}{dz}} \\ \cancel{\frac{df_z}{dr}} + \frac{df_r}{dz} \\ \cancel{\frac{df_\theta}{dr}} - \cancel{\frac{1}{r} \frac{df_r}{d\theta}} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{df_r}{dz} \\ -\frac{1}{r} \frac{df_r}{d\theta} \end{bmatrix}$$



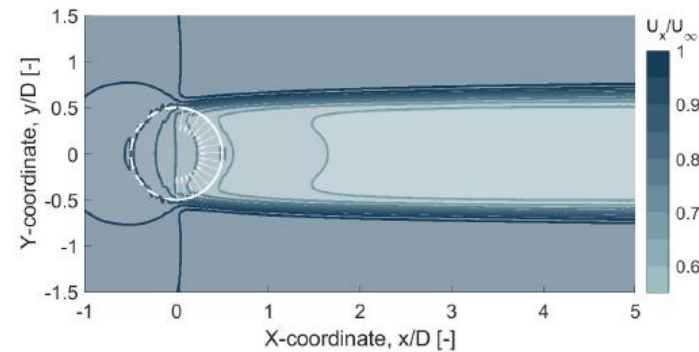
VAWT VORTEX SYSTEM



(a) All load upwind



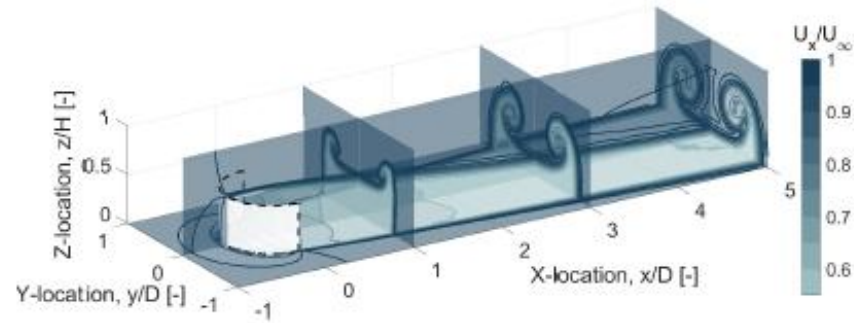
(b) 1/2 load upwind, 1/2 load downwind



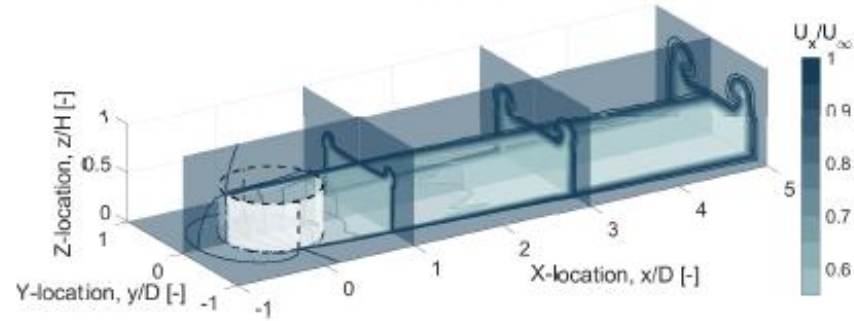
(c) All load downwind

⚠ In 2D all wakes would be exactly the same since trailing vortices are infinitely far.

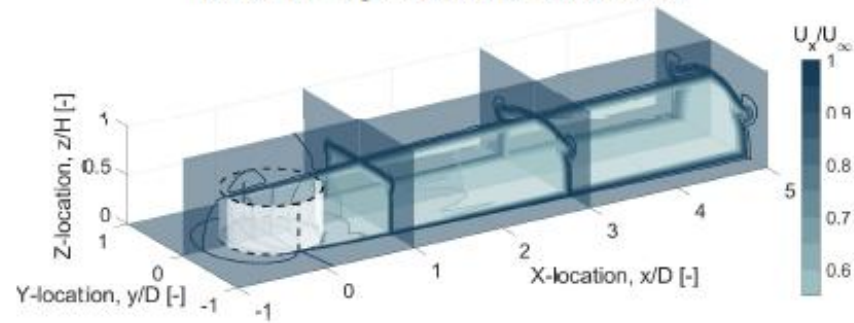
VAWT VORTEX SYSTEM



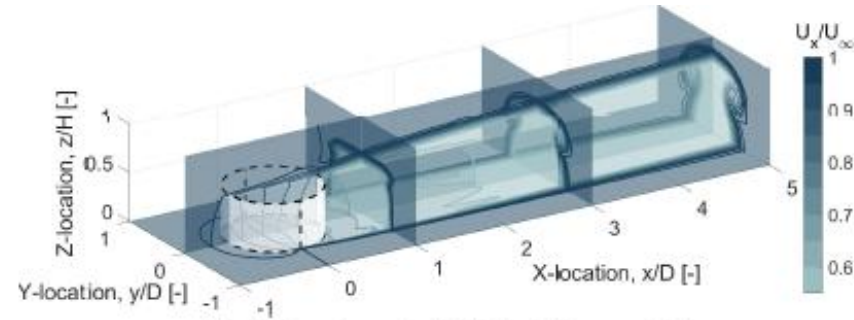
(a) All load upwind



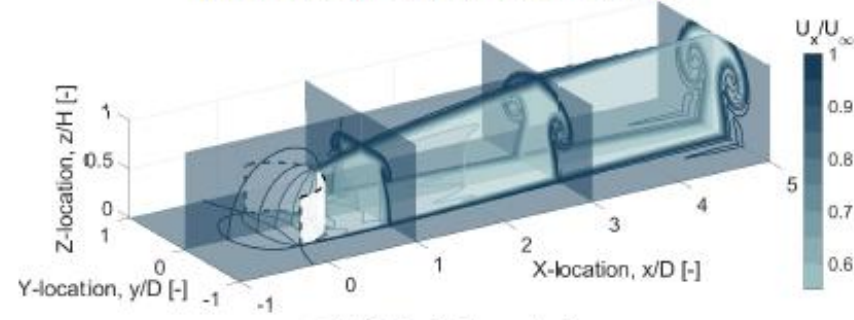
(b) 2/3 load upwind, 1/3 load downwind



(c) 1/2 load upwind, 1/2 load downwind



(d) 1/3 load upwind, 2/3 load downwind



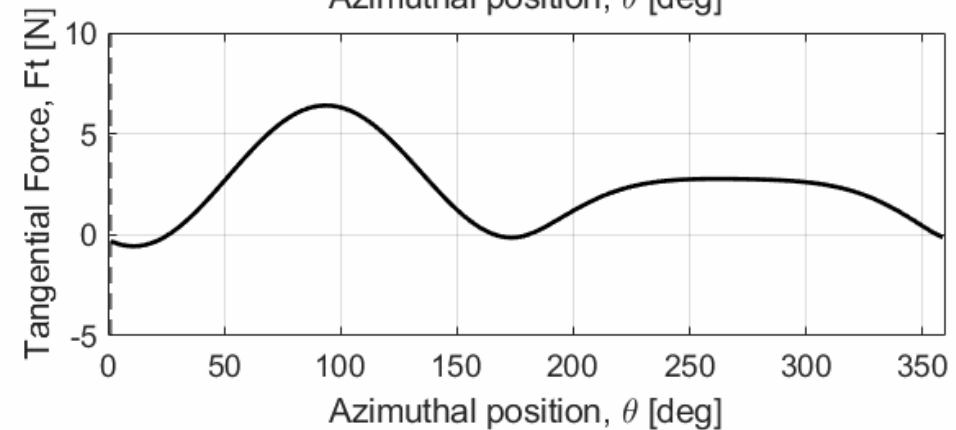
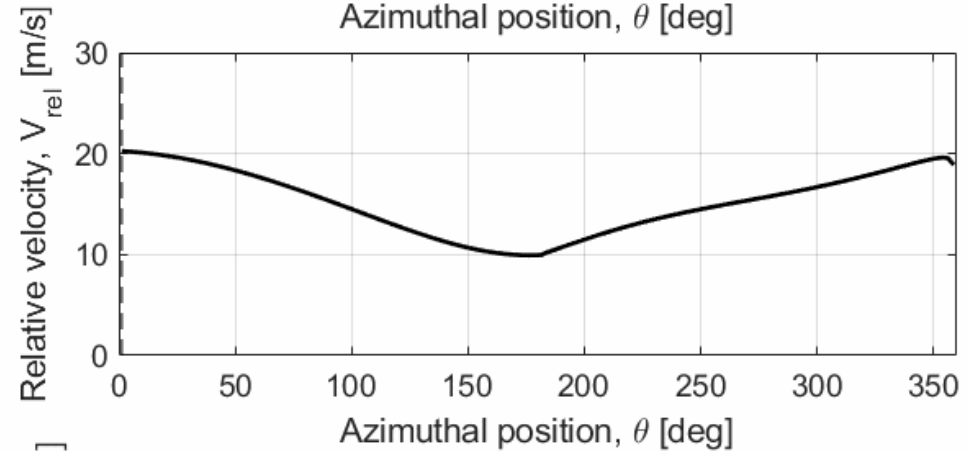
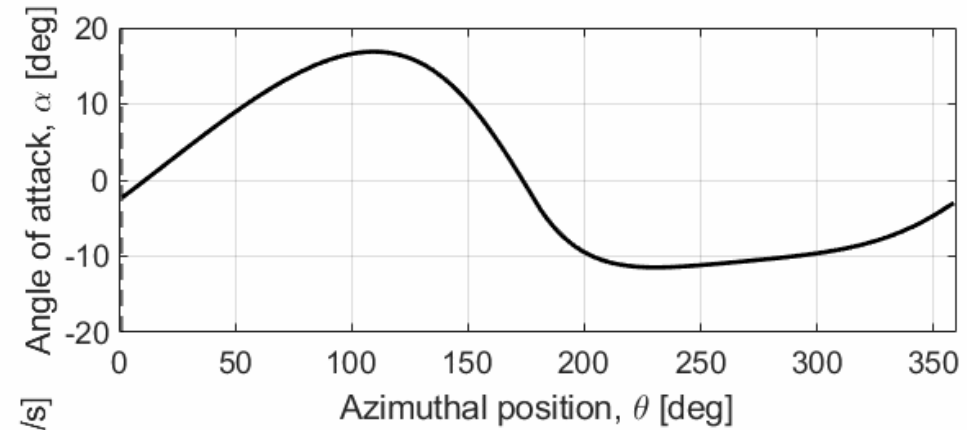
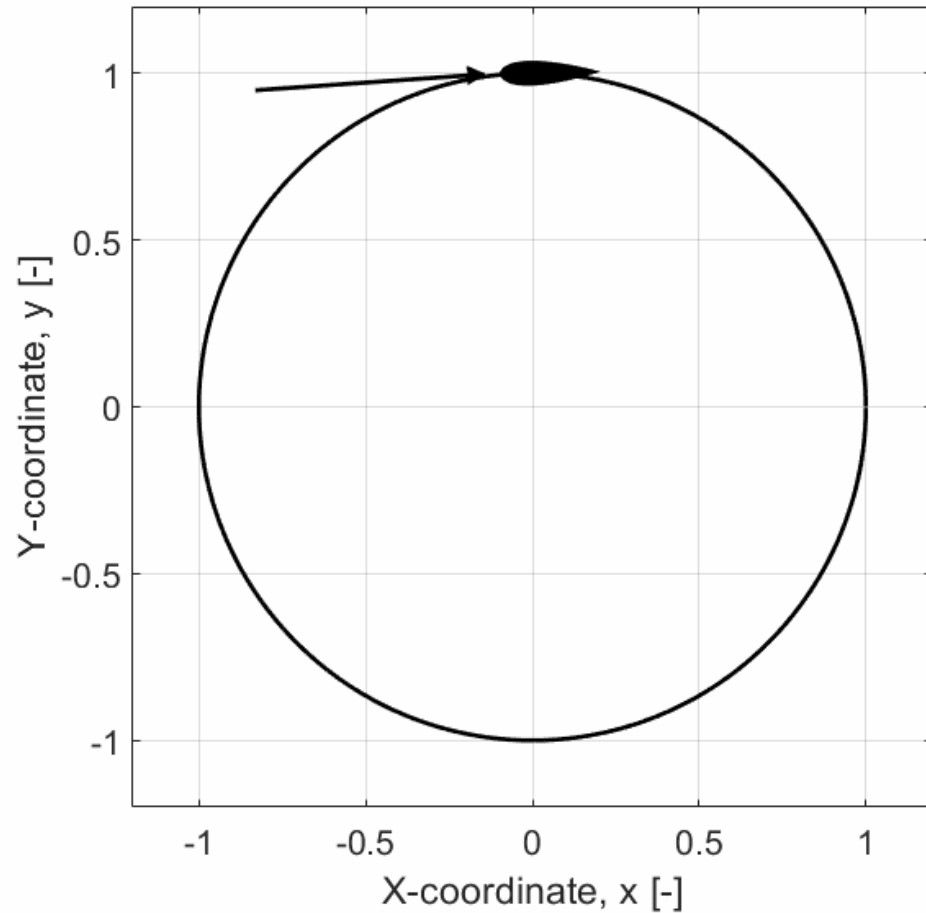
(e) All load downwind

X-velocity in the wake behind a 3D rotor with various load configurations. The loads are uniform and normal to the actuator.

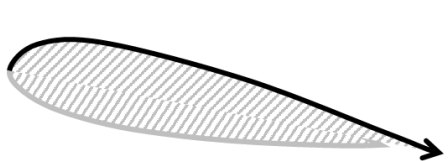
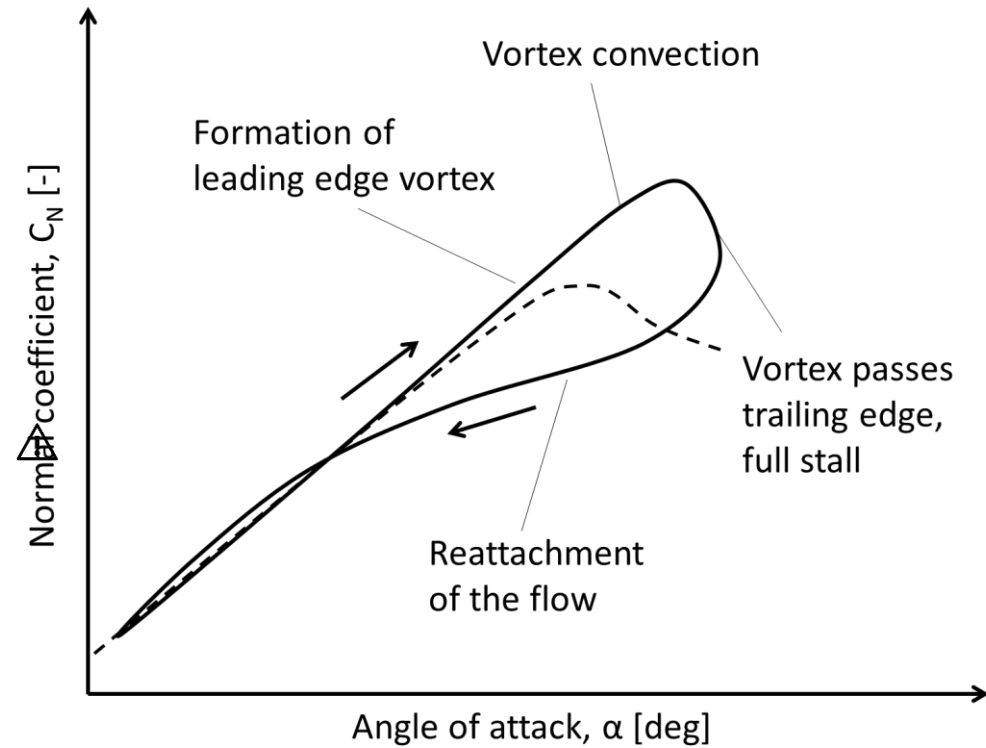
⚠ In 2D all wakes would be exactly the same since trailing vortices are infinitely far.

UNSTEADY AERODYNAMICS

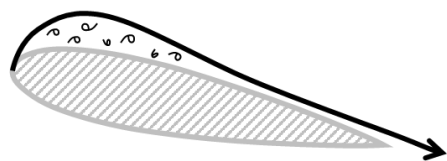
UNSTEADY AERODYNAMICS



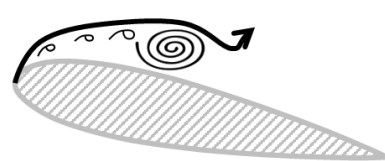
DYNAMIC STALL



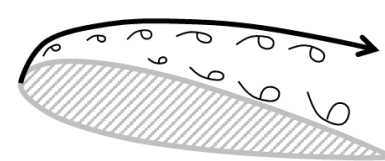
1. Attached flow, quasi static loading



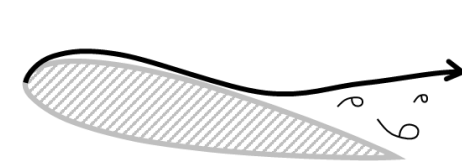
2. Post-static stall, formation of separation region near leading edge



3. LE vortex formation, growth and roll-up as it convect downstream. Lift, drag and pitch moment

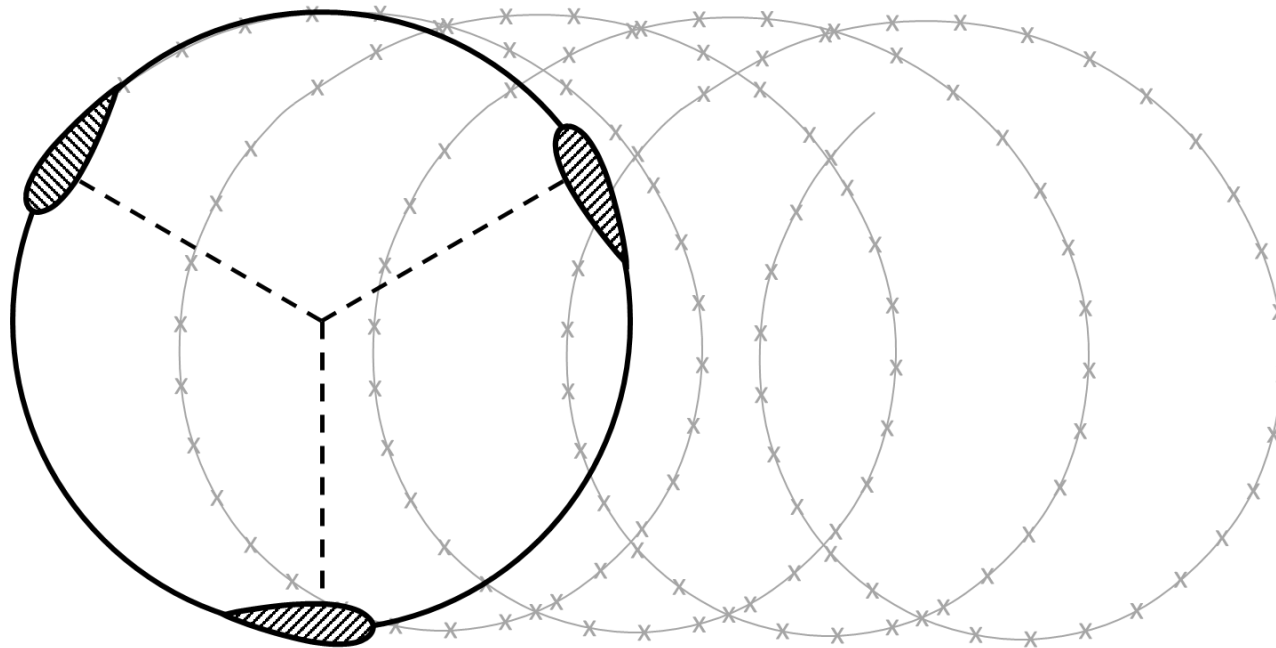


4. Flow separation, lift stall and pitch down moment



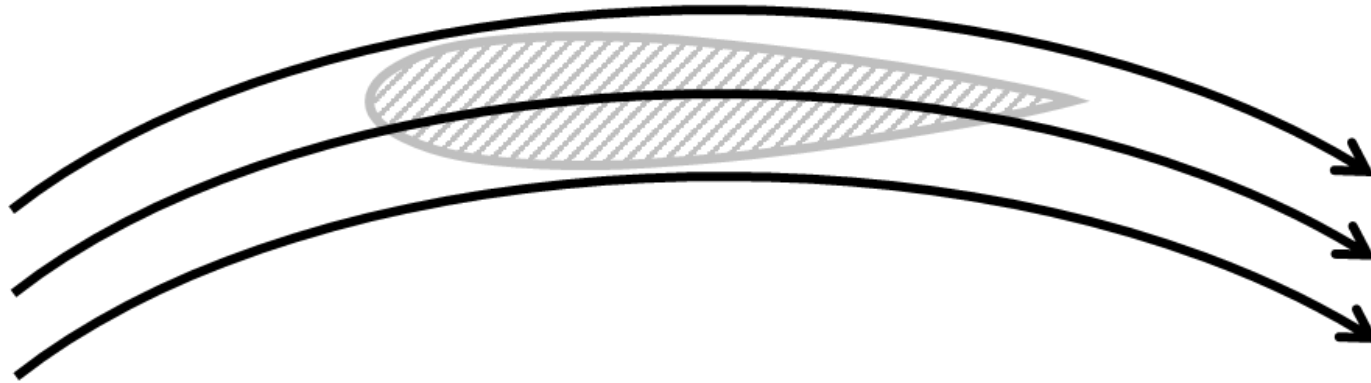
5. Flow reattachment, aerodynamic loads return to pre-stalled value

BLADE-VORTEX INTERACTION



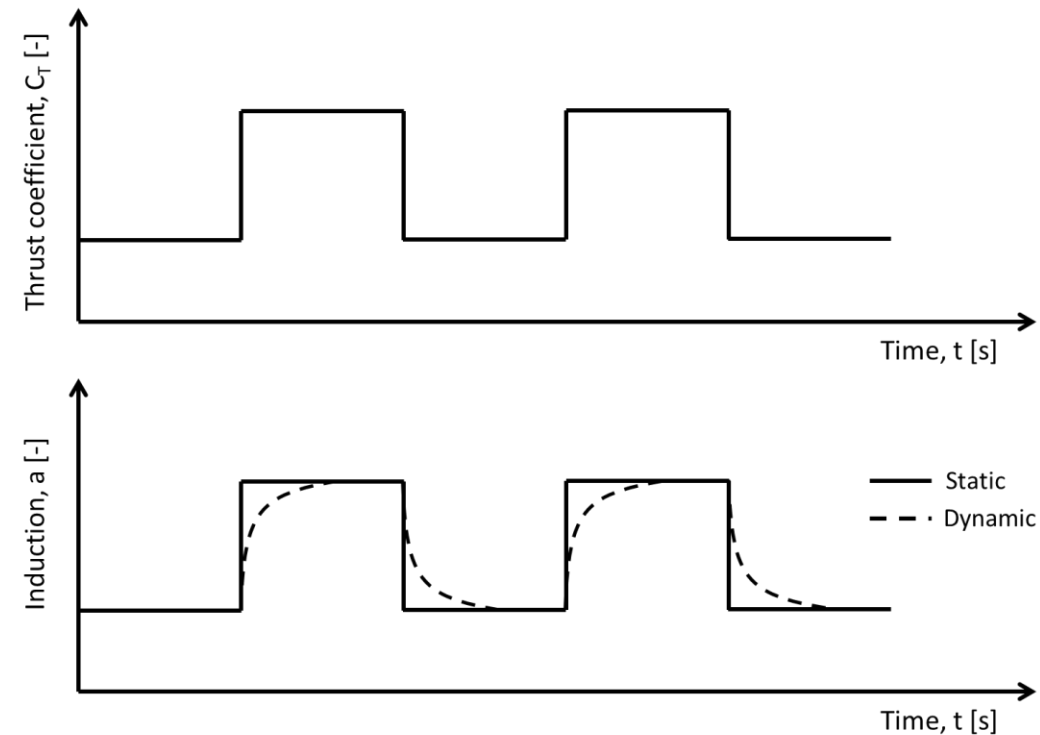
- Blades rotate in own wake
- Unsteady loads in downwind part of rotor

FLOW CURVATURE



- Curvilinear inflow due to rotation of rotor
- Transform geometric airfoil to rectilinear flow
- Additional camber and incidence angle

DYNAMIC INFLOW



- Unsteady relation between induction and unsteady loading and/or momentum of the flow
- Delay in response

UNSTEADY AERODYNAMICS

Remark:

Depending on the modelling choice, unsteady aerodynamics might be modelled implicitly. This is mostly true if the velocity and force field are calculated in space and time. If not, it needs to be included explicitly using extra engineering models.

SUMMARY

LEARNING OBJECTIVES

At the end of this lecture, you should be able to:

- Describe the history and future of VAWTs
- Define the basics of VAWT aerodynamics, incl. 2D velocity and force triangle
- Distinguish different modelling techniques and identify the difference between HAWT and VAWT wake dynamics
- Identify the role of unsteady aerodynamics in VAWT behaviour

Keywords: (1) History & future, (2) Basic VAWT aerodynamics , (3) Rotor representation, (4) VAWT modelling, (5) Wake dynamics, (6) Unsteady aerodynamics

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